A MILNOR-TYPE FORMULA FOR CONSTRUCTIBLE SHEAVES WITH NON-ISOLATED SINGULARITIES

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Abstract. Let $X \xrightarrow{f} Y \xrightarrow{g} S$ be morphisms of smooth schemes over a perfect field k of positive characteristic. Let $\mathcal{F}$ be a constructible complex on X. Assume that g is smooth, $g \circ f$ is $SS(\mathcal{F})$-transversal and f is $SS(\mathcal{F})$-transversal outside a closed subscheme Z. The cohomological non-acyclicity class is introduced in [23], and its geometric counterpart, called the geometric non-acyclicity class, is defined in [3]. In this paper, we prove a conjecture formulated in [3] that the image of the geometric non-acyclicity class under the cycle class map is the cohomological non-acyclicity class. This comparison yields a relative Milnor-type formula for non-isolated singularities. The proof uses the microlocalization of characteristic cycles developed in [18].

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1. Introduction

1.1. In complex singularity theory, the classical Milnor formula relates the topology of a nearby fibre to the algebraic information of an isolated critical point. Concretely, let $f : (\mathbb{C}^n, 0) \to (\mathbb{C}, 0)$ be a holomorphic function with an isolated critical point at the origin. Milnor [14] proved that its Milnor fibre M_f has the homotopy type of a bouquet of $\mu(f, 0)$ spheres of dimension $n-1$, where

$$\mu(f,0) := \dim_{\mathbb{C}} \mathcal{O}_{\mathbb{C}^n,0}/(\partial f/\partial z_1, \ldots, \partial f/\partial z_n) = \dim_{\mathbb{Q}} \widetilde{H}^{n-1}(M_f, \mathbb{Q}) \tag{1.1.1}$$

is called the *Milnor number*. For the positive characteristic case, Deligne established an analogue for the constant étale sheaf in [7, Théorème 2.4]. Let k be a perfect field of characteristic $p > 0$, and let Λ be a finite local ring whose residue field has characteristic $\ell \neq p$. Let X be a smooth scheme of pure dimension n over k, and let $f : X \to Y$ be a flat morphism to a smooth curve, with an isolated closed singular point x with respect to f. Deligne's Milnor formula gives

$$(-1)^n \mu(X/Y, x) = -\operatorname{dimtot} R\Phi_{\bar{x}}(\Lambda, f). \tag{1.1.2}$$

Here $\mu(X/Y, x) = (T^*_X X, df)_{T^*X,x}$ is the local intersection multiplicity of the zero section of T^*X with df, here df is regarded as a local section of T^*X. The complex $R\Phi(\Lambda, f)$ is the complex of vanishing cycles of the constant sheaf, and dimtot denotes dim + Sw.

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In general, Deligne [8] proposed a general form of the Milnor formula for a constructible complex $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$. Later, Beilinson and Saito constructed the singular support $SS(\mathcal{F})$ and the characteristic cycle $CC(\mathcal{F})$, respectively, see [4, 16]. Using the characteristic cycle, Saito confirmed the following general form of Milnor formula: for a morphism $f : X \to Y$ to a smooth curve and an isolated closed characteristic point x of f with respect to $\mathcal{F}$, one has

$$(CC(\mathcal{F}), df)_{T^*X,x} = -\operatorname{dimtot} R\Phi_{\bar{x}}(\mathcal{F}, f), \tag{1.1.3}$$

cf. [16, Theorem 5.9].

1.2. Consider morphisms

$$X \xrightarrow{f} Y \xrightarrow{g} S, \qquad h = g \circ f, \qquad \tau : Z \hookrightarrow X, \tag{1.2.1}$$

where S is a noetherian scheme on which ℓ is invertible, X and Y are separated schemes of finite type over S, and τ is a closed immersion. Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$ such that h is $SS(\mathcal{F})$-transversal and $f|_{X\setminus Z}$ is $SS(\mathcal{F}|_{X\setminus Z})$-transversal. These conditions imply that h is universally locally acyclic relatively to $\mathcal{F}$ and that f is universally locally acyclic relatively to $\mathcal{F}$ outside Z; see Proposition 2.9. The third named author and Y. Zhao [23] introduced the *cohomological non-acyclicity class* $\widetilde{C}^Z_{X/Y/S}(\mathcal{F}) \in H^0_Z(X, \mathcal{K}_{X/Y/S})$ where $\mathcal{K}_{X/Y/S}$ fits into the distinguished triangle

$$\mathcal{K}_{X/Y} \longrightarrow \mathcal{K}_{X/S} \xrightarrow{\alpha} \mathcal{K}_{X/Y/S} \xrightarrow{+1}, \tag{1.2.2}$$

see Section 2.13. When $S = \operatorname{Spec} k$ be the perfect field, Y is a smooth curve and $Z = \{x\}$ is a closed point, we have the cohomological Milnor formula

$$\widetilde{C}^Z_{X/Y/S}(\mathcal{F}) = -\operatorname{dimtot} R\Phi_{\bar{x}}(\mathcal{F}, f) \qquad \text{in} \qquad H^0_x(X, \mathcal{K}_{X/S}) \simeq H^0_x(X, \mathcal{K}_{X/Y/S}) \simeq \Lambda, \tag{1.2.3}$$

which is a cohomological version of (1.1.3).

In [3], we constructed the geometric incarnation of $\widetilde{C}^Z_{X/Y/S}(\mathcal{F})$, called the *geometric non-acyclicity class*, in the case where X, Y, S are smooth separated schemes of finite type and of pure dimension over the perfect field k, S is connected of dimension s, g is smooth of pure relative dimension $r > 0$. To be precise, let $q : \mathbb{P}(f^*T^*Y) \to X$ be the projectivization of cotangent bundle, and let $\overline{df}$ be the section of $q^*T^*X \otimes \mathcal{O}(1)$ induced by $df : f^*T^*Y \to T^*X$. Then the geometric non-acyclicity class is defined as

$$cc^Z_{X/Y/S}(\mathcal{F}) = (-1)^s q_{Z*}\left[c_{r-1}\left(\frac{q^*f^*T^*(Y/S)}{\mathcal{O}(-1)}\right) \cap \overline{df}^! CC(\mathcal{F})_{\mathcal{O}(1)}\right] \in \mathrm{CH}_s(Z). \tag{1.2.4}$$

Here the quotient bundle is taken on the reduced support $\mathbb{P}((df)^{-1}SS(\mathcal{F}))_{\mathrm{red}}$ of $\overline{df}^! CC(\mathcal{F})_{\mathcal{O}(1)}$, q_Z is its projection to Z, and $CC(\mathcal{F})_{\mathcal{O}(1)}$ is the twist of the conical cycle $q^*CC(\mathcal{F})$ by $\mathcal{O}(1)$, as in (2.14.6). See Section 2.14 for details.

It is natural to propose the following conjecture:

Conjecture 1.3 ([3, Conjecture 3.16])**.** *In the above setting, we have*

$$\widetilde{C}^Z_{X/Y/S}(\mathcal{F}) = \widetilde{\mathrm{cl}}(cc^Z_{X/Y/S}(\mathcal{F})). \tag{1.3.1}$$

Here, $\mathrm{CH}_s(Z) \xrightarrow{\widetilde{\mathrm{cl}}} H^0_Z(X, \mathcal{K}_{X/Y/S})$ is the cycle class map, see (5.1.2). This comparison can be viewed as a Milnor formula for non-isolated singularities. In the previous work [3, Theorem 1.6], we confirmed this conjecture in the case where k is an algebraically closed field, X is quasi-affine, $Y = \mathbb{A}^1_k$ and $S = \operatorname{Spec} k$.

1.4. Recently, T. Saito and the second and third named authors [18] gave a microlocal description of $SS(\mathcal{F})$ and $CC(\mathcal{F})$, which we recall below. For a constructible complex $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$, the (cohomological) characteristic class is defined by the composition

$$\delta_!\Lambda \xrightarrow{\mathrm{coev}} R\mathcal{H}om(\mathrm{pr}_1^*\mathcal{F}, \mathrm{pr}_2^!\mathcal{F}) \simeq D_{X/k}\mathcal{F} \boxtimes_k \mathcal{F} \xrightarrow{\mathrm{ev}} \delta_*\mathcal{K}_{X/k}, \tag{1.4.1}$$

where $\delta : X \to X \times_k X$ is the diagonal, pr_1, pr_2 are the natural projections. For an indeterminate t, let R be the henselization of $k[t]_{(t)}$ and let K be the fraction field of R. Let $\eta = \operatorname{Spec} K$, $s = \operatorname{Spec} k$, the category $D_c^b(X \overline{\times}_s \eta, \Lambda)$ is obtained from $D_c^b(X, \Lambda)$ by inflating the Galois action along the surjection $G(\overline{\eta}/\eta) \to G(\overline{s}/s)$, see 4.1. Following Verdier [21] and Laumon [11], we have the specialization functor

$$\nu_{X/X\times_k X} : D_c^b(X \times_k X, \Lambda) \longrightarrow D_c^b(TX \overline{\times}_s \eta, \Lambda) \tag{1.4.2}$$

and the Fourier transform functor

$$\mathscr{F}_{\psi, TX} : D_c^b(TX \overline{\times}_s \eta, \Lambda) \longrightarrow D_c^b(T^*X \overline{\times}_s \eta, \Lambda). \tag{1.4.3}$$

Put $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F}) = \mathscr{F}_{\psi,TX} \circ \nu_{X/X\times_k X} R\mathcal{H}om(\mathrm{pr}_1^*\mathcal{F}, \mathrm{pr}_2^!\mathcal{F})$. Here, TX and T^*X are the tangent and cotangent bundles of X, respectively, and π, $\pi^\vee$ are their projections to X; see Section 4.9 and 3 for details. Applying the functor $\mathscr{F}_{\psi,TX} \circ \nu_{X/X\times_k X}$ to the composition (1.4.1), we obtain

$$\Lambda \longrightarrow \mu\mathcal{H}om(\mathcal{F}, \mathcal{F}) \longrightarrow \pi^{\vee *}\mathcal{K}_{X/k}. \tag{1.4.4}$$

Since this is a G_k-invariant cohomology class, it descends to $CC_\mu(\mathcal{F}) \in H^0(T^*X, \pi^{\vee *}\mathcal{K}_{X/k})$. Samely, the support of $\mu\mathcal{H}om(\mathcal{F}, \mathcal{F})$ also descends to a closed subset of T^*X, which we denote by $SS_\mu(\mathcal{F})$. We have the following comparison theorem.

Theorem 1.5 ([18, Theorem 1.0.1 and 1.0.3])**.** *Assume X is smooth over k of pure dimension d. Then we have*

$$SS(\mathcal{F}) = SS_\mu(\mathcal{F}), \qquad \mathrm{cl}\,(CC(\mathcal{F})) = CC_\mu(\mathcal{F}). \tag{1.5.1}$$

Here, $\mathrm{cl} : \mathrm{Z}_d\,(SS(\mathcal{F})) \to H^0_{SS_\mu(\mathcal{F})}(T^*X, \pi^{\vee *}\mathcal{K}_{X/k})$ is the cycle class map. With this bridge between cycle classes and cohomological classes, we prove Conjecture 1.3. This is the main result of the paper.

Theorem 1.6 (Theorem 5.2)**.** *Under the assumptions of Conjecture 1.3, we have*

$$\widetilde{C}^Z_{X/Y/S}(\mathcal{F}) = \widetilde{\mathrm{cl}}(cc^Z_{X/Y/S}(\mathcal{F})) \quad \textit{in } H^0_Z(X, \mathcal{K}_{X/Y/S}). \tag{1.6.1}$$

For $S = \operatorname{Spec} k$ and $Y = \mathbb{A}^1_k$, Proposition 2.15 identifies the geometric class $cc^Z_{X/Y/S}(\mathcal{F})$ with $(df)^! CC(\mathcal{F})$, where df is the section of T^*X defined by f^*dt. Thus, the theorem implies

$$\widetilde{C}^Z_{X/\mathbb{A}^1_k/k}(\mathcal{F}) = \widetilde{\mathrm{cl}}((df)^! CC(\mathcal{F})). \tag{1.6.2}$$

In the case of isolated characteristic points, i.e. $\dim Z = 0$, it recovers the previous Milnor-type formulae (1.1.3) and (1.2.3).

1.7. We have the following applications of Theorem 1.6. Firstly, let $h : X \to S$ be a morphism between smooth schemes of pure dimension over k, with S connected of dimension s. Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$ be a constructible complex and assume that h is $SS(\mathcal{F})$-transversal. The differential morphism $dh : X \times_S T^*S \longrightarrow T^*X$ gives the geometric relative characteristic class

$$cc_{X/S}(\mathcal{F}) := (-1)^s (dh)^! CC(\mathcal{F}) \in \mathrm{CH}_s(X), \tag{1.7.1}$$

The refined pullback is supported on the zero section of $X \times_S T^*S$.

On the cohomology side, we have the characteristic class $C_{X/S}(\mathcal{F}) \in H^0(X, \mathcal{K}_{X/S})$ defined by the relative trace, cf. [22] and Section 2.12. We have the following comparison.

Theorem 1.8 (Theorem 6.2)**.** *Under these assumptions, we have*

$$C_{X/S}(\mathcal{F}) = \mathrm{cl}(cc_{X/S}(\mathcal{F})) \quad \text{in } H^0(X, \mathcal{K}_{X/S}). \tag{1.8.1}$$

This result confirms [22, Conjecture 3.8] and [3, Conjecture 3.6]. The proof is by applying Theorem 1.6 to the special case $X \xrightarrow{(h,0)} \mathbb{A}^1_S \to S$ and $Z = X$.

The second application is the conductor formula of Bloch in equal characteristic. Let Y be a smooth connected curve over k, let $y \in Y$ be a closed point, and let $f : X \to Y$ be a separated flat morphism, where X is smooth of pure dimension n over k. Assume that f is smooth outside $X_y = f^{-1}(y)$. Then the class $\widetilde{C}^{X_y}_{X/Y/k}(\Lambda)$ admits a unique lift $C^{X_y}_{X/Y/k}(\Lambda)$ under the map

$$H^0(X_y, \mathcal{K}_{X_y/k}) = H^0_{X_y}(X, \mathcal{K}_{X/k}) \xrightarrow{\alpha} H^0_{X_y}(X, \mathcal{K}_{X/Y/k}). \tag{1.8.2}$$

Theorem 1.9 (Theorem 6.6)**.** *In the above setting, we have*

$$C^{X_y}_{X/Y/k}(\Lambda) = (-1)^n \,\mathrm{cl}\left(c^X_{n,X_y}(\Omega^1_{X/Y}) \cap [X]\right) \quad \text{in} \quad H^0(X_y, \mathcal{K}_{X_y/k}), \tag{1.9.1}$$

where $c^X_{n,X_y}(\Omega^1_{X/Y})$ is the localized Chern class.

The proof uses the injectivity of α and identifies the refined intersection obtained from $CC(\Lambda) = (-1)^n[T^*_X X]$ with the localized Chern class; see [5] and [17, Lemma 2.1.4]. If f is proper, Corollary 6.7 gives

$$f_* C^{X_y}_{X/Y/k}(\Lambda) = (-1)^n \deg\left(c^X_{n,X_y}(\Omega^1_{X/Y}) \cap [X]\right) = -a_y(Rf_*\Lambda) \tag{1.9.2}$$

in $H^0_y(Y, \mathcal{K}_{Y/k}) = \Lambda$, where $a_y(Rf_*\Lambda) = \mathrm{rank}(Rf_*\Lambda)_{\bar\eta} - \mathrm{rank}(Rf_*\Lambda)_{\bar y} + \mathrm{Sw}_y(Rf_*\Lambda)$ is the Artin conductor. Here $\bar\eta$ is a geometric generic point of the strict localization at $\bar y$.

We also show that the localized characteristic class $C^0_Z(\mathcal{F})$ of Abbes-Saito can be represented using $CC(\mathcal{F})$, see Theorem 6.9.

1.10. The paper is organized as follows. In Section 2 we recall transversality, characteristic classes and non-acyclicity classes. Section 3 contains the properties of Fourier transform used in the proof. In Section 4 we study specialization and microlocalization. Section 5 is devoted to the comparison theorem. Applications of the main theorem are given in Section 6.

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Notation and conventions.

(1) Let S be a noetherian scheme and Sch_S the category of separated schemes of finite type over S.
(2) Let Λ be a noetherian ring such that $m\Lambda = 0$ for some integer m invertible on S unless otherwise stated explicitly.
(3) For any scheme $X \in \mathrm{Sch}_S$, we denote by $D^b_c(X, \Lambda)$ the derived category of complexes of Λ-modules with bounded constructible cohomology sheaves on X, and denote by $D_{\mathrm{ctf}}(X, \Lambda)$ the derived category of complexes of Λ-modules of finite Tor-dimension with constructible cohomology sheaves on X.

(4) For any separated morphism $f : X \to Y$ in Sch_S, we use the following duality notations

$$\mathcal{K}_{X/Y} = Rf^!\Lambda, \quad D_{X/Y}(-) = R\mathcal{H}om(-, \mathcal{K}_{X/Y}). \tag{1.10.1}$$

(5) For $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$ and $\mathcal{G} \in D_{\mathrm{ctf}}(Y, \Lambda)$ on S-schemes X and Y respectively, $\mathcal{F} \boxtimes^L_S \mathcal{G}$ denotes $\mathrm{pr}_1^*\mathcal{F} \otimes^L \mathrm{pr}_2^*\mathcal{G}$ on $X \times_S Y$.

(6) To simplify our notation, we omit R or L denoting the derived functors, unless stated explicitly or for $R\mathcal{H}om$ and $R\Gamma$.

(7) We use the standard stable ∞-categorical enhancement of $D^b_c(X, \Lambda)$. Cofibres and induced morphisms between them are constructed naturally in this enhancement; hence even in the ordinary derived category, we use the term *cofibre* rather than *mapping cone*. The corresponding cofibre sequences give distinguished triangles by the passage to the homotopy category $h\mathcal{D}^b_c(X, \Lambda) \simeq D^b_c(X, \Lambda)$.

(8) In chains of morphisms and diagrams, we use b.c., p.f., and adj. to denote base change, projection formula, and adjunction morphisms, respectively.

2. Transversality conditions and non-acyclicity classes

We recall the transversality conditions which characterize the smoothness of a morphism with respect to a constructible complex.

2.1. Let $f\colon X \to S$ be a separated morphism of finite type between noetherian schemes, and let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$. For a geometric point $\bar{x} \to X$, put $\bar{s} = f(\bar{x})$ and write $X_{(\bar{x})}$ and $S_{(\bar{s})}$ for the corresponding strict localizations. For an algebraic geometric point $\bar{t} \to S_{(\bar{s})}$, set

$$X_{(\bar{x})\bar{t}} := X_{(\bar{x})} \times_{S_{(\bar{s})}} \bar{t}. \tag{2.1.1}$$

Following [13, Section 2.1], we call f *locally acyclic relatively to* $\mathcal{F}$ if, for every such pair $(\bar{x}, \bar{t})$, the canonical morphism

$$\mathcal{F}_{\bar{x}} \longrightarrow R\Gamma\big(X_{(\bar{x})\bar{t}}, \mathcal{F}|_{X_{(\bar{x})\bar{t}}}\big) \tag{2.1.2}$$

is an isomorphism. We also say that $\mathcal{F}$ is *locally acyclic over* S. Moreover, we call f *universally locally acyclic relatively to* $\mathcal{F}$ if after every base change $S' \to S$, the morphism $f'\colon X' = X \times_S S' \to S'$ is locally acyclic relatively to $p^*\mathcal{F}$, where $p\colon X' \to X$ is the projection. Equivalently, we say that $\mathcal{F}$ is *universally locally acyclic over* S.

2.2. Consider the following cartesian diagram in Sch_S:

$$\begin{array}{ccc} X & \xrightarrow{i} & Y \\ {\scriptstyle p}\downarrow & \square & \downarrow{\scriptstyle f} \\ W & \xrightarrow{\delta} & T. \end{array} \tag{2.2.1}$$

Let $\mathcal{F} \in D_{\mathrm{ctf}}(Y, \Lambda)$ and $\mathcal{G} \in D_{\mathrm{ctf}}(T, \Lambda)$. Let $c_{\delta,f,\mathcal{F},\mathcal{G}}$ be the composition

$$\begin{aligned} c_{\delta,f,\mathcal{F},\mathcal{G}} : i^*\mathcal{F} \otimes^L p^*\delta^!\mathcal{G} &\xrightarrow{id\otimes \mathrm{b.c.}} i^*\mathcal{F} \otimes^L i^!f^*\mathcal{G} \\ &\xrightarrow{\mathrm{adj.}} i^!i_!(i^*\mathcal{F} \otimes^L i^!f^*\mathcal{G}) \\ &\xrightarrow[\simeq]{\mathrm{p.f.}} i^!(\mathcal{F} \otimes^L i_!i^!f^*\mathcal{G}) \xrightarrow{\mathrm{adj.}} i^!(\mathcal{F} \otimes^L f^*\mathcal{G}). \end{aligned} \tag{2.2.2}$$

In particular, we put $c_{\delta,f,\mathcal{F}} := c_{\delta,f,\mathcal{F},\Lambda} : i^*\mathcal{F} \otimes^L p^*\delta^!\Lambda \to i^!\mathcal{F}$. If $c_{\delta,f,\mathcal{F}}$ is an isomorphism, then we say that the morphism δ is $\mathcal{F}$-*transversal*. If $c_{i,\mathrm{id},\mathcal{F}}$ is an isomorphism, then we say i is $\mathcal{F}$-transversal.

In fact, the transversality condition is equivalent to local acyclicity by the following lemma.

Lemma 2.3 ([3, Lemma 2.2])**.** *Let $f : X \to S$ be a separated morphism of finite type between noetherian schemes and $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$. Then the following are equivalent:*

(1) *The morphism f is locally acyclic relatively to $\mathcal{F}$.*
(2) *The morphism f is universally locally acyclic relatively to $\mathcal{F}$.*
(3) *For any $\mathcal{G} \in D_{\mathrm{ctf}}(X, \Lambda)$, the canonical map*

$$D_{X/S}(\mathcal{G}) \boxtimes^L_S \mathcal{F} \to R\mathcal{H}om_{X\times_S X}(\mathrm{pr}_1^*\mathcal{G}, \mathrm{pr}_2^!\mathcal{F}) \tag{2.3.1}$$

is an isomorphism in $D_{\mathrm{ctf}}(X \times_S X, \Lambda)$, where $\mathrm{pr}_1 : X \times_S X \to X$ and $\mathrm{pr}_2 : X \times_S X \to X$ are the projections.
(4) *The canonical map*

$$D_{X/S}(\mathcal{F}) \boxtimes^L_S \mathcal{F} \to R\mathcal{H}om_{X\times_S X}(\mathrm{pr}_1^*\mathcal{F}, \mathrm{pr}_2^!\mathcal{F}) \tag{2.3.2}$$

is an isomorphism.
(5) *For any cartesian diagram*

$$\begin{array}{ccc} Y \times_S X & \xrightarrow{\mathrm{pr}_2} & X \\ {\scriptstyle \mathrm{pr}_1}\downarrow & \square & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[\delta]{} & S \end{array} \tag{2.3.3}$$

such that all schemes are noetherian and δ is separated of finite type, the morphism δ is $\mathcal{F}$-transversal.
(6) *For any cartesian diagram (2.3.3) and any $\mathcal{G} \in D_{\mathrm{ctf}}(S, \Lambda)$, the morphism $c_{\delta,f,\mathcal{F},\mathcal{G}}$ is an isomorphism.*

2.4. Keep the notation in the preceding square (2.2.1), and assume that $\delta^!\Lambda$ has finite Tor dimension. Following [23, § 2], we define the functor

$$\delta^\Delta_f \colon D_{\mathrm{ctf}}(Y, \Lambda) \longrightarrow D_{\mathrm{ctf}}(X, \Lambda), \qquad \delta^\Delta_f(\mathcal{G}) := \mathrm{cofib}\left(i^*\mathcal{G} \otimes^L p^*\delta^!\Lambda \xrightarrow{c_{\delta,f,\mathcal{G}}} i^!\mathcal{G}\right). \tag{2.4.1}$$

Recall that here the cofibre is taken as the cofibre in the fixed stable enhancements of the derived categories, thus δ^Δ_f is an exact functor equipped with a natural distinguished triangle

$$i^*\mathcal{G} \otimes^L p^*\delta^!\Lambda \xrightarrow{c_{\delta,f,\mathcal{G}}} i^!\mathcal{G} \xrightarrow{\partial_{\mathcal{G}}} \delta^\Delta_f(\mathcal{G}) \xrightarrow{+1} . \tag{2.4.2}$$

In particular, $\delta^\Delta_f(\mathcal{G}) = 0$ precisely when $c_{\delta,f,\mathcal{G}}$ is an isomorphism. For simplicity of notation, we suppress the subscript and write δ^Δ in the sequel whenever there is no risk of confusion.

2.5. If δ is a closed immersion, let $j \colon T \setminus W \hookrightarrow T$ be its open complement. The localization triangle

$$\delta_*\delta^!\Lambda \longrightarrow \Lambda \longrightarrow j_*\Lambda \xrightarrow{+1} \tag{2.5.1}$$

gives a natural identification

$$\delta^\Delta(\mathcal{G}) \simeq i^!(\mathcal{G} \otimes^L f^*j_*\Lambda), \tag{2.5.2}$$

under which the morphism $\partial_{\mathcal{G}}$ in (2.4.2) is induced by the adjunction unit $\Lambda \to j_*j^*\Lambda \simeq j_*\Lambda$. In fact, in the following we only use the case in which δ is a closed immersion.

When δ is a regular immersion, absolute purity provides an isomorphism $\delta^!\Lambda \simeq \Lambda(-r)[-2r]$ (cf. [10]) and hence the first term of (2.4.2) becomes $i^*\mathcal{G}(-r)[-2r]$.

We have the following functoriality results.

Lemma 2.6. *Retain the previous setting and the cartesian square*

$$\begin{array}{ccc} X & \xrightarrow{i} & Y \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle f} \\ W & \xrightarrow[\delta]{} & T, \end{array} \tag{2.6.1}$$

and assume that δ is a closed immersion and that $\delta^!\Lambda$ has finite Tor dimension.

(1) *Let $h\colon Y' \to Y$ be a separated morphism of finite type, and form the following diagram*

$$\begin{array}{ccc} X' & \xrightarrow{i'} & Y' \\ {\scriptstyle h_X}\downarrow & & \downarrow{\scriptstyle h} \\ X & \xrightarrow{i} & Y \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle f} \\ W & \xrightarrow[\delta]{} & T \end{array} \tag{2.6.2}$$

with cartesian squares. Then there are natural morphisms

$$\delta_f^{\Delta} h_* \longrightarrow h_{X*}\delta_{fh}^{\Delta}, \tag{2.6.3}$$

$$h_X^* \delta_f^{\Delta} \longrightarrow \delta_{fh}^{\Delta} h^*, \tag{2.6.4}$$

of functors $D_{\mathrm{ctf}}(Y', \Lambda) \to D_{\mathrm{ctf}}(X, \Lambda)$ and $D_{\mathrm{ctf}}(Y, \Lambda) \to D_{\mathrm{ctf}}(X', \Lambda)$, respectively. Moreover, (2.6.3) is an isomorphism when h is proper and (2.6.4) is an isomorphism when h is smooth.

(2) *Consider the diagram*

$$\begin{array}{ccc} X & \xrightarrow{i} & Y \\ {\scriptstyle p'}\downarrow & & \downarrow{\scriptstyle g} \\ W' & \xrightarrow{\delta'} & T' \\ {\scriptstyle b}\downarrow & & \downarrow{\scriptstyle t} \\ W & \xrightarrow[\delta]{} & T \end{array} \tag{2.6.5}$$

with cartesian squares, where $f = tg$ and $p = bp'$. Then there is a natural isomorphism

$$\delta_{tg}^{\Delta} \longrightarrow (\delta')_g^{\Delta} \tag{2.6.6}$$

of functors $D_{\mathrm{ctf}}(Y, \Lambda) \to D_{\mathrm{ctf}}(X, \Lambda)$. If the coefficient exchange $b^*\delta^!\Lambda \to (\delta')^!\Lambda$ is an isomorphism, then (2.6.5) is an isomorphism; this holds in particular if t is smooth.

(3) *Let $\varepsilon\colon Z \hookrightarrow W$ be a closed immersion, and consider the diagram*

$$\begin{array}{ccccc} X_Z & \xrightarrow{k} & X & \xrightarrow{i} & Y \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle f} \\ Z & \xrightarrow[\varepsilon]{} & W & \xrightarrow[\delta]{} & T \end{array} \tag{2.6.7}$$

with cartesian squares. Assume that $\varepsilon^!\Lambda$ has finite Tor dimension and that ε is $\delta^!\Lambda$-transversal. Then there are natural distinguished triangles of functors

$$k^*\delta_f^\Delta(-)\otimes^L q^*\varepsilon^!\Lambda \longrightarrow (\delta\varepsilon)_f^\Delta \longrightarrow \varepsilon_p^\Delta i^! \xrightarrow{+1} \tag{2.6.8}$$

and

$$\varepsilon_p^\Delta(i^*(-)\otimes^L p^*\delta^!\Lambda) \longrightarrow (\delta\varepsilon)_f^\Delta \longrightarrow k^!\delta_f^\Delta \xrightarrow{+1}. \tag{2.6.9}$$

(4) *Suppose that δ and i in (2.6.1) are regular immersions of codimensions r and s, respectively, and that the purity maps $\Lambda(-r)[-2r]\to\delta^!\Lambda$ and $\Lambda(-s)[-2s]\to i^!\Lambda$ are isomorphisms. Denote their normal bundles by $N_{W/T}$ and $N_{X/Y}$, and put $E:=p^*N_{W/T}/N_{X/Y}$ with $e:=\operatorname{rank}E=r-s$. Put $\mathcal{C}_E:=\operatorname{cofib}(\Lambda(-e)[-2e]\xrightarrow{c_e(E)}\Lambda)$. Then there is a natural distinguished triangle*

$$i^*(-)(-s)[-2s]\otimes^L\mathcal{C}_E\longrightarrow\delta_f^\Delta\longrightarrow i_{\mathrm{id}_Y}^\Delta\xrightarrow{+1}. \tag{2.6.10}$$

(5) *Assume that S is smooth over a field k in which m is invertible. Assume that Y is separated of finite type over S. Suppose that $\delta^!\Lambda$ is tensor-invertible. Then there is a natural isomorphism*

$$\delta_f^\Delta(D_{Y/S}(-))\xrightarrow{\simeq}(D_{X/S}(\delta_f^\Delta(-))\otimes^L p^*\delta^!\Lambda)[1] \tag{2.6.11}$$

of functors $D_{\mathrm{ctf}}(Y,\Lambda)\to D_{\mathrm{ctf}}(X,\Lambda)$.

Proof. (1)-(3). Using [23, Lemma 2.2 and 2.3] and (2.4.2) we get the desired natural morphisms. The isomorphic assertions in (1) and (2) are from base change.

(4). By [15, Proposition 2.3.2] and the purity isomorphism, $c_{\delta,f,\mathcal{G}}$ factors as

$$i^*\mathcal{G}(-r)[-2r]\xrightarrow{\mathrm{id}\otimes c_e(E)}i^*\mathcal{G}(-s)[-2s]\xrightarrow{c_{i,\mathrm{id}_Y,\mathcal{G}}}i^!\mathcal{G}. \tag{2.6.12}$$

The cofibre of the first arrow is $i^*\mathcal{G}(-s)[-2s]\otimes^L\mathcal{C}_E$. Taking cofibres gives (2.6.10).

(5). Using (2.2.2), compatibility with evaluation and $D_{X/S}i^!\simeq i^*D_{Y/S}$, $D_{X/S}i^*\simeq i^!D_{Y/S}$, we have $D_{X/S}(c_{\delta,f,\mathcal{G}})\otimes^L\mathrm{id}_{p^*\delta^!\Lambda}=c_{\delta,f,D_{Y/S}\mathcal{G}}$. Dualising the defining triangle (2.4.2), tensoring with $p^*\delta^!\Lambda$, and taking cofibres give (2.6.11). □

2.7. Let X be a scheme smooth over a field k, V be a vector bundle over X with zero section $0(X)$. For a closed subset C of V, we say it is *conical* if it is stable under the natural action of the multiplicative group $\mathbb{G}_{m,k}$, and call $C\cap 0(X)$ the *base* of C. We denote by $T^*_XX\subseteq T^*X$ the zero section of the cotangent bundle T^*X of X.

Now we recall the transversality condition in the geometrical sense, cf. Beilinson [4, 1.2] and Saito [16, Definitions 3.3, 3.5, and 7.1]. In the following definition we fix a closed conical subset C of T^*X.

(1) Let $h\colon W\to X$ be a morphism from a smooth scheme W over k. We say that h is *C-transversal* if $(C\times_X W)\cap dh^{-1}(T^*_WW)$ is contained in the zero section $T^*_XX\times_X W\subseteq T^*X\times_X W$, where $dh\colon T^*X\times_X W\to T^*W$ is the cotangent map induced by h.

If h is C-transversal, we define $h^\circ C$ to be the image of $C\times_X W$ under the map $dh\colon T^*X\times_X W\to T^*W$. By [16, Lemma 3.1], $h^\circ C$ is a closed conical subset of T^*W.

(2) Assume that X and C are purely of dimension d and that W is purely of dimension m. We say that a C-transversal map $h\colon W\to X$ is *properly C-transversal* if each irreducible component of $C\times_X W$ is of dimension m.

(3) We say that a morphism $f\colon X \to Y$ to a smooth scheme Y over k is *C-transversal* if the inverse image $df^{-1}(C)$ is contained in the zero section $T^*_Y Y \times_Y X \subseteq T^*Y \times_Y X$, where $df\colon T^*Y \times_Y X \to T^*X$ is the cotangent map induced by f.

If f is proper over the *base* $C \cap T^*_X X$ of C, we define $f_\circ C$ to be the image of $(df)^{-1}(C)$ under the projection $\mathrm{pr} : T^*Y \times_Y X \to T^*Y$. In this case, $f_\circ C$ is also a closed conical subset of T^*Y.

A point $x \in X$ is called a *C-characteristic point* of f if f is not C-transversal at x.

2.8. Let X be a smooth scheme of pure dimension d over a perfect field k, let Λ be a finite local ring such that the characteristic ℓ of the residue field of Λ is invertible in k, and let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$. We say that $\mathcal{F}$ is *microsupported* on a closed conical subset $C \subset T^*X$ if, for every pair of morphisms $X \xleftarrow{h} W \xrightarrow{f} Y$ of smooth schemes over k, where h is C-transversal and f is $h^\circ C$-transversal, the morphism f is locally acyclic relatively to $h^*\mathcal{F}$. By Beilinson [4, Theorem 1.3], there exists a smallest such $C = SS(\mathcal{F})$, called the *singular support* of $\mathcal{F}$. Every irreducible component of $SS(\mathcal{F})$ has dimension d.

We have the following properties for $SS(\mathcal{F})$.

Proposition 2.9 (cf. [4, 16])**.** *Let X, $\mathcal{F}$ be as above.*

(1) *We have $SS(\mathcal{F}) \cap T^*_X X = \mathrm{supp}(\mathcal{F})$.*
(2) *Let $f : X \to Y$ be a morphism to a smooth scheme Y over k. If f is $SS(\mathcal{F})$-transversal, then f is universally locally acyclic relatively to $\mathcal{F}$.*
(3) *Let $f : W \to X$ be a morphism from a smooth scheme W over k. If f is $SS(\mathcal{F})$-transversal, then f is $\mathcal{F}$-transversal.*
(4) *Let $f : X \to S$ be a morphism to a smooth scheme S over k. If f is $SS(\mathcal{F})$-transversal, then f is smooth on an open neighbourhood of $SS(\mathcal{F}) \cap T^*_X X = \mathrm{supp}(\mathcal{F})$.*

Proof. Assertions (1) and (2) follow from [4, Lemma 2.3(iii)] and [4, Lemma 2.1(ii)], respectively. Assertion (3) is [16, Proposition 8.13(1)]. For (4), apply the observation in [4, §1.2] that a C-transversal morphism is smooth on a neighbourhood of the base of C. □

Furthermore, if we assume k is perfect, Saito [16, Theorems 5.9 and 5.18] constructs a unique d-cycle $CC(\mathcal{F}) = \sum_a m_a[C_a], m_a \in \mathbb{Z}$, called the *characteristic cycle* where the C_a are the irreducible components of $SS(\mathcal{F})$. This cycle is uniquely characterized by the *Milnor formula*: for any étale morphism $j : U \to X$ and morphism $f : U \to \mathbf{A}^1_k$, and any isolated $j^\circ SS(\mathcal{F})$-characteristic point $u \in U$, we have

$$(j^*CC(\mathcal{F}), df)_{T^*U,u} = -\operatorname{dimtot}(R\Phi_f(j^*\mathcal{F})_{\bar{u}}). \tag{2.9.1}$$

Here $\bar{u}$ is a geometric point above u, the left-hand side denotes the local intersection multiplicity with the section df, cf. [16, Theorem 5.9].

2.10. Let X be a scheme separated of finite type, $\tau : Z \hookrightarrow X$ be a closed immersion with open complement $j : U := X \setminus Z \to X$. Let $\mathcal{G}, \mathcal{K} \in D^b_c(X, \Lambda)$ such that $j^*\mathcal{G} = 0$, then by the localization triangle we have $\tau_*\tau^!\mathcal{G} \xrightarrow{\simeq} \mathcal{G}$. Assume we have a composition of morphisms

$$\Lambda \xrightarrow{u} \mathcal{G} \xrightarrow{v} \mathcal{K}, \tag{2.10.1}$$

then it induces

$$\Lambda \xrightarrow{u} \mathcal{G} \simeq \tau_*\tau^!\mathcal{G} \xrightarrow{\tau_*\tau^! v} \tau_*\tau^!\mathcal{K} \tag{2.10.2}$$

which defines an element $C \in H^0_Z(X, \mathcal{K})$

If we furthermore demand $\mathcal{G},\ \mathcal{K} \in D_{\mathrm{ctf}}(X,\Lambda)$, put $\mathcal{G}^d = R\mathcal{H}om(\mathcal{G},\mathcal{K})$, then we have natural morphisms

$$v^d : \Lambda \longrightarrow R\mathcal{H}om(\mathcal{G},\mathcal{K}) = \mathcal{G}^d, \qquad u^d : \mathcal{G}^d = R\mathcal{H}om(\mathcal{G},\mathcal{K}) \longrightarrow R\mathcal{H}om(\Lambda,\mathcal{K}) \simeq \mathcal{K} \tag{2.10.3}$$

induced by v and u, respectively.

Lemma 2.11. *Retain the previous constructions, then $\mathcal{G}^d$ also satisfies $j^*\mathcal{G}^d = 0$, and the composition*

$$\Lambda \xrightarrow{v^d} \mathcal{G}^d \xrightarrow{u^d} \mathcal{K} \tag{2.11.1}$$

is equal to C as an element in $H^0_Z(X,\mathcal{K})$.

Proof. By [2, Exposé XVIII, Proposition 3.1.10] and the support condition of $\mathcal{G}$, we have

$$R\mathcal{H}om(\mathcal{G},\mathcal{K}) \simeq \tau_* R\mathcal{H}om(\tau^!\mathcal{G},\tau^!\mathcal{K}), \tag{2.11.2}$$

hence $\mathcal{G}^d$ is supported on Z. We verify the commutativity of the following diagram

$$\begin{array}{ccccc} \Lambda & & \xrightarrow{\tau^* u} & & \tau^!\mathcal{G} \\ & {\scriptstyle \tau^! v}\searrow & & {\scriptstyle \tau^! v\otimes \mathrm{id}}\downarrow & \quad\searrow{\scriptstyle \tau^! v} \\ R\mathcal{H}om(\tau^!\mathcal{G},\tau^!\mathcal{K}) & & \xrightarrow[\mathrm{id}\otimes\tau^! u]{} & R\mathcal{H}om(\tau^!\mathcal{G},\tau^!\mathcal{K})\otimes^L \tau^!\mathcal{G} & \xrightarrow[\mathrm{ev}]{} \tau^!\mathcal{K}. \end{array} \tag{2.11.3}$$

The rectangle is commutative by functoriality of the tensor product, and the triangle is commutative by adjunction. Using (2.11.2) and adjunction, the composition (2.11.1) is equivalent to

$$\Lambda \xrightarrow{\tau^* v^d} R\mathcal{H}om(\tau^!\mathcal{G},\tau^!\mathcal{K}) \xrightarrow{\tau^! u^d} \tau^!\mathcal{K} \tag{2.11.4}$$

which is the lower path of the diagram (2.11.3). The composition (2.10.2) defining C is the upper path of (2.11.3), hence by the commutativity we get the desired consequence.

□

2.12. In this subsection we assume S is a noetherian scheme, Λ is a noetherian ring such that $m\Lambda = 0$ for some integer m invertible on S. Let $h : X \to S$ be a scheme separated of finite type over S, let pr_1, pr_2: $X \times_S X \to X$ be the projections. We put

$$\mathcal{H}_{\mathcal{F}/S} := R\mathcal{H}om_{X\times_S X}(\mathrm{pr}_2^*\mathcal{F}, \mathrm{pr}_1^!\mathcal{F}), \qquad \mathcal{T}_{\mathcal{F}/S} := \mathcal{F}\boxtimes_S D_{X/S}\mathcal{F}. \tag{2.12.1}$$

Let $\delta_{X/S} : X \to X \times_S X$ be the diagonal morphism; by [6, Corollaire 3.1.12.2], we have a natural isomorphism

$$R\mathcal{H}om(\mathcal{F},\mathcal{F}) \xrightarrow{\simeq} \delta^!_{X/S}\mathcal{H}_{\mathcal{F}/S}, \tag{2.12.2}$$

hence we may define the coevaluation morphism

$$\mathrm{coev} : \Lambda \xrightarrow{\mathrm{id}} R\mathcal{H}om(\mathcal{F},\mathcal{F}) \xrightarrow{\simeq} \delta^!_{X/S}\mathcal{H}_{\mathcal{F}/S}. \tag{2.12.3}$$

On the other hand, we may define the evaluation morphism as follows:

$$\mathrm{ev} : \delta^*_{X/S}\mathcal{T}_{\mathcal{F}/S} \xrightarrow{\simeq} \mathcal{F}\otimes^L D_{X/S}\mathcal{F} \longrightarrow \mathcal{K}_{X/S}, \tag{2.12.4}$$

where the first morphism is a natural isomorphism by definition and the second is the adjoint of

$$\mathrm{id} \in R\mathcal{H}om(D_{X/S}\mathcal{F}, D_{X/S}\mathcal{F}) = R\mathcal{H}om\left(D_{X/S}\mathcal{F}, R\mathcal{H}om(\mathcal{F},\mathcal{K}_{X/S})\right). \tag{2.12.5}$$

Now assume $h : X \to S$ is locally acyclic relatively to $\mathcal{F}$, recall we have a natural isomorphism $\mathcal{T}_{\mathcal{F}/S} \xrightarrow{\simeq} \mathcal{H}_{\mathcal{F}/S}$ by Lemma 2.3. Hence we may form the following composition

$$\Lambda \xrightarrow{\text{coev}} \delta^!_{X/S}\mathcal{H}_{\mathcal{F}/S} \simeq \delta^!_{X/S}\mathcal{T}_{\mathcal{F}/S} \longrightarrow \delta^*_{X/S}\mathcal{T}_{\mathcal{F}/S} \xrightarrow{\text{ev}} \mathcal{K}_{X/S}, \tag{2.12.6}$$

where the middle morphism is

$$\delta^!_{X/S} = \delta^*_{X/S}\delta_{X/S*}\delta^!_{X/S} \xrightarrow{\text{adj.}} \delta^*_{X/S}. \tag{2.12.7}$$

The composition (2.12.6) defines a cohomology class $C_{X/S}(\mathcal{F}) \in H^0(X, \mathcal{K}_{X/S})$, which is called *cohomological characteristic class* of $\mathcal{F}$ relative to h. In the following discussion, we also use the adjoint form

$$\delta_{X/S!}\Lambda \xrightarrow{\text{coev}} \mathcal{H}_{\mathcal{F}/S} \simeq \mathcal{T}_{\mathcal{F}/S} \xrightarrow{\text{ev}} \delta_{X/S*}\mathcal{K}_{X/S} \tag{2.12.8}$$

of (2.12.6) as the definition of $C_{X/S}(\mathcal{F})$.

2.13. Retain the constructions of Section 2.12, we recall the definition of cohomological non-acyclicity class [23, Section 4]. Consider a commutative diagram in Sch_S

$$\begin{array}{ccccc} Z & \xrightarrow{\tau} & X & \xrightarrow{f} & Y, \\ & & \searrow^{h} & \swarrow_{g} & \\ & & S & & \end{array} \tag{2.13.1}$$

where $\tau : Z \to X$ is a closed immersion and g is a smooth morphism. We have the following commutative diagram, where the two squares are cartesian:

$$\begin{array}{ccccc} & & X & = & X \\ & & \downarrow{\scriptstyle \delta_{X/Y}} & \square & \downarrow{\scriptstyle \delta_{X/S}} \\ f & & X \times_Y X & \xrightarrow{i} & X \times_S X \\ & & \downarrow{\scriptstyle p} & \square & \downarrow{\scriptstyle f\times f} \\ & & Y & \xrightarrow{\delta_{Y/S}} & Y \times_S Y, \end{array} \tag{2.13.2}$$

where $\delta_{Y/S}$, $\delta_{X/Y}$, $\delta_{X/S}$ are diagonal morphisms, $i : X \times_Y X \to X \times_S X$ is a closed immersion; cf. [23, (4.2)]. Put

$$\mathcal{K}_{X/Y/S} := \delta^{\Delta}_{Y/S}\mathcal{K}_{X/S} \simeq \delta^*_{X/Y}\delta^{\Delta}_{Y/S}\delta_{X/S*}\mathcal{K}_{X/S} \in D_{\mathrm{ctf}}(X, \Lambda). \tag{2.13.3}$$

By [23, Section 4.2] it sits into a distinguished triangle

$$\mathcal{K}_{X/Y} \longrightarrow \mathcal{K}_{X/S} \xrightarrow{\alpha} \mathcal{K}_{X/Y/S} \xrightarrow{+1} . \tag{2.13.4}$$

Let $\mathcal{F}$ be a constructible complex in $D_{\mathrm{ctf}}(X, \Lambda)$ such that $h : X \to S$ is locally acyclic relatively to $\mathcal{F}$ and $f|_{X\setminus Z} : X \setminus Z \to Y$ is locally acyclic relatively to $\mathcal{F}$; in this case we still have the natural isomorphism $\mathcal{T}_{\mathcal{F}/S} \simeq \mathcal{H}_{\mathcal{F}/S}$. We may consider the following composition

$$\begin{aligned} \Lambda \xrightarrow{\text{coev}} \delta^!_{X/S}\mathcal{H}_{\mathcal{F}/S} \simeq \delta^!_{X/Y}i^!\mathcal{H}_{\mathcal{F}/S} \to \delta^*_{X/Y}i^!\mathcal{H}_{\mathcal{F}/S} \\ \xrightarrow{\partial} \delta^*_{X/Y}\delta^{\Delta}_{Y/S}\mathcal{H}_{\mathcal{F}/S} \xrightarrow{\text{ev}} \delta^*_{X/Y}\delta^{\Delta}_{Y/S}\delta_{X/S*}\mathcal{K}_{X/S} = \mathcal{K}_{X/Y/S}, \end{aligned} \tag{2.13.5}$$

here the morphism ∂ sits in the defining distinguished triangle

$$i^*\mathcal{H}_{\mathcal{F}/S} \otimes^L p^*\delta^!_{Y/S}\Lambda \xrightarrow{c_{\delta_{Y/S}, f\times f, \mathcal{H}_{\mathcal{F}/S}}} i^!\mathcal{H}_{\mathcal{F}/S} \xrightarrow{\partial} \delta^{\Delta}_{Y/S}\mathcal{H}_{\mathcal{F}/S} \xrightarrow{+1} . \tag{2.13.6}$$

By [23, Section 4.3], $c_{\delta_{Y/S},f\times f,\mathcal{H}_{\mathcal{F}/S}}$ is an isomorphism on $X\setminus Z$, hence $\delta^*_{X/Y}\delta^{\Delta}_{Y/S}\mathcal{H}_{\mathcal{F}/S}$ is supported on Z and the composition (2.13.5) defines a cohomological class $\widetilde{C}^Z_{X/Y/S}(\mathcal{F})\in H^0_Z(X,\mathcal{K}_{X/Y/S})$, as (2.10.2). We call $\widetilde{C}^Z_{X/Y/S}$ the *cohomological non-acyclicity class*. This class is compactible with arbitary pullback and proper pushforward, cf. [23, Section 4].

2.14. We introduce the geometric counterpart of non-acyclicity class now, see [3, Section 3] for details. Let k be a perfect field of characteristic $p>0$ and Λ be a finite local ring whose residue field is of characteristic $\ell\neq p$. Let Sm_k be the category of smooth schemes over k. Again we consider the diagram (2.13.1), but here we additionally demand that X, Y, S are in Sm_k. Assume X, Y, S are of pure dimension, S is connected of dimension s, g is smooth of relative dimension $r>0$, $n=\dim X$. Let $\mathcal{F}\in D_{\mathrm{ctf}}(X,\Lambda)$ be a constructible complex such that h is $SS(\mathcal{F})$-transversal and $f|_{X\setminus Z}$ is $SS(\mathcal{F}|_{X\setminus Z})$-transversal.

Consider the vector bundle $T^*Y\times_Y X$ over X which induces a projective bundle

$$q:\mathbb{P}(T^*Y\times_Y X)\longrightarrow X, \tag{2.14.1}$$

which is equipped with the tautological line bundle $\mathcal{O}(-1)$ as a subbundle of $q^*(T^*Y\times_Y X)$. The composition

$$\mathcal{O}(-1)\hookrightarrow q^*(T^*Y\times_Y X)\xrightarrow{q^*df} q^*T^*X \tag{2.14.2}$$

defines a section

$$\overline{df}\in H^0\left(\mathbb{P}(T^*Y\times_Y X),q^*T^*X\otimes\mathcal{O}(1)\right) \tag{2.14.3}$$

of the vector bundle $q^*T^*X\otimes\mathcal{O}(1)$ under the natural isomorphism

$$\mathcal{H}om(\mathcal{O}(-1),q^*T^*X)\simeq q^*T^*X\otimes\mathcal{O}(1). \tag{2.14.4}$$

Let $D\subset T^*X$ be a conical closed subscheme represented by its sheaf of graded algebra $D=\mathrm{Spec}_X(\bigoplus_{a\geq 0}\mathcal{A}_a)$. We put

$$D_{\mathcal{O}(1)}:=\mathrm{Spec}_{\mathbb{P}(T^*Y\times_Y X)}\left(\bigoplus_{a\geq 0}q^*\mathcal{A}_a\otimes\mathcal{O}(-a)\right). \tag{2.14.5}$$

which is again a conical closed subscheme. For $CC(\mathcal{F})=\sum_a m_a[C_a]$, set

$$CC(\mathcal{F})_{\mathcal{O}(1)}:=\sum_a m_a[(C_a)_{\mathcal{O}(1)}]. \tag{2.14.6}$$

Consider the following cartesian diagram

$$\begin{array}{ccc}\mathbb{P}((df)^{-1}SS(\mathcal{F})) & \longrightarrow & SS(\mathcal{F})_{\mathcal{O}(1)}\\ \downarrow & & \downarrow\\ \mathbb{P}(T^*Y\times_Y X) & \xrightarrow[\overline{df}]{} & q^*T^*X\otimes\mathcal{O}(1).\end{array} \tag{2.14.7}$$

By the transversality conditions we can show that on the closed subscheme $\mathbb{P}((df)^{-1}SS(\mathcal{F}))$ of $\mathbb{P}(T^*Y\times_Y X)$, $\mathcal{O}(-1)$ is a subbundle of $q^*f^*T^*(Y/S)$, see [3, Section 3.8]. Moreover, the projection q sends $\mathbb{P}\left((df)^{-1}SS(\mathcal{F})\right)$ into the closed subscheme Z of X. Hence we may define the *geometric non-acyclicity class* of $\mathcal{F}$ by

$$cc^Z_{X/Y/S}(\mathcal{F}):=(-1)^s q_{Z*}\left[c_{r-1}\left(\frac{q^*f^*T^*(Y/S)}{\mathcal{O}(-1)}\right)\cap\overline{df}^!CC(\mathcal{F})_{\mathcal{O}(1)}\right]\in\mathrm{CH}_s(Z). \tag{2.14.8}$$

Here, the quotient bundle is taken on $\mathbb{P}((df)^{-1}SS(\mathcal{F}))$. For the special case $S=\operatorname{Spec}k$ and $Y=\mathbb{A}^1_k$, we have the following calculation.

Proposition 2.15 ([3, Proposition 3.14])**.** *Retain the previous setting. Let $S=\operatorname{Spec}k$, $Y=\mathbb{A}^1_k$ with coordinate t. With slight abuse of notation, we write $df: X\longrightarrow X\times_{\mathbb{A}^1_k}T^*\mathbb{A}^1_k\longrightarrow T^*X$ for the section defined by f^*dt, then we have*

$$cc^Z_{X/\mathbb{A}^1_k/k}(\mathcal{F})=(df)^!CC(\mathcal{F})\quad \text{in } \mathrm{CH}_0(Z). \tag{2.15.1}$$

Proof. Notice that the bundle $T^*\mathbb{A}^1_k$ is trivial, and the section $dt: X\longrightarrow T^*\mathbb{A}^1_k\times_{\mathbb{A}^1_k}X$ identifies $\mathbb{P}(T^*\mathbb{A}^1_k\times_{\mathbb{A}^1_k}X)$ with X. Hence the line bundle $\mathcal{O}(-1)$ is trivial, $\overline{df}=df$. In (2.14.8), $r=1$, $s=0$ and the Chern class involved is $c_0=1$, hence the consequence follows. □

We give an expression of $cc^Z_{X/Y/S}(\mathcal{F})$ using the relative cotangent bundle. Consider the setting at the beginning of Section 2.14. By Proposition 2.9 the morphism $h: X\to S$ is smooth on an open neighbourhood of $\operatorname{supp}(\mathcal{F})$, we restrict to this neighbourhood and assume h is smooth. Let $p_{X/S}: T^*X\longrightarrow T^*(X/S)$, $q_S:\mathbb{P}(f^*T^*(Y/S))\longrightarrow X$ be the natural projections, then the restriction of $p_{X/S}$ on $SS(\mathcal{F})$ is finite. The relative differential $\mathcal{O}(-1)\to q_S^*f^*T^*(Y/S)\xrightarrow{d_Sf}q_S^*T^*(X/S)$ defines a section

$$\overline{d_Sf}:\mathbb{P}(f^*T^*(Y/S))\longrightarrow q_S^*T^*(X/S)\otimes\mathcal{O}(1). \tag{2.15.2}$$

Lemma 2.16 ([3, Lemma 3.21])**.** *With the above setting, we have*

$$cc^Z_{X/Y/S}(\mathcal{F})=(-1)^s(q_S)_*\left[c_{r-1}\left(\frac{q_S^*f^*T^*(Y/S)}{\mathcal{O}(-1)}\right)\cap\overline{d_Sf}^!\left(p_{X/S*}CC(\mathcal{F})\right)_{\mathcal{O}(1)}\right], \tag{2.16.1}$$

here $p_{X/S*}$ is taken on the characteristic-cycle support, and $(q_S)_*$ is taken on the reduced intersection support over Z.

2.17. We record some calculations on projective bundles. Let X be a scheme separated and of finite type over k with closed subscheme $\tau: Z\hookrightarrow X$, let E be a vector bundle over X of rank r with zero section $X\xrightarrow{0}E$, projection $E\xrightarrow{\pi}X$ and the projectivization $\mathbb{P}(E)\xrightarrow{q}X$. Put $E^\times:=E\setminus 0(X)$ which is equipped with natural morphism $p: E^\times\to\mathbb{P}(E)$, put $\pi^\times:=q\circ p$, $\overline{\tau}:\overline{Z}:=Z\times_X\mathbb{P}(E)\hookrightarrow\mathbb{P}(E)$. Let $\xi=c_1(\mathcal{O}_{\mathbb{P}(E)}(1))\in H^2(\mathbb{P}(E),\Lambda(1))$ be the (étale) first Chern class of the canonical line bundle of $\mathbb{P}(E)$.

$$\begin{array}{ccc} E^\times & \xrightarrow{p} & \mathbb{P}(E) \\ & \searrow^{\pi^\times} & \downarrow q \\ & & X \end{array} \tag{2.17.1}$$

Let $\mathcal{K}\in D^b_c(X,\Lambda)$ be a constructible complex, we fix a cohomology class $c\in H^m_{\overline{Z}}(\mathbb{P}(E),q^*\mathcal{K})$. By the projective bundle formula (cf. [15, Théorème 1.2])

$$\sum_{j=0}^{r-1}\xi^j\cup q^*(-):\bigoplus_{j=0}^{r-1}\mathcal{K}(-j)[-2j]\xrightarrow{\simeq}q_*q^*\mathcal{K}, \tag{2.17.2}$$

we have a natural isomorphism

$$\bigoplus_{j=0}^{r-1}R\Gamma_Z\left(X,\mathcal{K}(-j)[-2j]\right)\xrightarrow{\simeq}R\Gamma_{\overline{Z}}\left(\mathbb{P}(E),q^*\mathcal{K}\right), \tag{2.17.3}$$

hence we have a unique expression

$$c=\sum_{j=0}^{r-1}\xi^j\cup q^*c_j,\qquad c_j\in H_Z^m\left(X,\mathcal{K}(-j)[-2j]\right). \tag{2.17.4}$$

Notice that $p^*\mathcal{O}_{\mathbb{P}(E)}(1)$ is trivial, hence $p^*\xi=0$, we have

$$p^*c=\left(\pi^{\times}\right)^*c_0. \tag{2.17.5}$$

By the canonical exact sequence

$$0\longrightarrow\mathcal{O}_{\mathbb{P}(E)}(-1)\longrightarrow q^*E\longrightarrow q^*E/\mathcal{O}_{\mathbb{P}(E)}(-1)\longrightarrow 0 \tag{2.17.6}$$

and the Whitney formula, we have

$$c_{r-1}\left(q^*E/\mathcal{O}(-1)\right)=\sum_{j=0}^{r-1}q^*c_j(E)\cup\xi^{r-1-j} \tag{2.17.7}$$

and hence

$$q_*\left(c_{r-1}\left(q^*E/\mathcal{O}(-1)\right)\cup\xi^k\right)=\sum_{j=0}^{k}c_j(E)s_{k-j}(E)=\begin{cases}1,k=0;\\0,\text{else},\end{cases} \tag{2.17.8}$$

where $s_{k-j}(E)$ is the Segre class. Combining with projection formula, we get

$$q_*\left(c_{r-1}\left(q^*E/\mathcal{O}_{\mathbb{P}(E)}(-1)\right)\cup c\right)=c_0. \tag{2.17.9}$$

2.18. Retain the construction of the previous section. Let $j:E^{\times}\hookrightarrow E$ and $i:0(X)\hookrightarrow E$ be the open and closed immersion, respectively. Applying $\pi_!$ to the (rotated) localization triangle

$$\mathrm{id}\longrightarrow i_*i^*\longrightarrow j_!j^*[1]\xrightarrow{+1}, \tag{2.18.1}$$

we get a morphism $b:\Lambda\to\pi_!^{\times}\Lambda[1]$. Combining with the morphism $c:\Lambda\to\overline{\tau}_*\overline{\tau}^!q^*\mathcal{K}[m]$ induced by $c\in H_{\overline{Z}}^m(\mathbb{P}(E),q^*\mathcal{K})$, we get a composition

$$\begin{aligned}\Lambda\xrightarrow{b}\pi_!^{\times}\Lambda[1]&\xrightarrow{c}\pi_!^{\times}\left(p^*\overline{\tau}_*\overline{\tau}^!q^*\mathcal{K}[m]\right)[1]\\&\xrightarrow{\text{b.c.}}\pi_!^{\times}\left(\pi^{\times *}\tau_*\tau^!\mathcal{K}[m]\right)[1]\\&\xrightarrow{\text{p.f.}}\tau_*\tau^!\mathcal{K}[m]\otimes^L\pi_!^{\times}\Lambda[1]\\&\simeq\tau_*\tau^!\left(\mathcal{K}\otimes^L\pi_!^{\times}\Lambda[1]\right)[m].\end{aligned} \tag{2.18.2}$$

Moreover, we can define another composition

$$\begin{aligned}\Lambda\simeq\Lambda\otimes^L\Lambda&\xrightarrow{c_0\otimes b}\tau_*\tau^!\mathcal{K}[m]\otimes^L\pi_!^{\times}\Lambda[1]\\&\xrightarrow{\simeq}\tau_*\tau^!\left(\mathcal{K}\otimes^L\pi_!^{\times}\Lambda[1]\right)[m].\end{aligned} \tag{2.18.3}$$

The equality (2.17.5) of cohomology classes implies the two compositions (2.18.2) and (2.18.3) are equal.

3. Fourier transform

In this section we fix a field k of characteristic $p>0$. Let Λ be a finite local ring with residue characteristic $\ell\neq p$ for some prime number ℓ. Choose a nontrivial character $\psi:\mathbb{F}_p\to\Lambda^{\times}$ after a finite étale faithfully flat coefficient extension if necessary (see [18, Lemma 2.1.1]), and put $\psi^{\vee}(t)=\psi(-t)$. Let $\mathcal{L}_{\psi}$ be the locally constant constructible sheaf of free Λ-modules of rank 1 on $\mathbb{A}_k^1$ corresponding to the character ψ.

3.1. Let X be a noetherian scheme over k, let V be a vector bundle over X of rank r, let $V^\vee$ be the dual bundle of V. Denote the two projections from $V \times_X V^\vee$ by pr_1, pr_2 and the pairing by $\mu = \langle -, - \rangle : V \times_X V^\vee \to \mathbb{A}^1_k$. We define the *Fourier transform* by

$$\mathscr{F}_{\psi,V} : D^b_c(V, \Lambda) \to D^b_c(V^\vee, \Lambda), \quad \mathscr{F}_{\psi,V}(-) := \mathrm{pr}_{2!}\left(\mathrm{pr}_1^*(-) \otimes^L \mu^* \mathcal{L}_\psi\right). \tag{3.1.1}$$

Here we use the unshifted convention of [18, Definition 2.1.2].

We recall some standard properties of Fourier transform.

Proposition 3.2 ([18, Lemma 2.1.3, Proposition 2.1.6, 2.1.9, Corollary 2.1.18])**.** *Retain the previous notation, let $0 : X \to V$ and $0^\vee : X \to V^\vee$ be the zero sections, let $\pi : V \to X$ and $\pi^\vee : V^\vee \to X$ be the natural projections.*

(1) *We have canonical isomorphisms*

$$(0^\vee)^* \circ \mathscr{F}_{\psi,V} \simeq \pi_!, \qquad \mathscr{F}_{\psi,V} \circ 0_* \simeq (\pi^\vee)^* \tag{3.2.1}$$

of functors $D^b_c(V, \Lambda) \to D^b_c(X, \Lambda)$ and $D^b_c(X, \Lambda) \to D^b_c(V^\vee, \Lambda)$, respectively.

(2) *We have canonical isomorphism*

$$\mathscr{F}_{\psi,V} \circ \pi^! \simeq 0^\vee_* \tag{3.2.2}$$

of functors $D^b_c(X, \Lambda) \to D^b_c(V^\vee, \Lambda)$.

(3) *We have canonical isomorphism*

$$\mathscr{F}_{\psi,V} \circ \pi^* \simeq 0^\vee_*(-r)[-2r] \tag{3.2.3}$$

of functors $D^b_c(X, \Lambda) \to D^b_c(V^\vee, \Lambda)$.

(4) *We have canonical isomorphism*

$$\mathscr{F}_{\psi^\vee,V^\vee} \circ \mathscr{F}_{\psi,V}(-) \simeq - \otimes_\Lambda \pi^* \pi^\vee_! \Lambda \tag{3.2.4}$$

of functors $D^b_c(V, \Lambda) \to D^b_c(V, \Lambda)$. Moreover, we have an adjoint pair $\mathscr{F}_{\psi^\vee,V^\vee}(-)(r)[2r] \dashv \mathscr{F}_{\psi,V}(-)$ with isomorphic counit

$$\mathscr{F}_{\psi^\vee,V^\vee}(r)[2r] \circ \mathscr{F}_{\psi,V} \simeq \mathrm{id} \tag{3.2.5}$$

as functors $D^b_c(V, \Lambda) \to D^b_c(V, \Lambda)$.

(5) *For a linear map $q : W \to V$ of vector bundles and its dual map $q^\vee : V^\vee \to W^\vee$, we have canonical isomorphism*

$$\mathscr{F}_{\psi,V} q_! \simeq (q^\vee)^* \mathscr{F}_{\psi,W} \tag{3.2.6}$$

of functors $D^b_c(V, \Lambda) \to D^b_c(V^\vee, \Lambda)$.

(6) *For a subbundle $i : V' \hookrightarrow V$ of codimension s, we have canonical isomorphisms*

$$\mathscr{F}_{\psi,V'}(i^*(-))(-s)[-2s] \simeq i^\vee_! \mathscr{F}_{\psi,V}(-), \tag{3.2.7}$$

$$\mathscr{F}_{\psi,V'}(i^!(-)) \simeq i^\vee_* \mathscr{F}_{\psi,V}(-) \tag{3.2.8}$$

of functors $D^b_c(V', \Lambda) \to D^b_c(V'^\vee, \Lambda)$.

(7) *For addition morphism $\mathrm{add} : V \times_X V \to V$, we have canonical isomorphism*

$$\mathscr{F}_{\psi,V} \mathrm{add}_!(\mathcal{A} \boxtimes_X \mathcal{B}) \simeq \mathscr{F}_{\psi,V} \mathcal{A} \otimes^L \mathscr{F}_{\psi,V} \mathcal{B}, \tag{3.2.9}$$

where $\mathcal{A},\ \mathcal{B} \in D^b_c(V, \Lambda)$.

(8) *Suppose we have morphism $f : X \to S$ with its dualizing complex $\mathcal{K}_{X/S} := f^!\Lambda$, then we have canonical isomorphism*

$$\begin{aligned} \mathscr{F}_{\psi^\vee,V} D_{V/S} &\simeq D_{V^\vee/S}(\mathscr{F}_{\psi,V}(-))(-r)[-2r] \\ &\simeq R\mathcal{H}om(\mathscr{F}_{\psi,V}(-), \pi^{\vee *}\mathcal{K}_{X/S}) \end{aligned} \tag{3.2.10}$$

of functors $D_c^b(V,\Lambda) \to D_c^b(V^\vee,\Lambda)$.

All these results can be found in [18, § 2.1], for completeness we record the proof below.

Proof. (1). Consider the following cartesian diagrams

$$\begin{array}{ccc} V & \xrightarrow{\mathrm{id}_V \times_X 0^\vee} & V \times_X V^\vee \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_2} \\ X & \xrightarrow[0^\vee]{} & V^\vee \end{array} \qquad \begin{array}{ccc} V^\vee & \xrightarrow{0 \times_X \mathrm{id}_{V^\vee}} & V \times_X V^\vee \\ {\scriptstyle\pi^\vee}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_1} \\ X & \xrightarrow[0]{} & V. \end{array} \tag{3.2.11}$$

The restrictions of $\mu^*\mathcal{L}_\psi$ to the two upper-left corners are trivial, hence base change and the projection formula give

$$\begin{aligned} (0^\vee)^*\mathscr{F}_{\psi,V} &\simeq \pi_!\big((\mathrm{id}_V \times_X 0^\vee)^* \mathrm{pr}_1^*(-) \otimes^L (\mathrm{id}_V \times_X 0^\vee)^*\mu^*\mathcal{L}_\psi\big) \simeq \pi_!, \\ \mathscr{F}_{\psi,V} 0_* &\simeq \mathrm{pr}_{2!}(0 \times_X \mathrm{id}_{V^\vee})_*(\pi^\vee)^* \simeq (\pi^\vee)^*. \end{aligned} \tag{3.2.12}$$

(2)–(3). For $j : V^\vee \setminus 0^\vee(X) \hookrightarrow V^\vee$, we have $j^*\mathscr{F}_{\psi,V}\pi^! = 0$. Thus (1), the trace isomorphism and smooth purity give

$$\begin{aligned} \mathscr{F}_{\psi,V}\pi^! &\simeq 0^\vee_*(0^\vee)^*\mathscr{F}_{\psi,V}\pi^! \simeq 0^\vee_*\pi_!\pi^! \simeq 0^\vee_*, \\ \mathscr{F}_{\psi,V}\pi^* &\simeq \mathscr{F}_{\psi,V}\pi^!(-r)[-2r] \simeq 0^\vee_*(-r)[-2r]. \end{aligned} \tag{3.2.13}$$

(4). On $V \times_X V^\vee \times_X V$, let pr_{ij} denote the ordered factor projections; in particular, $\mathrm{pr}_{32}(v,\xi,w) = (w,\xi)$. Let $\mathrm{pr}_1', \mathrm{pr}_2' : V \times_X V \to V$ be the projections, and define

$$\begin{aligned} &\delta : V \longrightarrow V \times_X V, && v \longmapsto (v,v), \\ &e : V \times_X V^\vee \longrightarrow V \times_X V^\vee \times_X V, && (v,\xi) \longmapsto (v,\xi,v), \\ &h : V \times_X V^\vee \times_X V \longrightarrow \mathbb{A}^1_k, && (v,\xi,w) \longmapsto \langle v-w,\xi\rangle. \end{aligned} \tag{3.2.14}$$

We have the following cartesian squares

$$\begin{array}{ccc} V \times_X V^\vee \times_X V & \xrightarrow{\mathrm{pr}_{12}} & V \times_X V^\vee \\ {\scriptstyle\mathrm{pr}_{32}}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_2} \\ V \times_X V^\vee & \xrightarrow[\mathrm{pr}_2]{} & V^\vee, \end{array} \tag{3.2.15}$$

$$\begin{array}{ccc} V \times_X V^\vee & \xrightarrow{e} & V \times_X V^\vee \times_X V \\ {\scriptstyle\mathrm{pr}_1}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_{13}} \\ V & \xrightarrow[\delta]{} & V \times_X V \end{array} \qquad \begin{array}{ccc} V \times_X V^\vee & \xrightarrow{\mathrm{pr}_2} & V^\vee \\ {\scriptstyle\mathrm{pr}_1}\downarrow & & \downarrow{\scriptstyle \pi^\vee} \\ V & \xrightarrow[\pi]{} & X. \end{array} \tag{3.2.16}$$

We have $\mathrm{pr}_{13!}\, h^*\mathcal{L}_\psi$ supported on $\delta(V)$ by definition of $\mathcal{L}_\psi$. Since $e^*h^*\mathcal{L}_\psi \simeq \Lambda$, base change and the projection formula yield

$$\begin{aligned}
\mathscr{F}_{\psi^\vee,V^\vee}\mathscr{F}_{\psi,V} &\simeq \mathrm{pr}'_{2!}\,\mathrm{pr}_{13!}(\mathrm{pr}_{13}^*(\mathrm{pr}'_1)^*(-)\otimes^L h^*\mathcal{L}_\psi)\\
&\simeq \mathrm{pr}'_{2!}((\mathrm{pr}'_1)^*(-)\otimes^L \mathrm{pr}_{13!}\, h^*\mathcal{L}_\psi)\\
&\simeq \mathrm{pr}'_{2!}((\mathrm{pr}'_1)^*(-)\otimes^L \delta_*\,\mathrm{pr}_{1!}\,\Lambda)\\
&\simeq \mathrm{pr}'_{2!}\,\delta_*(-\otimes^L \pi^*\pi_!^\vee\Lambda)\\
&\simeq -\otimes^L \pi^*\pi_!^\vee\Lambda \simeq \mathrm{id}(-r)[-2r].
\end{aligned} \tag{3.2.17}$$

The isomorphism (3.2.5) follows directly.
(5). Let q_1, q_2 be the projections of $W\times_X V^\vee$, $\mathrm{pr}_{W1}, \mathrm{pr}_{W2}$ for those of $W\times_X W^\vee$, and $\mu_W : W\times_X W^\vee \to \mathbb{A}^1_k$ for the pairing. Consider the cartesian diagram

$$\begin{array}{ccc} W\times_X V^\vee & \xrightarrow{q\times \mathrm{id}_{V^\vee}} & V\times_X V^\vee \\ {\scriptstyle q_1}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_1} \\ W & \xrightarrow[q]{} & V \end{array} \qquad \begin{array}{ccc} W\times_X V^\vee & \xrightarrow{\mathrm{id}_W\times q^\vee} & W\times_X W^\vee \\ {\scriptstyle q_2}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_{W2}} \\ V^\vee & \xrightarrow[q^\vee]{} & W^\vee. \end{array} \tag{3.2.18}$$

Since $\mu\circ(q\times \mathrm{id}_{V^\vee}) = \mu_W\circ(\mathrm{id}_W\times q^\vee)$, these squares give

$$\begin{aligned}
\mathscr{F}_{\psi,V}q_! &\simeq q_{2!}(q_1^*(-)\otimes^L (q\times \mathrm{id}_{V^\vee})^*\mu^*\mathcal{L}_\psi)\\
&\simeq q_{2!}(\mathrm{id}_W\times q^\vee)^*(\mathrm{pr}_{W1}^*(-)\otimes^L \mu_W^*\mathcal{L}_\psi)\\
&\simeq (q^\vee)^*\,\mathrm{pr}_{W2!}(\mathrm{pr}_{W1}^*(-)\otimes^L \mu_W^*\mathcal{L}_\psi) = (q^\vee)^*\mathscr{F}_{\psi,W}.
\end{aligned}$$

(6). Since $\mathrm{rk}\, V' = r-s$, using (4), (5) for $i^\vee$ we have

$$\begin{aligned}
i_!^\vee\mathscr{F}_{\psi,V} &\simeq \mathscr{F}_{\psi,V'}\mathscr{F}_{\psi^\vee,(V')^\vee}i_!^\vee\mathscr{F}_{\psi,V}(r-s)[2r-2s]\\
&\simeq \mathscr{F}_{\psi,V'}i^*\mathscr{F}_{\psi^\vee,V^\vee}\mathscr{F}_{\psi,V}(r-s)[2r-2s]\\
&\simeq \mathscr{F}_{\psi,V'}i^*(-s)[-2s].
\end{aligned} \tag{3.2.19}$$

The second isomorphism is similar.
(7). Here pr_{ij} and pr_3 denote the projections from $V\times_X V\times_X V^\vee$. Consider the cartesian diagram

$$\begin{array}{ccc} V\times_X V\times_X V^\vee & \xrightarrow{\mathrm{add}\times \mathrm{id}_{V^\vee}} & V\times_X V^\vee \\ {\scriptstyle \mathrm{pr}_{12}}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_1} \\ V\times_X V & \xrightarrow[\mathrm{add}]{} & V \end{array} \qquad \begin{array}{ccc} V\times_X V\times_X V^\vee & \xrightarrow{\mathrm{pr}_{23}} & V\times_X V^\vee \\ {\scriptstyle \mathrm{pr}_{13}}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_2} \\ V\times_X V^\vee & \xrightarrow[\mathrm{pr}_2]{} & V^\vee. \end{array} \tag{3.2.20}$$

The first square and additivity of $\mathcal{L}_\psi$, followed by Künneth for the second square, give the following isomorphisms of bifunctors:

$$\begin{aligned}
\mathscr{F}_{\psi,V}\,\mathrm{add}_!(-\boxtimes_X -) &\simeq \mathrm{pr}_{3!}\Big(\mathrm{pr}_{13}^*(\mathrm{pr}_1^*(-)\otimes^L \mu^*\mathcal{L}_\psi)\otimes^L \mathrm{pr}_{23}^*(\mathrm{pr}_1^*(-)\otimes^L \mu^*\mathcal{L}_\psi)\Big)\\
&\simeq \mathrm{pr}_{2!}(\mathrm{pr}_1^*(-)\otimes^L \mu^*\mathcal{L}_\psi)\otimes^L \mathrm{pr}_{2!}(\mathrm{pr}_1^*(-)\otimes^L \mu^*\mathcal{L}_\psi)\\
&= \mathscr{F}_{\psi,V}(-)\otimes^L \mathscr{F}_{\psi,V}(-).
\end{aligned} \tag{3.2.21}$$

(8). Relative duality, smooth purity and [18, (2.1.11)] give

$$\begin{aligned} D_{V^\vee/S}\mathscr{F}_{\psi,V} &\simeq \mathrm{pr}_{2*}(-\otimes^L \mu^*\mathcal{L}_{\psi^\vee})\,\mathrm{pr}_1^!\, D_{V/S} \\ &\simeq \mathrm{pr}_{2*}(-\otimes^L \mu^*\mathcal{L}_{\psi^\vee})\,\mathrm{pr}_1^*\, D_{V/S}(r)[2r] \\ &\simeq \mathrm{pr}_{2!}(-\otimes^L \mu^*\mathcal{L}_{\psi^\vee})\,\mathrm{pr}_1^*\, D_{V/S}(r)[2r] \\ &= \mathscr{F}_{\psi^\vee,V} D_{V/S}(r)[2r]. \end{aligned} \tag{3.2.22}$$

Consequently, we have

$$\begin{aligned} \mathscr{F}_{\psi^\vee,V} D_{V/S} &\simeq D_{V^\vee/S}\mathscr{F}_{\psi,V}(-r)[-2r] \\ &\simeq R\mathcal{H}om(\mathscr{F}_{\psi,V}(-), (\pi^\vee)^!\mathcal{K}_{X/S})(-r)[-2r] \\ &\simeq R\mathcal{H}om(\mathscr{F}_{\psi,V}(-), (\pi^\vee)^*\mathcal{K}_{X/S}). \end{aligned} \tag{3.2.23}$$

□

3.3. Retain the setting of Section 3.1, let $i : V' \hookrightarrow V$ be a subbundle of codimension s, with dual homomorphism $i^\vee : V^\vee \to V'^\vee$. Using [18, (2.1.28)], for $\mathcal{G} \in D_{\mathrm{ctf}}(V^\vee, \Lambda)$ we have a commutative diagram

$$\begin{array}{ccc} \mathscr{F}_{\psi,V'}\left(i^*\mathscr{F}_{\psi^\vee,V^\vee}\mathcal{G}(r-s)[2r-2s]\right) & \xrightarrow{(3.2.7)} & i_!^\vee\mathscr{F}_{\psi,V}\mathscr{F}_{\psi^\vee,V^\vee}\mathcal{G}(r)[2r] \\ {\scriptstyle (3.2.6)}\downarrow\simeq & & \simeq\downarrow{\scriptstyle \mathrm{adj.}} \\ \mathscr{F}_{\psi,V'}\mathscr{F}_{\psi^\vee,V'^\vee} i_!^\vee\mathcal{G} & \xrightarrow[\mathrm{adj.}]{} & i_!^\vee\mathcal{G}. \end{array} \tag{3.3.1}$$

We have the following compatibility with transversality morphisms.

Proposition 3.4. *Retain the setting of Sections 3. Let $i : V' \to V$ be a linear morphism of vector bundles over X which fits into the following cartesian square*

$$\begin{array}{ccc} V' & \xrightarrow{i} & V \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle f} \\ W & \xrightarrow[\delta]{} & T, \end{array} \tag{3.4.1}$$

with dual $i^\vee : V^\vee \to V'^\vee$, then for $\mathcal{N} \in D_{\mathrm{ctf}}(V, \Lambda)$, the following diagram is commutative:

$$\begin{array}{ccc} \mathscr{F}_{\psi,V'}(i^*\mathcal{N}\otimes^L p^*\delta^!\Lambda) & \xrightarrow{c_{\delta,f,\mathcal{N}}} & \mathscr{F}_{\psi,V'}(i^!\mathcal{N}) \\ \downarrow & & \downarrow\simeq \\ i_!^\vee\mathscr{F}_{\psi,V}\mathcal{N} & \xrightarrow[\mathrm{adj.}]{} & i_*^\vee\mathscr{F}_{\psi,V}\mathcal{N}, \end{array} \tag{3.4.2}$$

where the left vertical morphism is

$$\begin{aligned} \mathscr{F}_{\psi,V'}(i^*\mathcal{N}\otimes^L p^*\delta^!\Lambda) &\xrightarrow{\mathrm{b.c.}} \mathscr{F}_{\psi,V'}(i^*\mathcal{N}\otimes^L i^!\Lambda) \\ &\xrightarrow{\simeq} i_!^\vee\mathscr{F}_{\psi,V}\mathcal{N}, \end{aligned} \tag{3.4.3}$$

Proof. By (2.2.2), $c_{\delta,f,\mathcal{N}}$ is equal to the composition

$$i^*\mathcal{N}\otimes^L p^*\delta^!\Lambda \xrightarrow{\mathrm{b.c.}} i^*\mathcal{N}\otimes^L i^!\Lambda \xrightarrow{c_{i,\mathrm{id},\mathcal{N}}} i^!\mathcal{N}, \tag{3.4.4}$$

thus it suffices to prove the assertion for $c_{i,\mathrm{id},\mathcal{N}}$. Put $r = \operatorname{rank} V$ and $r' = \operatorname{rank} V'$. By Proposition 3.2(4), we may write $\mathcal{N} = \mathscr{F}_{\psi^\vee,V^\vee}\mathcal{G}(r)[2r]$ for $\mathcal{G} \in D_{\mathrm{ctf}}(V^\vee,\Lambda)$. Let $\mathrm{pr}_1, \mathrm{pr}_2$ be the projections from $V \times_X V^\vee$, let $\mathrm{pr}'_1 : V' \times_X V^\vee \to V'$ be the first projection, and put $\iota = i \times \mathrm{id}_{V^\vee}$, $\mathcal{H} = \mathrm{pr}_2^*\mathcal{G} \otimes^L \mu^*\mathcal{L}_{\psi^\vee}$. Smooth purity for the projections to $V^\vee$ gives $\iota^!\mathcal{H} \simeq \iota^*\mathcal{H} \otimes^L (\mathrm{pr}'_1)^* i^!\Lambda$. Compatibility of base change and the projection formula with adjunction identifies the composition

$$i^* \mathrm{pr}_{1!}\mathcal{H} \otimes^L i^!\Lambda \xrightarrow[\text{b.c., p.f.}]{\simeq} \mathrm{pr}'_{1!}\iota^!\mathcal{H} \xrightarrow{\text{b.c.}} \mathrm{pr}'_{1*}\iota^!\mathcal{H} \xrightarrow[\text{b.c.}]{\simeq} i^!\mathrm{pr}_{1*}\mathcal{H} \tag{3.4.5}$$

with

$$i^* \mathrm{pr}_{1!}\mathcal{H} \otimes^L i^!\Lambda \xrightarrow{c_{i,\mathrm{id},\mathrm{pr}_{1!}}\mathcal{H}} i^!\mathrm{pr}_{1!}\mathcal{H} \xrightarrow{\text{b.c.}} i^!\mathrm{pr}_{1*}\mathcal{H}. \tag{3.4.6}$$

By [18, (2.1.11)], the last morphism in (3.4.6) is an isomorphism. Factor pr'_1 through $\mathrm{id}_{V'} \times i^\vee : V' \times_X V^\vee \to V' \times_X V'^\vee$. The equality $\langle i(v'), \xi\rangle = \langle v', i^\vee(\xi)\rangle$, the projection formula and base change identify (3.4.5) with

$$\mathscr{F}_{\psi^\vee,V'^\vee}(i^\vee_!\mathcal{G})(r')[2r'] \xrightarrow{\text{b.c.}} \mathscr{F}_{\psi^\vee,V'^\vee}(i^\vee_*\mathcal{G})(r')[2r'] \tag{3.4.7}$$

after twisting by $(r)[2r]$. For the target we again use [18, (2.1.11)]. Under the Fourier inversion isomorphism of Proposition 3.2(4), the left identification is the one characterized by [18, Corollary 2.1.10(1), (2.1.28)], and the right identification is obtained by adjunction from [18, Proposition 2.1.9]. Applying $\mathscr{F}_{\psi,V'}$ therefore gives the required square for $c_{i,\mathrm{id},\mathcal{N}}$. □

Corollary 3.5. *For any linear morphism $i : V' \to V$ of vector bundles over X and $\mathcal{N} \in D_{\mathrm{ctf}}(V,\Lambda)$, the following diagram is commutative, and both vertical arrows are isomorphisms:*

$$\begin{array}{ccc} \mathscr{F}_{\psi,V'}(i^*\mathcal{N} \otimes^L i^!\Lambda) & \xrightarrow{c_{i,\mathrm{id},\mathcal{N}}} & \mathscr{F}_{\psi,V'}(i^!\mathcal{N}) \\ \downarrow \simeq & & \downarrow \simeq \\ i^\vee_!\mathscr{F}_{\psi,V}\mathcal{N} & \xrightarrow[\text{adj}]{} & i^\vee_*\mathscr{F}_{\psi,V}\mathcal{N}. \end{array} \tag{3.5.1}$$

3.6. Let X be a noetherian scheme over k, $V \to X$ be a vector bundle of rank r. Let π, 0 and $\pi^\vee, 0^\vee$ be the projection and zero section of V and $V^\vee$, respectively, and put $j^\vee : (V^\vee)^\times := V^\vee \setminus 0^\vee(X) \hookrightarrow V^\vee$, $\pi^{\vee,\times} = \pi^\vee \circ j^\vee$.

For the vector bundle V, denote its natural $\mathbb{G}_{m,k}$-action by $\rho : \mathbb{G}_{m,k} \times_k V \to V$. Let $x : \operatorname{Spec}\Omega \to V$ be any geometric point, consider its orbit under $\mathbb{G}_{m,k}$-action:

$$w_x : \mathbb{G}_{m,\Omega} = \mathbb{G}_{m,k} \times_k \operatorname{Spec}\Omega \longrightarrow \mathbb{G}_{m,k} \times_k V \xrightarrow{\rho} V. \tag{3.6.1}$$

For constructible complex $\mathcal{N} \in D^b_c(V,\Lambda)$, we say it is *monodromic* if for any geometric point $x \in V$, $w_x^*\mathcal{N}$ is a locally constant complex on $\mathbb{G}_{m,\Omega}$ with tame ramification along 0 and ∞. This definition is due to Verdier [21].

Lemma 3.7 ([21, Lemma 6.1])**.** *Let $\mathcal{N} \in D^b_c(V,\Lambda)$ be a monodromic complex, then we have natural isomorphisms*

$$0^!\mathcal{N} \xrightarrow{\simeq} \pi_!\mathcal{N}, \qquad \pi_*\mathcal{N} \xrightarrow{\simeq} 0^*\mathcal{N}. \tag{3.7.1}$$

Proof. Let $j : V \setminus 0(X) \hookrightarrow V$ be the open immersion. By [21, Lemma 6.1], we have $\pi_! j_* j^* = 0$, $\pi_* j_! j^* = 0$ for the monodromic complex $\mathcal{N}$. Applying $\pi_!$ and π_* to the two localization triangles induced by 0 and j, we have

$$\begin{aligned} \pi_!0_*0^!\mathcal{N} \longrightarrow \pi_!\mathcal{N} \longrightarrow 0 \xrightarrow{+1}, \\ 0 \longrightarrow \pi_*\mathcal{N} \longrightarrow \pi_*0_*0^*\mathcal{N} \xrightarrow{+1}. \end{aligned} \tag{3.7.2}$$

Since $\pi \circ 0 = \mathrm{id}_X$, these yield the asserted natural isomorphisms. □

Recall that for $\mathcal{N} \in D_c^b(V, \Lambda)$, we have the transversality morphism $c_{0,\mathrm{id},\mathcal{N}}$ by (2.2.2), which fits into the distinguished triangle

$$0^*\mathcal{N} \otimes^L 0^!\Lambda \xrightarrow{c_{0,\mathrm{id},\mathcal{N}}} 0^!\mathcal{N} \longrightarrow 0^\Delta\mathcal{N} \xrightarrow{+1} . \tag{3.7.3}$$

Lemma 3.8. *Retain the previous notation. Assume $\mathcal{N} \in D_c^b(V, \Lambda)$ is monodromic, then we have a natural isomorphism*

$$\vartheta_{\mathcal{N}} : 0^\Delta\mathcal{N} \xrightarrow{\simeq} \pi_!^{\vee,\times} j^{\vee *} \mathscr{F}_{\psi,V}\mathcal{N}[1]. \tag{3.8.1}$$

Proof. By [18, Proposition 2.1.11(2), (2.1.38)], we have the following commutative diagram:

$$\begin{array}{ccc} \mathscr{F}_{\psi,V^\vee}\Lambda & \xrightarrow{\text{res.}} & \mathscr{F}_{\psi,V^\vee}(0_*^\vee\Lambda) \\ {\scriptstyle (-1)^r\sim}\downarrow & & \downarrow{\scriptstyle \sim} \\ 0_*\Lambda(-r)[-2r] & \xrightarrow[\text{adj.}]{} & \Lambda, \end{array} \tag{3.8.2}$$

here the morphisms res. and adj. are both induced by adjunction. Now apply the functor $\pi_!(\mathcal{N}\otimes^L -)$ to the diagram, we calculate the resulting terms. For the upper-left term, using projection formula, we have

$$\begin{aligned} \pi_!\left(\mathcal{N} \otimes^L \mathscr{F}_{\psi,V^\vee}\Lambda\right) &= \pi_!\left(\mathcal{N} \otimes^L \mathrm{pr}_{1!}\,\mu^*\mathcal{L}_\psi\right) \\ &\simeq \pi_!\,\mathrm{pr}_{1!}\left(\mathrm{pr}_1^*\mathcal{N} \otimes^L \mu^*\mathcal{L}_\psi\right) \\ &\simeq \pi_!^\vee\,\mathrm{pr}_{2!}\left(\mathrm{pr}_1^*\mathcal{N} \otimes^L \mu^*\mathcal{L}_\psi\right) = \pi_!^\vee\mathscr{F}_{\psi,V}(\mathcal{N}). \end{aligned} \tag{3.8.3}$$

For the upper-right term, using Proposition 3.2 (1) we have

$$\pi_!\left(\mathcal{N} \otimes^L \mathscr{F}_{\psi,V^\vee}(0_*^\vee\Lambda)\right) \simeq \pi_!\mathcal{N} \simeq 0^{\vee *}\mathscr{F}_{\psi,V}\mathcal{N}. \tag{3.8.4}$$

For the lower-left term, using projection formula we have

$$\pi_!\left(\mathcal{N} \otimes^L 0_*\Lambda(-r)[-2r]\right) \simeq \pi_!0_!\left(0^*\mathcal{N} \otimes^L \Lambda(-r)[-2r]\right) = 0^*\mathcal{N}(-r)[-2r]. \tag{3.8.5}$$

For the lower-right term, Lemma 3.7 implies $\pi_!\mathcal{N} \simeq 0^!\mathcal{N}$. Hence we have a commutative diagram:

$$\begin{array}{ccc} \pi_!^\vee\mathscr{F}_{\psi,V}\mathcal{N} & \xrightarrow{\text{res}} & 0^{\vee *}\mathscr{F}_{\psi,V}\mathcal{N} \\ {\scriptstyle (-1)^r\sim}\downarrow & & \downarrow{\scriptstyle \sim} \\ 0^*\mathcal{N}(-r)[-2r] & \xrightarrow[\pi_!(\mathrm{id}_\mathcal{N}\otimes^L\mathrm{adj})]{} & 0^!\mathcal{N}. \end{array} \tag{3.8.6}$$

One may verify that the lower horizontal morphism is the transversality morphism $c_{0,\mathrm{id},\mathcal{N}}$, under the absolute purity isomorphism $0^!\Lambda \simeq \Lambda(-r)[-2r]$.

Recall that we have the open-close decomposition $j^\vee : (V^\vee)^\times \hookrightarrow V^\vee \hookleftarrow X : 0^\vee$ which induces the localization triangle (of functors $D_c^b(V^\vee, \Lambda) \to D_c^b(V^\vee, \Lambda)$)

$$\mathrm{id} \longrightarrow 0^\vee{}_*0^{\vee *} \longrightarrow j_!^\vee j^{\vee *}[1] \xrightarrow{+1} . \tag{3.8.7}$$

Applying $\pi_!^\vee$ to (3.8.7), we get a distinguished triangle of functors $D_c^b(V^\vee, \Lambda) \to D_c^b(X, \Lambda)$

$$\pi_!^\vee \longrightarrow 0^{\vee *} \longrightarrow \pi_!^{\vee,\times} j^{\vee *}[1] \xrightarrow{+1} . \tag{3.8.8}$$

Taking cofibres of the two horizontal morphisms of (3.8.6), using (3.8.8) we get the following diagram of distinguished triangles

$$\begin{array}{ccccccc} \pi_!^{\vee}\mathscr{F}_{\psi,V}\mathcal{N} & \xrightarrow{\mathrm{res}} & 0^{\vee *}\mathscr{F}_{\psi,V}\mathcal{N} & \longrightarrow & \pi_!^{\vee,\times} j^{\vee *}\mathscr{F}_{\psi,V}\mathcal{N}[1] & \xrightarrow{+1} & \\ {\scriptstyle (-1)^r\sim}\downarrow & & \downarrow{\scriptstyle \sim} & & {\scriptstyle \sim}\downarrow{\scriptstyle \vartheta_{\mathcal{N}}^{-1}} & & \\ 0^*\mathcal{N}(-r)[-2r] & \xrightarrow[c_{0,\mathrm{id},\mathcal{N}}]{} & 0^!\mathcal{N} & \xrightarrow[\partial]{} & 0^{\Delta}\mathcal{N} & \xrightarrow[+1]{} & \end{array} \tag{3.8.9}$$

which gives the natural morphism

$$\pi_!^{\vee,\times}(j^{\vee})^*\mathscr{F}_{\psi,V}\mathcal{N}[1] \longrightarrow 0^{\Delta}\mathcal{N}. \tag{3.8.10}$$

Proposition 3.2 implies the two left vertical morphisms in (3.8.9) are isomorphisms, hence (3.8.10) is a natural isomorphism. Denote its inverse by $\vartheta_{\mathcal{N}}$, the consequence follows. □

3.9. Retain the notation of Lemma 3.8. Apply (3.8.8) to the constant sheaf Λ, we get a distinguished triangle

$$\Lambda(-r)[-2r] \xrightarrow{c_r(V^{\vee})} \Lambda \xrightarrow{b_{V^{\vee}}} \pi_!^{\vee,\times}\Lambda[1] \xrightarrow{+1}. \tag{3.9.1}$$

By definition one can show the first morphism of (3.9.1) is equal to the étale-theoretical Chern class $c_r(V^{\vee})$. Denote the second morphism by $b_{V^{\vee}}$.

Let $\mathcal{N}_0 \in D_c^b(X,\Lambda)$ be a constructible complex, we consider the monodromic sheaf $\mathcal{N} = 0_*\mathcal{N}_0$. By Proposition 3.2(1) and the projection formula, we have

$$\begin{aligned} \pi_!^{\vee,\times} j^{\vee *}\mathscr{F}_{\psi,V}(0_*\mathcal{N}_0)[1] &\simeq \pi_!^{\vee,\times} j^{\vee *}(\pi^{\vee})^*\mathcal{N}_0[1] \\ &\simeq \pi_!^{\vee,\times}((\pi^{\vee,\times})^*\mathcal{N}_0 \otimes^L \Lambda)[1] \\ &\simeq \mathcal{N}_0 \otimes^L \pi_!^{\vee,\times}\Lambda[1]. \end{aligned} \tag{3.9.2}$$

On the other hand, by (2.5.2) and the projection formula we have

$$\begin{aligned} 0^{\Delta}(0_*\mathcal{N}_0) &\simeq 0^!\left(0_*\mathcal{N}_0 \otimes^L j_*\Lambda\right) \\ &\simeq 0^!0_*\left(\mathcal{N}_0 \otimes^L 0^*j_*\Lambda\right) \\ &\simeq \mathcal{N}_0 \otimes^L 0^*j_*\Lambda. \end{aligned} \tag{3.9.3}$$

We have the following expression for $\vartheta_{0_*\mathcal{N}_0}$ in this case.

Corollary 3.10. *Retain the previous setting. Under the identification* (3.9.2) *and* (3.9.3) *for the source and target of* $\vartheta_{0_*\mathcal{N}_0}$*, we have*

$$\vartheta_{0_*\mathcal{N}_0} = \mathrm{id}_{\mathcal{N}_0} \otimes^L \vartheta_{0_*\Lambda} : \mathcal{N}_0 \otimes^L 0^*j_*\Lambda \xrightarrow{\simeq} \mathcal{N}_0 \otimes^L \pi_!^{\vee,\times}\Lambda[1]. \tag{3.10.1}$$

Proof. Under the identifications (3.9.2) and (3.9.3), the diagram (3.8.9) defining $\vartheta_{0_*\mathcal{N}_0}$ is obtained from the diagram defining $\vartheta_{0_*\Lambda}$ by applying $\mathcal{N}_0 \otimes^L (-)$. Taking cofibres gives (3.10.1). □

4. Specialization and microlocalization

4.1. Let $S = \operatorname{Spec} R$ be the spectrum of a henselian discrete valuation ring, with closed point $s = \operatorname{Spec} k$ and generic point $\eta = \operatorname{Spec} K$. Let Λ be a finite commutative local ring whose characteristic is invertible on S. Choose a separable closure $\bar{\eta} = \operatorname{Spec} K^{\mathrm{sep}}$ of η and a compatible separable closure $\bar{s} = \operatorname{Spec} k^{\mathrm{sep}}$ of s. Put $G_\eta = \mathrm{Gal}(\bar{\eta}/\eta)$ and $G_s = \mathrm{Gal}(\bar{s}/s)$, and let I_η be the inertia group which fits in the following exact sequence

$$1 \longrightarrow I_\eta \longrightarrow G_\eta \xrightarrow{q} G_s \longrightarrow 1. \tag{4.1.1}$$

For any scheme X, we denote its étale topos by $X_{\text{ét}}$, which is the category of sheaves of sets on the (small) étale site of X by definition. For schemes X_1, X_2 over a scheme T, we denote the fibre product of topoi $X_{1\text{ét}} \times_{T_{\text{ét}}} X_{2\text{ét}}$ by $X_1 \overline{\times}_T X_2$; this product is taken in the 2-category of topoi.

In particular, for any field L, $(\operatorname{Spec} L)_{\text{ét}}$ is equivalent to the category Set_{G_L} of sets with continuous G_L-action, where G_L is the absolute Galois group of L. Hence the surjection $q : G_\eta \to G_s$ in (4.1.1) gives a canonical morphism of topoi

$$\sigma : \eta_{\text{ét}} \to s_{\text{ét}}, \tag{4.1.2}$$

where $\sigma^* : \operatorname{Set}_{G_s} \to \operatorname{Set}_{G_\eta}$ is given by the standard group action inflation along q, $\sigma_* : \operatorname{Set}_{G_\eta} \to \operatorname{Set}_{G_s}$ is given by taking fixed part $B \mapsto B^{I_\eta}$.

Let Y be a separated scheme of finite type over $s = \operatorname{Spec} k$, put $Y_{\bar{s}} = Y \times_s \bar{s}$. With the aid of σ, we may define the fibre product of topoi $Y \overline{\times}_s \eta$ which is called the *Deligne topos*. By definition, objects of $D_c^b(Y \overline{\times}_s \eta, \Lambda)$ correspond to the objects of $D_c^b(Y_{\bar{s}}, \Lambda)$ with continuous action of G_η.

Remark 4.2. We shall remind that here $Y \overline{\times}_s \eta$ is *not* a fibre product of schemes, and the construction does not require a morphism of schemes $\eta \to s$; even when such a morphism exists, the topos $Y \overline{\times}_s \eta$ is different from $(Y \times_s \eta)_{\text{ét}}$ in general, see [12, Remark 1.5].

Consider the morphism of topoi

$$\begin{aligned} u_Y : (Y_{\bar{s}})_{\text{ét}} &\simeq Y_{\text{ét}} \times_{s_{\text{ét}}} \bar{s}_{\text{ét}} \\ &\simeq Y_{\text{ét}} \times_{s_{\text{ét}}} \overline{\eta}_{\text{ét}} \\ &\simeq (Y_{\text{ét}} \times_{s_{\text{ét}}} \eta_{\text{ét}}) \times_{\eta_{\text{ét}}} \overline{\eta}_{\text{ét}} \longrightarrow Y_{\text{ét}} \times_{s_{\text{ét}}} \eta_{\text{ét}}, \end{aligned} \tag{4.2.1}$$

where the last projection is the base change of $q_\eta : \overline{\eta}_{\text{ét}} \to \eta_{\text{ét}}$. Its inverse image functor defines

$$u_Y^* : D_c^b(Y \overline{\times}_s \eta, \Lambda) \longrightarrow D_c^b(Y_{\bar{s}}, \Lambda). \tag{4.2.2}$$

For any object of $D_c^b(Y \overline{\times}_s \eta, \Lambda)$ viewed as $\mathcal{F} \in D_c^b(Y_{\bar{s}}, \Lambda)$ with continuous G_η-action, u_Y^* acts on it by forgetting the G_η-action. In the sequel, we suppress this forgetful functor if no confusion is caused.

4.3. Retain the setting of the previous subsection, let X be a separated scheme of finite type over k. The projection morphism $p_X : X \overline{\times}_s \eta \to X$ induces the adjoint pair

$$D_c^b(X, \Lambda) \underset{p_{X*}}{\overset{p_X^*}{\rightleftarrows}} D_c^b(X \overline{\times}_s \eta, \Lambda), \qquad p_X^* \dashv p_{X*}. \tag{4.3.1}$$

For $\mathcal{F} \in D_c^b(X \overline{\times}_s \eta, \Lambda)$, if we regard it as $\mathcal{F} \in D_c^b(X_{\bar{s}}, \Lambda)$ with continuous G_η-action, we may describe $p_{X*}\mathcal{F}$ as $\mathcal{F}^{I_\eta}$ which is a complex in $D_c^b(X_{\bar{s}}, \Lambda)$ with continuous G_s-action, and hence descends to a complex in $D_c^b(X, \Lambda)$. For $\mathcal{G} \in D_c^b(X, \Lambda)$ regarded as $\mathcal{G}_{\bar{s}}$ with continuous G_s-action as descent data, p_X^* can be described as taking the inflation of this action along the quotient q in (4.1.1).

We have the following descent result.

Proposition 4.4. *For every $\mathcal{K} \in D^b(X, \Lambda)$, we have a natural isomorphism*

$$\mathcal{K} \oplus \mathcal{K}(-1)[-1] \xrightarrow{\simeq} p_{X*}p_X^*\mathcal{K}. \tag{4.4.1}$$

In particular, for every closed subscheme $Z \hookrightarrow X$, (4.4.1) induces an isomorphism

$$R\Gamma_Z(X, \mathcal{K}) \oplus R\Gamma_Z(X, \mathcal{K}(-1)[-1]) \simeq R\Gamma_Z(X \overline{\times}_s \eta, p_X^*\mathcal{K}). \tag{4.4.2}$$

Proof. Let ℓ be the residue characteristic of Λ, and choose $N = \ell^r$ such that $N \cdot \Lambda = 0$. The Kummer exact sequence

$$0 \longrightarrow \mu_N \longrightarrow \mathbb{G}_{m,\eta} \xrightarrow{\cdot N} \mathbb{G}_{m,\eta} \longrightarrow 0 \tag{4.4.3}$$

induces the following composition

$$\{-\} : H^0(\eta, \mathbb{G}_{m,\eta}) \longrightarrow H^1(\eta, \mu_N) \xrightarrow{-\otimes_{\mathbb{Z}/N}\Lambda} H^1(\eta, \Lambda(1)) \tag{4.4.4}$$

which gives the *Kummer class* $\{t\} \in H^1(\eta, \Lambda(1))$ for any $t \in H^0(\eta, \mathbb{G}_{m,\eta}) = K^\times$. Let $p_\eta : X\overline{\times}_s\eta \to \eta$ be the second projection, we have the following cup product morphism

$$\cup\{t\} : p_X^*\mathcal{K}(-1)[-1] \simeq p_X^*\mathcal{K} \otimes^L p_\eta^*\Lambda(-1)[-1] \xrightarrow{\mathrm{id}\otimes p_\eta^*\{t\}} p_X^*\mathcal{K}, \tag{4.4.5}$$

which gives a natural morphism $\mathcal{K}(-1)[-1] \to p_{X*}p_X^*\mathcal{K}$ under adjunction. Now put $t = \varpi$, a uniformizer of R; together with the adjunction $\mathcal{K} \to p_{X*}p_X^*\mathcal{K}$, we get the natural morphism (4.4.1). We show it is an isomorphism.

By taking truncation, it suffices to consider the case $\mathcal{K}$ is a sheaf. Taking stalks at an arbitrary geometric point $\overline{x}$ of X, by the previous description of p_X^* and p_{X*}, we have $(p_{X*}p_X^*\mathcal{K})_{\overline{x}} \simeq R\Gamma(I_\eta, \mathcal{K}_{\overline{x}})$. Hence if we put $M = \mathcal{K}_{\overline{x}}$ as a G_s-module, it suffices to prove the morphism

$$M \oplus M(-1)[-1] \longrightarrow R\Gamma(I_\eta, M) \tag{4.4.6}$$

induced by (4.4.1) is an isomorphism, where the right side is the continuous group cohomology, and I_η-action on M is the trivial action. Using the classical calculation [12, Section 2.2, Lemma 2.9 and Remark 2.11]

$$H^j(I_\eta, M) \simeq H^j(\mathbb{Z}_\ell(1), M) = \begin{cases} M, & j = 0, \\ M(-1), & j = 1, \\ 0, & j \neq 0, 1, \end{cases} \tag{4.4.7}$$

the consequence follows. □

Taking cohomology, we have the following result.

Corollary 4.5. *Retain the previous setting, then we have*

$$H^m_Z(X, \mathcal{K}) \oplus H^{m-1}_Z(X, \mathcal{K}(-1)) \simeq H^m_Z(X\overline{\times}_s\eta, p_X^*\mathcal{K}) \tag{4.5.1}$$

for each integer m.

The adjunction morphism $\mathcal{K} \to p_{X*}p_X^*\mathcal{K}$ gives the 'inflation map'

$$H^m_Z(X, \mathcal{K}) \longrightarrow H^m_Z(X\overline{\times}_s\eta, p_X^*\mathcal{K}), \tag{4.5.2}$$

which coincides with the inclusion of the first summand in (4.5.1); therefore, it is split injective and is an isomorphism if and only if $H^{m-1}_Z(X, \mathcal{K}(-1)) = 0$. The splitting depends on the chosen $t \in K^\times$, whereas inflation (4.5.2) does not.

4.6. We retain the setting in Section 4.1, such that R is a henselian discrete valuation ring and $S = \operatorname{Spec} R$ with closed point s and generic point η. Let $f : X \to S$ be a separated morphism of finite type, put $X_s := X \times_S s$ and $X_\eta := X \times_S \eta$. Let $S^{\mathrm{sh}} = S_{(\bar{s})}$ be the strict localization of S at $\bar{s}$. Consider the following diagram with cartesian rectangles:

$$\begin{array}{ccccccc} X_{\bar{s}} & \xrightarrow{\bar{i}} & X \times_S S^{\mathrm{sh}} & \xleftarrow{\bar{j}} & X_{\overline{\eta}} & \xrightarrow{q_X} & X_\eta \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \bar{s} & \longrightarrow & S^{\mathrm{sh}} & \longleftarrow & \overline{\eta} & \longrightarrow & \eta. \end{array} \tag{4.6.1}$$

Define the *nearby cycles functor*

$$\Psi_X : D^b_c(X_\eta, \Lambda) \longrightarrow D^b_c(X_s\overline{\times}_s\eta, \Lambda) \tag{4.6.2}$$

by $\Psi_X(\mathcal{F}) := \bar{i}^*\bar{j}_*q_X^*\mathcal{F}$, with the natural continuous G_η-action. In particular, applying the forgetful functor $u_{X_s}^*$ we have

$$u_{X_s}^*\Psi_X(\mathcal{F}) \simeq \bar{i}^*\bar{j}_*q_X^*\mathcal{F} \in D_c^b(X_{\bar{s}}, \Lambda). \tag{4.6.3}$$

Denote by $i : X_s \hookrightarrow X$ the closed immersion, for $\mathcal{F} \in D_c^b(X, \Lambda)$ we have the following natural morphism by adjunction:

$$p_{X_s}^*(i^*\mathcal{F}) \longrightarrow \Psi_X(\mathcal{F}), \tag{4.6.4}$$

which is called the *specialization* morphism.

Let $f : X \to Y$ be a morphism of schemes separated of finite type over S, we have the following natural morphism of functors (cf. [18, (2.3.3)-(2.3.6)])

$$f_{\bar{s}}^*\Psi_Y \to \Psi_X f_{\bar{\eta}}^*, \tag{4.6.5}$$

$$\Psi_Y f_{\bar{\eta}*} \to f_{\bar{s}*}\Psi_X, \tag{4.6.6}$$

$$f_{\bar{s}!}\Psi_Y \to \Psi_X f_{\bar{\eta}!}, \tag{4.6.7}$$

$$\Psi_Y f_{\bar{\eta}}^! \to f_{\bar{s}}^!\Psi_X. \tag{4.6.8}$$

The morphisms (4.6.5) and (4.6.6), (4.6.7) and (4.6.8) are adjoint to each other, respectively. If f is proper, (4.6.6) is an isomorphism; if f is smooth, (4.6.5) is an isomorphism.

Lemma 4.7. *Retain the preceding notation. For $\mathcal{F}, \mathcal{G} \in D_{\mathrm{ctf}}(X_\eta, \Lambda)$, there is a canonical morphism*

$$\Psi_X\mathcal{F} \otimes^L \Psi_X\mathcal{G} \longrightarrow \Psi_X(\mathcal{F} \otimes^L \mathcal{G}), \tag{4.7.1}$$

natural in both variables. Moreover, let $\mathrm{sp}_\Lambda : \Lambda \to \Psi_X\Lambda$ be the specialization morphism, the composite

$$\begin{aligned} \Psi_X\mathcal{F} &\simeq \Psi_X\mathcal{F} \otimes^L \Lambda \\ &\xrightarrow{1\otimes \mathrm{sp}_\Lambda} \Psi_X\mathcal{F} \otimes^L \Psi_X\Lambda \\ &\xrightarrow{(4.7.1)} \Psi_X(\mathcal{F} \otimes^L \Lambda) \simeq \Psi_X\mathcal{F} \end{aligned} \tag{4.7.2}$$

is the identity.

Proof. The first part is the tensor compatibility of the nearby-cycle functor, cf. [18, Lemma 2.3.1]. The open immersion $\bar{j} : X_{\bar{\eta}} \hookrightarrow X \times_S S^{\mathrm{sh}}$ gives a natural morphism

$$\bar{j}^*\left(\bar{j}_*q_X^*\mathcal{F} \otimes^L \bar{j}_*\Lambda\right) \simeq \bar{j}^*\bar{j}_*q_X^*\mathcal{F} \otimes^L \bar{j}^*\bar{j}_*\Lambda \xrightarrow{\text{adj.}} q_X^*\mathcal{F} \otimes^L \Lambda, \tag{4.7.3}$$

which gives the natural morphism

$$\bar{j}_*q_X^*\mathcal{F} \otimes^L \bar{j}_*\Lambda \longrightarrow \bar{j}_*q_X^*\mathcal{F} \tag{4.7.4}$$

by adjunction. Hence one may verify the composition

$$\bar{j}_*q_X^*\mathcal{F} \simeq \bar{j}_*q_X^*\mathcal{F} \otimes^L \Lambda \xrightarrow{\text{adj.}} \bar{j}_*q_X^*\mathcal{F} \otimes^L \bar{j}_*\Lambda \xrightarrow{(4.7.4)} \bar{j}_*q_X^*\mathcal{F} \tag{4.7.5}$$

is the identity by considering its adjoint. Now apply $\bar{i}^*$ to (4.7.5), the morphism marked by adj. becomes $1 \otimes \mathrm{sp}_\Lambda$ and the morphism marked by (4.7.4) becomes (4.7.1), therefore we get the second result. □

4.8. Let $i : Z \hookrightarrow X$ be a closed immersion of separated schemes of finite type over k, with ideal $\mathcal{I} \subset \mathcal{O}_X$. Recall that the *deformation to the normal cone* is given by

$$A_Z X := \operatorname{Spec}_X \mathcal{O}_X[t, \mathcal{I}t^{-1}], \tag{4.8.1}$$

as a scheme over $\mathbb{A}^1_k$, where t is an indeterminate and $\mathcal{O}_X[t, \mathcal{I}t^{-1}] \subset \mathcal{O}_X[t, t^{-1}]$. By definition, $A_Z X$ is equipped with a flat morphism $A_Z X \to \mathbb{A}^1_k = \operatorname{Spec} k[t]$, with fibres

$$(A_Z X)|_{\mathbb{G}_{m,k}} \simeq X \times_k \mathbb{G}_{m,k}, \qquad (A_Z X)_0 \simeq C_Z X := \operatorname{Spec}_Z \left(\bigoplus_{n \geq 0} \mathcal{I}^n / \mathcal{I}^{n+1} \right). \tag{4.8.2}$$

Here the latter space $C_Z X$ is called the *normal cone* of Z in X. The ideal generated by $\mathcal{I}t^{-1}$ gives the quotient

$$\mathcal{O}_X[t, \mathcal{I}t^{-1}] / \left(\mathcal{I}t^{-1} \right) \simeq (\mathcal{O}_X / \mathcal{I})[t] \tag{4.8.3}$$

which defines a closed subscheme $\widetilde{i} : Z \times_k \mathbb{A}^1_k \hookrightarrow A_Z X$. On the fibre over 0, $\widetilde{i}$ restricts to the *base* $Z \hookrightarrow C_Z X$ of the normal cone; on the fibres over $\mathbb{G}_{m,k}$, $\widetilde{i}$ restricts to the original closed immersion $Z \hookrightarrow X$. If X and Z are smooth over k, then $A_Z X \to \mathbb{A}^1_k$ is smooth. See [9], [18, Definition 2.2.4].

4.9. From this section on, we take $R = k[t]^h_{(t)}$ to be the henselization of $k[t]_{(t)}$, $S = \operatorname{Spec} k[t]^h_{(t)}$ with closed point $s = \operatorname{Spec} k$ and generic point $\eta_0 = \operatorname{Spec} K$. We use the notation $\mathcal{G}_{\eta_0} := p_Y^* \mathcal{G} \in D^b_c(Y \overline{\times}_s \eta_0, \Lambda)$ for $Y \in \mathrm{Sch}_k$ and $\mathcal{G} \in D^b_c(Y, \Lambda)$. In some cases we also suppress p_Y^* or $(-)_{\eta_0}$ to simplify the notations if no confusion is caused.

Let $\mathrm{pr}_1 : X \times_k \mathbb{G}_m \to X$ be the projection and we denote by $\Psi_{A_Z X}$ the nearby cycles functor for $A_Z X \times_{\mathbb{A}^1_k} S$. *Verdier's specialization* is defined as

$$\nu_i = \nu_{Z/X} := \Psi_{A_Z X} \circ \mathrm{pr}_1^* : D^b_c(X, \Lambda) \to D^b_c(C_Z X \overline{\times}_s \eta_0, \Lambda), \tag{4.9.1}$$

cf. [21, Section 8] and [18, (2.3.26)–(2.3.27)]. For $\mathcal{G} \in D^b_c(Y \overline{\times}_s \eta_0, \Lambda)$, by Galois descent there exists a unique closed subset $\operatorname{Supp}_k \mathcal{G} \subset Y$ satisfying $(\operatorname{Supp}_k \mathcal{G})_{\bar{s}} = \operatorname{Supp} \mathcal{G}$; see [18, (3.1.13)].

Proposition 4.10. *For $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$, we have the following properties for specialization.*

(1) *Consider the commutative square*

$$\begin{array}{ccc} Z & \xrightarrow{i} & X \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ W & \xrightarrow{j} & Y \end{array} \tag{4.10.1}$$

of smooth schemes over k with regular immersions i, j. Then we have natural morphisms

$$A_g(f) : A_Z X \to A_W Y, \qquad Tf : N_{Z/X} \to N_{W/Y}, \tag{4.10.2}$$

with natural base change morphisms (cf. [18, (2.3.35)-(2.3.38)])

$$(Tf)^* \nu_j \longrightarrow \nu_i f^*, \tag{4.10.3}$$

$$\nu_j f_* \longrightarrow (Tf)_* \nu_i, \tag{4.10.4}$$

$$(Tf)_! \nu_i \longrightarrow \nu_j f_! \tag{4.10.5}$$

$$\nu_i f^! \longrightarrow (Tf)^! \nu_j. \tag{4.10.6}$$

When $A_g(f)$ is smooth, (4.10.3) is an isomorphism; when $A_g(f)$ is proper, (4.10.4) and (4.10.5) are isomorphisms.

(2) *The complex $\nu_i\mathcal{F}$ is monodromic. Let $0 : Z \hookrightarrow C_ZX$ be the inclusion of the base, then we have natural isomorphisms*

$$0^*\nu_i\mathcal{F} \xrightarrow{\simeq} i^*\mathcal{F}, \qquad i^!\mathcal{F} \xrightarrow{\simeq} 0^!\nu_i\mathcal{F}. \tag{4.10.7}$$

In particular, for $\mathcal{G} \in D_{\mathrm{ctf}}(Z, \Lambda)$ we have

$$\nu_i(i_*\mathcal{G}) \simeq 0_*\mathcal{G}. \tag{4.10.8}$$

(3) *If $\mathrm{Supp}\,\mathcal{F} \subset W \subset X$, with W a closed subscheme of X, then*

$$\mathrm{Supp}_k\,\nu_i\mathcal{F} \subset \overline{W \times \mathbb{G}_m} \cap C_ZX, \tag{4.10.9}$$

where the closure is taken in A_ZX.

(4) *If X is defined over a smooth k-scheme S, then*

$$\nu_i D_{X/S}\mathcal{F} \xrightarrow{\simeq} D_{C_ZX/S}(\nu_i\mathcal{F}). \tag{4.10.10}$$

Here, the duality functor on the right side is defined as

$$D_{C_ZX/S}(-) := R\mathcal{H}om(-, \mathcal{K}_{C_ZX/S}). \tag{4.10.11}$$

Proof. For (1)-(2), see [18, (2.3.35)–(2.3.38), Proposition 2.3.7] and [21, Section 8, (SP1)–(SP5)]. Formula (4.10.8) follows by proper base change. For (3), apply open restriction to the complement of the closure. Compatibility of nearby-cycle and duality gives (4), see [13, Construction 3.3, Corollaries 3.8–3.10]. □

Proposition 4.11. *Retain the previous setting. For $\mathcal{F}, \mathcal{G} \in D_{\mathrm{ctf}}(X, \Lambda)$, there is a canonical morphism*

$$\nu_i\mathcal{F} \otimes^L \nu_i\mathcal{G} \longrightarrow \nu_i(\mathcal{F} \otimes^L \mathcal{G}) \tag{4.11.1}$$

in $D_c^b(C_ZX\overline{\times}_s\eta_0, \Lambda)$, natural in both variables.

Proof. Apply Lemma 4.7 to $\mathrm{pr}_1^*\,\mathcal{F}$ and $\mathrm{pr}_1^*\,\mathcal{G}$ on the generic fibre of $A_ZX \times_{\mathbf{A}^1_k} S$, using the definition $\nu_i = \Psi_{A_ZX} \circ \mathrm{pr}_1^*$. □

Lemma 4.12. *Consider the commutative square*

$$\begin{array}{ccc} Z & \xrightarrow{i} & X \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ W & \xrightarrow{j} & Y \end{array} \tag{4.12.1}$$

of smooth schemes over k with regular immersions i, j. For constructible complexes $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$, $\mathcal{G} \in D_{\mathrm{ctf}}(Y, \Lambda)$, we have natural morphisms

$$\nu_i\mathcal{F} \otimes^L (Tf)^*\nu_j\mathcal{G} \longrightarrow \nu_i\mathcal{F} \otimes^L \nu_i f^*\mathcal{G} \longrightarrow \nu_i(\mathcal{F} \otimes^L f^*\mathcal{G}). \tag{4.12.2}$$

In particular, after applying 0^*, the morphisms (4.10.7) identify (4.12.2) with

$$i^*\mathcal{F} \otimes^L g^*j^*\mathcal{G} \xrightarrow{\simeq} i^*(\mathcal{F} \otimes^L f^*\mathcal{G}). \tag{4.12.3}$$

Proof. Use (4.10.3) and Proposition 4.11. □

Lemma 4.13. *Let $i : Z \hookrightarrow X$ be a regular immersion between smooth schemes over k, and write $j : X \setminus Z \hookrightarrow X$ and $j_T : (N_{Z/X})^\times \hookrightarrow N_{Z/X}$ be the open immersions, then we have an isomorphism*

$$\nu_i(j_*\Lambda) \xrightarrow{\simeq} j_{T*}\Lambda. \tag{4.13.1}$$

Proof. Recall that we have the zero section $0 : Z \to N_{Z/X}$. Using Proposition 4.10, we have

$$\nu_i(i_*i^!\Lambda) \simeq 0_*i^!\Lambda, \qquad \nu_i(\Lambda) \simeq \Lambda, \tag{4.13.2}$$

hence the localization triangle gives the natural isomorphism (4.13.1). □

4.14. Now we give a generalization of Lemma 3.8. Let k be a perfect field of characteristic $p > 0$, consider the following cartesian square

$$\begin{array}{ccc} X & \xrightarrow{i} & Y \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle f} \\ W & \xrightarrow{\delta} & T \end{array} \tag{4.14.1}$$

in Sch_k, with W, T smooth over k and δ a regular immersion of codimension r. Let $N_{W/T}$ be the normal bundle of W in T, $E := p^*N_{W/T}$ be a vector bundle of rank r over X, $\iota : C_XY \hookrightarrow E$ be the closed immersion. Let $0 : X \to E$, $\pi : E \to X$ and $0^\vee : X \to E^\vee$, $\pi^\vee : E^\vee \to X$ be the zero section and projection of E and $E^\vee$, respectively. Let $j^\vee : E^{\vee,\times} := E^\vee \setminus 0^\vee(X) \hookrightarrow E^\vee$ be the open immersion and $\pi^{\vee,\times} := \pi^\vee \circ j^\vee$.

Proposition 4.15. *Retain the previous construction. Let $\mathcal{F} \in D_{\mathrm{ctf}}(Y, \Lambda)$ be a constructible complex, then we have a natural isomorphism*

$$\delta^\Delta\mathcal{F} \xrightarrow{\simeq} \pi^{\vee,\times}_! j^{\vee *}\mathscr{F}_{\psi,E}\left(\iota_*\nu_{X/Y}\mathcal{F}\right)[1] \tag{4.15.1}$$

in $D_{\mathrm{ctf}}(X\overline{\times}_s\eta_0, \Lambda)$.

We postpone the proof of Proposition 4.15 to Section 4.19.

Lemma 4.16. *Consider the setting of Proposition 4.15, then $\iota_*\nu_{X/Y}\mathcal{F}$ is a monodromic complex over E, and we have natural isomorphisms*

$$0^*\iota_*\nu_{X/Y}(\mathcal{F}) \simeq i^*\mathcal{F}, \qquad i^!\mathcal{F} \simeq 0^!\iota_*\nu_{X/Y}(\mathcal{F}). \tag{4.16.1}$$

Proof. By Proposition 4.10, $\nu_{X/Y}(\mathcal{F})$ is a monodromic complex over C_XY. Since $\iota : C_XY \hookrightarrow E$ is a homogeneous closed immersion, $\iota_*\nu_{X/Y}\mathcal{F}$ is monodromic by definition.

Let $0_C : X \to C_XY$ be the base of the cone C_XY. By proper base change we have

$$0^*\iota_* \simeq 0_C^*, \qquad 0_C^! \simeq 0^!\iota_*, \tag{4.16.2}$$

combining with (4.10.7) we get (4.16.1). □

Lemma 4.17. *Let $A \xrightarrow{u} B \xrightarrow{v} C$ be two composable morphisms in a stable ∞-category $\mathcal{D}$, then we have a cofibre sequence*

$$\mathrm{cofib}(u) \longrightarrow \mathrm{cofib}(v \circ u) \longrightarrow \mathrm{cofib}(v). \tag{4.17.1}$$

Proof. Fix the zero object $0 \in \mathcal{D}$. By successively forming pushouts, we obtain a coherent diagram in $\mathcal{D}$

$$\begin{array}{ccccc} A & \xrightarrow{u} & B & \xrightarrow{v} & C \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathrm{cofib}(u) & \xrightarrow{\alpha} & \mathrm{cofib}(v \circ u) \\ & & \downarrow & & \downarrow{\scriptstyle \beta} \\ & & 0 & \longrightarrow & \mathrm{cofib}(v), \end{array} \tag{4.17.2}$$

where we may verify by definition that all three small squares are pushouts. Thus the lower-right square exhibits $\mathrm{cofib}(v)$ as the cofibre of α, giving the cofibre sequence (4.17.1).

□

Proposition 4.18. *Consider the setting of (4.14.1), but we do not require that δ is regular here. Let $\mathcal{Q}_\delta := \mathrm{cofib}(\Lambda \to \delta^!\Lambda(r)[2r]) \in D_{\mathrm{ctf}}(W,\Lambda)$ be the cofibre of the purity morphism, then we have the following distinguished triangle in $D_{\mathrm{ctf}}(X\overline{\times}_s\eta_0,\Lambda)$ with natural morphisms:*

$$i^*\mathcal{F}\otimes^L p^*\mathcal{Q}_\delta(-r)[-2r] \longrightarrow \pi_!^{\vee,\times} j^{\vee *}\mathscr{F}_{\psi,E}\left(\iota_*\nu_{X/Y}\mathcal{F}\right)[1] \longrightarrow \delta^\Delta\mathcal{F} \xrightarrow{+1}. \tag{4.18.1}$$

Proof. We verify that the diagram

$$\begin{array}{ccccc} 0^*\iota_*\nu_{X/Y}(\mathcal{F})(-r)[-2r] & & \xrightarrow{\quad\mathrm{pur}_0\quad} & & 0^!\iota_*\nu_{X/Y}(\mathcal{F}) \\ \downarrow\simeq & & & & \uparrow\simeq \\ i^*\mathcal{F}(-r)[-2r] & \xrightarrow{\mathrm{pur}_\delta} & i^*\mathcal{F}\otimes^L p^*\delta^!\Lambda & \xrightarrow{c_{\delta,f,\mathcal{F}}} & i^!\mathcal{F}, \end{array} \tag{4.18.2}$$

is commutative. Here, by slight abuse of notation, we denote both the absolute purity morphisms and their adjoints by pur_ and omit the pullback $(-)_{\eta_0}$ to $X\overline{\times}_s\eta_0$. Recall that $c_{\delta,f,\mathcal{F}}$ is the transversality morphism (2.2.2), the vertical isomorphisms are (4.16.1).

Let $0_N : W \to N_{W/T}$ be the zero section, $p_N : E = N_{W/T}\times_W X \to N_{W/T}$ be the canonical projection. The closed immersion $W\times_k \mathbb{A}^1_k \hookrightarrow A_W T$ induces a closed immersion $\tilde{\delta} : W\times_k S \hookrightarrow A_W T\times_{\mathbb{A}^1_k} S$ which restricts to 0_N over s and δ over η_0. By [15, Proposition 2.3.2], $\mathrm{pur}_{\tilde{\delta}}$ restricts to pur_{0_N} over s and pur_δ over η_0. By Proposition 4.10 and naturality we have a commutative diagram

$$\begin{array}{ccc} 0_{N*}\Lambda(-r)[-2r] & \xrightarrow{\mathrm{pur}_{0_N}} & \Lambda \\ \simeq\downarrow & & \downarrow\simeq \\ \nu_\delta\left(\delta_*\Lambda(-r)[-2r]\right) & \xrightarrow{\mathrm{pur}_\delta} & \nu_\delta\Lambda. \end{array} \tag{4.18.3}$$

We now verify that the following diagram commutes:

$$\begin{array}{ccc} 0_*0^*\iota_*\nu_{X/Y}\mathcal{F}(-r)[-2r] & \xrightarrow{\mathrm{pur}_0} & \iota_*\nu_{X/Y}\mathcal{F} \\ \simeq\downarrow & & \downarrow\simeq \\ \iota_*\nu_{X/Y}\mathcal{F}\otimes^L p_N^*\nu_\delta\left(\delta_*\Lambda(-r)[-2r]\right) & \xrightarrow{\mathrm{pur}_\delta} & \iota_*\nu_{X/Y}\mathcal{F}\otimes^L p_N^*\nu_\delta\Lambda \\ \downarrow & & \downarrow \\ \iota_*\nu_{X/Y}\left(\mathcal{F}\otimes^L f^*\delta_*\Lambda(-r)[-2r]\right) & \xrightarrow{\mathrm{pur}_\delta} & \iota_*\nu_{X/Y}\mathcal{F} \\ \simeq\downarrow & & \| \\ 0_*i^*\mathcal{F}(-r)[-2r] & \longrightarrow & \iota_*\nu_{X/Y}\mathcal{F}. \end{array} \tag{4.18.4}$$

By projection formula we have

$$\begin{aligned} 0_*0^*\iota_*\nu_{X/Y}\mathcal{F}(-r)[-2r] &= 0_*\left(0^*\iota_*\nu_{X/Y}\mathcal{F}\otimes^L\Lambda(-r)[-2r]\right) \\ &\simeq \iota_*\nu_{X/Y}\mathcal{F}\otimes^L 0_*\Lambda(-r)[-2r], \end{aligned} \tag{4.18.5}$$

and the top horizontal morphism in (4.18.4) is equivalent to

$$\iota_*\nu_{X/Y}\mathcal{F}\otimes^L 0_*\Lambda(-r)[-2r] \xrightarrow{\mathrm{id}\otimes\mathrm{pur}_0} \iota_*\nu_{X/Y}\mathcal{F} \tag{4.18.6}$$

under the identification (4.18.5). Therefore, pulling back (4.18.3) by p_N^* and using proper base change we get the commutativity of the upper square. Using Lemma 4.12 and base change, we get a natural morphism

$$\iota_*\nu_{X/Y}\mathcal{G}_1 \otimes^L p_N^*\nu_\delta\mathcal{G}_2 \longrightarrow \iota_*\nu_{X/Y}\left(\mathcal{G}_1 \otimes^L f^*\mathcal{G}_2\right) \tag{4.18.7}$$

for $\mathcal{G}_1 \in D_{\mathrm{ctf}}(Y,\Lambda)$, $\mathcal{G}_2 \in D_{\mathrm{ctf}}(T,\Lambda)$. It gives the two middle vertical morphisms, and its naturality gives the commutativity of the middle square. The lower left vertical morphism is defined by

$$\begin{aligned}\iota_*\nu_{X/Y}\left(\mathcal{F} \otimes^L f^*\delta_*\Lambda(-r)[-2r]\right) &\xrightarrow[\text{b.c.}]{\simeq} \iota_*\nu_{X/Y}\left(\mathcal{F} \otimes^L i_*\Lambda(-r)[-2r]\right)\\ &\xleftarrow[\text{p.f.}]{\simeq} \iota_*\nu_{X/Y}\left(i_*i^*\mathcal{F}(-r)[-2r]\right)\\ &\xrightarrow[(4.10.4),\text{b.c.}]{\simeq} 0_*\nu_{X/X}\left(i^*\mathcal{F}(-r)[-2r]\right)\\ &\xrightarrow[(4.10.3)]{\simeq} 0_*i^*\mathcal{F}(-r)[-2r].\end{aligned} \tag{4.18.8}$$

The lower horizontal morphism is defined by applying $\iota_*\nu_{X/Y}(-)$ to

$$\begin{aligned}i_*i^*\mathcal{F}(-r)[-2r] &\xrightarrow[\text{p.f.}]{\simeq} \mathcal{F} \otimes^L f^*\delta_*\Lambda(-r)[-2r]\\ &\xrightarrow{\mathrm{pur}_\delta} \mathcal{F} \otimes^L f^*\delta_*\delta^!\Lambda \xrightarrow{\text{adj.}} \mathcal{F},\end{aligned} \tag{4.18.9}$$

and the naturality gives the commutativity of the lower square. By Lemma 4.7, the right vertical composite is the identity. By the definition, the left vertical composite is equal to (apply 0_* on) the left vertical isomorphism of (4.18.2).

Consider the natural isomorphism $0_*\mathcal{G} \simeq \iota_*\nu_{X/Y}(i_*\mathcal{G})$ for $\mathcal{G} \in D_{\mathrm{ctf}}(X,\Lambda)$ induced by base change and (4.10.4). Its naturality gives the commutativity of the upper and middle squares of the following diagram:

$$\begin{array}{ccc}
0_*i^*\mathcal{F}(-r)[-2r] & \xrightarrow{\simeq} & \iota_*\nu_{X/Y}\left(i_*i^*\mathcal{F}(-r)[-2r]\right)\\
\downarrow{\scriptstyle \mathrm{pur}_\delta} & & \downarrow{\scriptstyle \mathrm{pur}_\delta}\\
0_*(i^*\mathcal{F} \otimes^L p^*\delta^!\Lambda) & \xrightarrow{\simeq} & \iota_*\nu_{X/Y}\left(i_*(i^*\mathcal{F} \otimes^L p^*\delta^!\Lambda)\right)\\
\downarrow{\scriptstyle c_{\delta,f,\mathcal{F}}} & & \downarrow{\scriptstyle c_{\delta,f,\mathcal{F}}}\\
0_*i^!\mathcal{F} & \xrightarrow{\simeq} & \iota_*\nu_{X/Y}(i_*i^!\mathcal{F})\\
\downarrow & & \downarrow{\scriptstyle \text{adj.}}\\
0_*0^!\iota_*\nu_{X/Y}\mathcal{F} & \xrightarrow[\text{adj.}]{} & \iota_*\nu_{X/Y}\mathcal{F}.
\end{array} \tag{4.18.10}$$

The lower left vertical morphism is the right vertical isomorphism of (4.18.2) whose definition gives the commutativity of the bottom square. Notice that the right composite of (4.18.10) is equal to

$$i_*i^*\mathcal{F}(-r)[-2r] \xrightarrow{\mathrm{pur}_\delta} i_*\left(i^*\mathcal{F} \otimes^L p^*\delta^!\Lambda\right) \xrightarrow{c_{\delta,f,\mathcal{F}}} i_*i^!\mathcal{F} \xrightarrow{\text{adj.}} \mathcal{F} \tag{4.18.11}$$

after applying $\iota_*\nu_{X/Y}(-)$, and the composite (4.18.11) is equal to (4.18.9) by the construction of $c_{\delta,f,\mathcal{F}}$. Therefore, combining (4.18.4), (4.18.10) and the adjunction pair $0_* \dashv 0^!$ we get the commutativity of (4.18.2).

Now apply Lemma 4.17 to

$$0^*\iota_*\nu_{X/Y}(\mathcal{F})(-r)[-2r] \simeq i^*\mathcal{F}(-r)[-2r] \xrightarrow{\mathrm{pur}_\delta} i^*\mathcal{F} \otimes^L p^*\delta^!\Lambda \xrightarrow{c_{\delta,f,\mathcal{F}}} i^!\mathcal{F} \simeq 0^!\iota_*\nu_{X/Y}(\mathcal{F}), \tag{4.18.12}$$

using (4.18.2), we get a distinguished triangle

$$i^*\mathcal{F}\otimes^L p^*\mathcal{Q}_\delta(-r)[-2r]\longrightarrow 0^\Delta\iota_*\nu_{X/Y}\mathcal{F}\longrightarrow\delta^\Delta\mathcal{F}\xrightarrow{+1}. \tag{4.18.13}$$

Applying Lemma 3.8 with $\mathcal{N}=\iota_*\nu_{X/Y}\mathcal{F}$, $V=E$, we have

$$0^\Delta\iota_*\nu_{X/Y}\mathcal{F}\simeq\pi_!^{\vee,\times}j^{\vee *}\mathscr{F}_{\psi,E}\left(\iota_*\nu_{X/Y}\mathcal{F}\right)[1] \tag{4.18.14}$$

and the desired result follows. □

4.19. *Proof of Proposition 4.15.* When δ is regular, the purity morphism $\mathrm{pur}_\delta:\Lambda\to\delta^!\Lambda(r)[2r]$ is an isomorphism and hence $\mathcal{Q}_\delta=0$. In this case, (4.18.1) gives the desired isomorphism.

4.20. We retain the setting of 4.9, and let k be a perfect field of characteristic $p>0$. Let $X\in\mathrm{Sm}_k$, $i:Z\hookrightarrow X$ be a regular immersion. Let Λ be a finite local ring with residue characteristic $\ell\neq p$ for some prime number ℓ, choose a nontrivial character $\psi:\mathbb{F}_p\to\Lambda^\times$. For constructible complex $\mathcal{F}\in D_{\mathrm{ctf}}(X,\Lambda)$, we define its *microlocalization* along i to be

$$\mu_{Z/X}(\mathcal{F}):=\mathscr{F}_{\psi,N_{Z/X}}\circ\nu_{Z/X}(\mathcal{F})\in D_{\mathrm{ctf}}(N^\vee_{Z/X}\overline{\times}_s\eta_0,\Lambda). \tag{4.20.1}$$

Moreover, we define a bifunctor

$$\mu\mathcal{H}om(-,-):D_{\mathrm{ctf}}(X,\Lambda)^{\mathrm{op}}\times D_{\mathrm{ctf}}(X,\Lambda)\to D_{\mathrm{ctf}}(T^*X\overline{\times}_s\eta_0,\Lambda) \tag{4.20.2}$$

by

$$\mu\mathcal{H}om(\mathcal{F}_1,\mathcal{F}_2):=\mathscr{F}_{\psi,TX}\circ\nu_{X/X\times_kX}R\mathcal{H}om_{X\times_kX}(\mathrm{pr}_1^*\mathcal{F}_1,\mathrm{pr}_2^!\mathcal{F}_2), \tag{4.20.3}$$

where $\mathrm{pr}_1,\mathrm{pr}_2$ are projections from $X\times_k X$ and the latter specialization is with respect to the diagonal regular immersion $\delta_{X/k}:X\hookrightarrow X\times_k X$, whose normal bundle is isomorphic to T^*X.

Let $Y\in\mathrm{Sch}_k$, recall that giving $\mathcal{G}\in D^b_c(Y\overline{\times}_s\eta_0,\Lambda)$ is equivalent to giving a constructible complex $\mathcal{G}\in D^b_c(Y_{\overline{s}},\Lambda)$ together with continuous action of G_{η_0}. Notice that the action of G_{η_0} is semilinear, with G_{η_0} acting on $Y_{\overline{s}}$ along the surjection $G_{\eta_0}\twoheadrightarrow G_s$. Therefore, the closed subset $\mathrm{Supp}\,\mathcal{G}\subset|Y_{\overline{s}}|$ descends to a closed subset $\mathrm{Supp}_k\mathcal{G}\subset|Y|$ such that $(\mathrm{Supp}_k\mathcal{G})_{\overline{s}}=\mathrm{Supp}\,\mathcal{G}$. Now for $X\in\mathrm{Sm}_k$ and $\mathcal{F}\in D_{\mathrm{ctf}}(X,\Lambda)$, we define

$$SS_\mu(\mathcal{F}):=\mathrm{Supp}_k\left(\mu\mathcal{H}om(\mathcal{F},\mathcal{F})\right) \tag{4.20.4}$$

as a closed subset of T^*X.

Put $\delta=\delta_{X/k}:X\to X\times_k X$, recall that by (2.12.8) we have a series of morphisms

$$\delta_!\Lambda\xrightarrow{\mathrm{coev}}R\mathcal{H}om(\mathrm{pr}_2^*\mathcal{F},\mathrm{pr}_1^!\mathcal{F})\xrightarrow{\mathrm{ev}}\delta_*\mathcal{K}_{X/k}. \tag{4.20.5}$$

Applying $D_{X\times_kX/k}(-)$, (4.20.5) becomes

$$\delta_!\Lambda\xrightarrow{\mathrm{coev}}R\mathcal{H}om(\mathrm{pr}_1^*\mathcal{F},\mathrm{pr}_2^!\mathcal{F})\xrightarrow{\mathrm{ev}}\delta_*\mathcal{K}_{X/k}. \tag{4.20.6}$$

which has the same composite as (4.20.5) by Lemma 2.11. Applying the functor $\mu_{X/X\times_kX}$ we get a series of morphisms

$$\Lambda\longrightarrow\mu\mathcal{H}om(\mathcal{F},\mathcal{F})\longrightarrow\pi^{\vee *}\mathcal{K}_{X/k}, \tag{4.20.7}$$

here $\pi^\vee:T^*X\to X$ is the projection from cotangent bundle and for the first and last terms we use Proposition 4.10 and 3.2. The composition (4.20.7) defines a class in $H^0_{SS_\mu(\mathcal{F})}(T^*X,\pi^{\vee *}\mathcal{K}_{X/k})$, we denote it by $CC_\mu(\mathcal{F})$.

We have the following comparison with $SS(\mathcal{F})$ and $CC(\mathcal{F})$.

Theorem 4.21 ([18, Theorems 1.0.1 and 1.0.3])**.** *Retain the construction above and assume X is of pure dimension d. Then we have*

$$SS(\mathcal{F}) = SS_{\mu}(\mathcal{F}), \qquad \mathrm{cl}\,(CC(\mathcal{F})) = CC_{\mu}(\mathcal{F}), \tag{4.21.1}$$

here $\mathrm{cl} : \mathrm{Z}_d\,(SS(\mathcal{F})) \to H^0_{SS_{\mu}(\mathcal{F})}(T^*X, \pi^{\vee *}\mathcal{K}_{X/k})$ *is the cycle class map.*

5. Proof of theorem

5.1. Let k be a perfect field of characteristic $p > 0$, and let Λ be a finite local ring of residue characteristic $\ell \neq p$. Consider the following commutative diagram

$$\begin{array}{ccccc} Z & \overset{\tau}{\hookrightarrow} & X & \xrightarrow{\quad f \quad} & Y, \\ & & & {\scriptstyle h}\searrow \quad \swarrow{\scriptstyle g} & \\ & & & S & \end{array} \tag{5.1.1}$$

where X, Y, S are smooth of pure dimension over k, $n = \dim X$, g is smooth of relative dimension $r > 0$, S is connected of dimension s and τ is a closed immersion. We have the following cycle class map:

$$\begin{aligned} \widetilde{\mathrm{cl}} : \mathrm{CH}_j(Z) \xrightarrow{\mathrm{cl}} H^0_Z\left(X, \mathcal{K}_{X/k}[-2j](-j)\right) &\simeq H^0_Z\left(X, \mathcal{K}_{X/S}[2(s-j)](s-j)\right) \\ &\xrightarrow{\alpha} H^0_Z\left(X, \mathcal{K}_{X/Y/S}[2(s-j)](s-j)\right), \end{aligned} \tag{5.1.2}$$

where the middle isomorphism follows from the smooth purity for S over k, the map α is from the distinguished triangle (2.13.4).

Theorem 5.2. *Retain the setting above. Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$, such that $f\,|_{X\setminus Z}$ is $SS(\mathcal{F}\,|_{X\setminus Z})$-transversal and h is $SS(\mathcal{F})$-transversal, then we have*

$$\widetilde{C}^Z_{X/Y/S} = \widetilde{\mathrm{cl}}\left(cc^Z_{X/Y/S}(\mathcal{F})\right) \tag{5.2.1}$$

in $H^0_Z(X, \mathcal{K}_{X/Y/S})$.

We devote the whole section to the proof of Theorem 5.2.

5.3. We first do some reductions.

(1) By Proposition 2.9, h is smooth on an open neighbourhood of $SS(\mathcal{F}) \cap T^*_X X = \mathrm{supp}(\mathcal{F})$. Using the open pullback compatibility of both sides of (5.2.1), we may assume h is smooth in the following discussion.

(2) By [18, Lemma 2.1.1], there exists a finite étale faithfully flat local Λ-algebra Λ' and a non-trivial character $\psi : \mathbb{F}_p \to (\Lambda')^{\times}$. Let Y be a scheme separated of finite type over k, let W be its closed subscheme and $\mathcal{G} \in D^b_c(Y, \Lambda)$, we have the injective coefficient extension

$$H^i_W(Y, \mathcal{G}) \hookrightarrow H^i_W(Y, \mathcal{G}) \otimes_{\Lambda} \Lambda' \xrightarrow{\simeq} H^i_W(Y, \mathcal{G} \otimes^L_{\Lambda} \Lambda'), \tag{5.3.1}$$

hence we may assume there exists a non-trivial character $\psi : \mathbb{F}_p \to \Lambda^{\times}$ in the following discussion.

5.4. We recall some constructions of tangent and cotangent bundles. By [18, Lemma 2.2.2 and 2.2.5], we have a regular immersion

$$\widetilde{i} : A_X(X \times_S X) \longrightarrow A_X(X \times_k X) \tag{5.4.1}$$

which has generic fibre $i_h : X \times_S X \hookrightarrow X \times_k X$ and special fibre $i_{h,0} : T(X/S) \hookrightarrow TX$. On the cotangent bundles, we have a surjection $p_{X/S} := i_{h,0}^{\vee} : T^*X \to T^*(X/S)$. Let $0_{Y/S} : Y \to T(Y/S)$ be the zero section, $\pi : TX \to X$, $\pi^{\vee} : T^*X \to X$, $\pi_{X/S} : T(X/S) \to X$, $\pi_{X/S}^{\vee} : T^*(X/S) \to X$ be the bundle projections. Let $\delta_{X/k} : X \to X \times_k X$, $\delta_{X/S} : X \to X \times_S X$, $\delta_{X/Y} : X \to X \times_Y X$ be the corresponding diagonal morphisms. By Proposition 2.9, h is locally acyclic relatively to $\mathcal{F}$, hence we may put $\mathcal{H}_{\mathcal{F}/k} = \mathcal{F} \boxtimes_k D_{X/k}\mathcal{F}$, $\mathcal{H}_{\mathcal{F}/S} = \mathcal{F} \boxtimes_S D_{X/S}\mathcal{F}$, $\mathcal{H}^{\vee}_{\mathcal{F}/k} = D_{X/k}\mathcal{F} \boxtimes_k \mathcal{F}$ and $\mathcal{H}^{\vee}_{\mathcal{F}/S} = D_{X/S}\mathcal{F} \boxtimes_S \mathcal{F}$ following Section 2.12.

For the relative tangential map $T_S f : T(X/S) \to f^*T(Y/S)$, we define

$$T(X/Y) := T(X/S) \times_{f^*T(Y/S)} X = \operatorname{Spec}_X \operatorname{Sym} \Omega^1_{X/Y} \tag{5.4.2}$$

as the following cartesian diagram:

$$\begin{array}{ccc} T(X/Y) & \xrightarrow{i_{f,0}} & T(X/S) \\ {\scriptstyle \pi_{X/Y}}\downarrow & & \downarrow{\scriptstyle T_S f} \\ X & \xrightarrow[0_{Y/S} \times \mathrm{id}_X]{} & f^*T(Y/S). \end{array} \tag{5.4.3}$$

In this case, we have closed immersion $i_{f,0} : T(X/Y) \to T(X/S)$, $0_{X/Y} : X \to T(X/Y)$ and projection $\pi_{X/Y} : T(X/Y) \to X$.

5.5. Retain the previous setting. By Proposition 4.10 and 4.11, we have a natural morphism

$$a_{\mathcal{F}} : i_{h,0}^* \nu_{\delta_{X/k}} \mathcal{H}^{\vee}_{\mathcal{F}/k} \otimes^L i_{h,0}^! \Lambda \longrightarrow \nu_{\delta_{X/S}} \left(i_h^* \mathcal{H}^{\vee}_{\mathcal{F}/k} \otimes^L i_h^! \Lambda \right) \xrightarrow{\simeq} \nu_{\delta_{X/S}} \mathcal{H}^{\vee}_{\mathcal{F}/S}, \tag{5.5.1}$$

where the last isomorphism is [23, Proposition 2.5]. By [18, Lemma 2.3.2], $a_{\mathcal{F}}$ fits into the following commutative diagram

$$\begin{array}{ccccc} i_{h,0}^* \nu_{\delta_{X/k}} \mathcal{H}^{\vee}_{\mathcal{F}/k} \otimes^L i_{h,0}^! \Lambda & \xrightarrow{a_{\mathcal{F}}} & \nu_{\delta_{X/S}} \left(i_h^* \mathcal{H}^{\vee}_{\mathcal{F}/k} \otimes^L i_h^! \Lambda \right) & \simeq & \nu_{\delta_{X/S}} \mathcal{H}^{\vee}_{\mathcal{F}/S} \\ {\scriptstyle c_{i_{h,0}, \mathrm{id}, \nu_{\delta_{X/k}} \mathcal{H}^{\vee}_{\mathcal{F}/k}}}\downarrow & & & & \downarrow{\scriptstyle c_{i_h, \mathrm{id}, \mathcal{H}^{\vee}_{\mathcal{F}/k}}} \\ i_{h,0}^! \nu_{\delta_{X/k}} \mathcal{H}^{\vee}_{\mathcal{F}/k} & \xleftarrow[(4.10.6)]{} & & & \nu_{\delta_{X/S}} i_h^! \mathcal{H}^{\vee}_{\mathcal{F}/k}. \end{array} \tag{5.5.2}$$

Lemma 5.6. *In the above setting, the morphisms $0^*_{X/S} a_{\mathcal{F}}$ and $0^!_{X/S} a_{\mathcal{F}}$ are both isomorphisms.*

Proof. By base change and Proposition 4.10, we have $0^*_{X/S} a_{\mathcal{F}} = \delta^*_{X/S} c_{i_h, \mathrm{id}, \mathcal{H}_{\mathcal{F}/k}}$, which is an isomorphism by Lemma 2.3.

The $SS(\mathcal{F})$-transversality of h implies the inclusion

$$SS(\mathcal{F}) \cap \ker p_{X/S} \subset T^*_X X, \tag{5.6.1}$$

hence by [4, proof of Lemma 1.2(ii)], $p_{X/S}$ is finite after restricting to $SS(\mathcal{F})$. Combining this with Theorem 4.21, we conclude that the natural morphism

$$p_{X/S!} \mu\mathcal{H}om(\mathcal{F}, \mathcal{F}) \longrightarrow p_{X/S*} \mu\mathcal{H}om(\mathcal{F}, \mathcal{F}) \tag{5.6.2}$$

is an isomorphism. By Corollary 3.5, the morphism $c_{i_{h,0},\mathrm{id},\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k}}$ is an isomorphism. Again, use Proposition 4.10, the upper right to lower left composition of (5.5.2)

$$\nu_{\delta_{X/S}}\mathcal{H}^\vee_{\mathcal{F}/S} \to i^!_{h,0}\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k} \tag{5.6.3}$$

is an isomorphism after applying $0^!_{X/S}$, hence $0^!_{X/S}a_{\mathcal{F}}$ is also an isomorphism since $c_{i_{h,0},\mathrm{id},\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k}}$ is an isomorphism. □

5.7. By Lemma 5.6 and the naturality of $a_{\mathcal{F}}$, we have a commutative diagram

$$\begin{array}{ccccc} 0_{X/S*}\Lambda & \xrightarrow{\mathrm{coev}} & i^*_{h,0}\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k}\otimes^L i^!_{h,0}\Lambda & \xrightarrow{\mathrm{ev}} & 0_{X/S*}\mathcal{K}_{X/S} \\ \| & & \downarrow{\scriptstyle a_{\mathcal{F}}} & & \| \\ 0_{X/S*}\Lambda & \xrightarrow[\mathrm{coev}]{} & \nu_{\delta_{X/S}}\mathcal{H}^\vee_{\mathcal{F}/S} & \xrightarrow[\mathrm{ev}]{} & 0_{X/S*}\mathcal{K}_{X/S}, \end{array} \tag{5.7.1}$$

where the upper evaluation morphism is the composition of $a_{\mathcal{F}}$ with lower evaluation. Applying $\mathscr{F}_{\psi,T(X/S)}$, using (5.6.2), we get a commutative diagram

$$\begin{array}{ccccc} p_{X/S*}\Lambda & \xrightarrow{\mathrm{coev}} & p_{X/S*}\mu\mathcal{H}om(\mathcal{F},\mathcal{F}) & & \\ \uparrow{\scriptstyle \mathrm{adj.}} & & \uparrow\simeq & & \\ & & p_{X/S!}\mu\mathcal{H}om(\mathcal{F},\mathcal{F}) & \xrightarrow{\mathrm{ev}} & p_{X/S!}(\pi^\vee)^*\mathcal{K}_{X/k} \\ & & \downarrow\simeq & & \downarrow{\scriptstyle (-1)^s\mathrm{Tr}_{p_{X/S}}} \\ \Lambda & \xrightarrow{\mathrm{coev}} & \mathscr{F}_{\psi,T(X/S)}\left(i^*_{h,0}\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k}\otimes^L i^!_{h,0}\Lambda\right) & \xrightarrow{\mathrm{ev}} & (\pi^\vee_{X/S})^*\mathcal{K}_{X/S} \\ \| & & \downarrow{\scriptstyle a_{\mathcal{F}}} & & \| \\ \Lambda & \xrightarrow{\mathrm{coev}} & \mathscr{F}_{\psi,T(X/S)}\left(\nu_{\delta_{X/S}}\mathcal{H}^\vee_{\mathcal{F}/S}\right) & \xrightarrow{\mathrm{ev}} & (\pi^\vee_{X/S})^*\mathcal{K}_{X/S}. \end{array} \tag{5.7.2}$$

The upper left square is commutative by Corollary 3.5, the upper right square is commutative by naturality of (3.2.7) and [18, 2.1.43]. The upper composition represents $(-1)^s p_{X/S*}CC_\mu(\mathcal{F})$, the lower composition can be viewed as a relative over S definition of $CC_\mu(\mathcal{F})$.

5.8. Retain the previous construction. As in Section 2.14 and 2.17, we define

$$\begin{aligned} q_S &: \mathbb{P}(f^*T^*(Y/S)) \longrightarrow X, \\ p &: (f^*T^*(Y/S))^\times \longrightarrow \mathbb{P}(f^*T^*(Y/S)) \end{aligned} \tag{5.8.1}$$

to be the projections, $\pi^\times = q_S \circ p$. Let $j : (f^*T^*(Y/S))^\times \to f^*T^*(Y/S)$ be the natural open immersion, and recall that $d_S f : f^*T^*(Y/S) \to T^*(X/S)$ is the relative differential. The composition

$$\mathcal{O}(-1) \to q_S^* f^*T^*(Y/S) \xrightarrow{d_S f} q_S^* T^*(X/S) \tag{5.8.2}$$

defines the section

$$\overline{d_S f} : \mathbb{P}(f^*T^*(Y/S)) \longrightarrow q_S^* T^*(X/S)\otimes\mathcal{O}(1) \tag{5.8.3}$$

as an element in $H^0(\mathbb{P}(f^*T^*(Y/S)), q_S^*T^*(X/S)\otimes\mathcal{O}(1))$. Since it is a section of a vector bundle of rank $n-s$, $\overline{d_S f}$ is a regular immersion of codimension $n-s$.

Lemma 5.9. *In the above setting, we have an inclusion*

$$\mathbb{P}((d_Sf)^{-1}(p_{X/S}SS(\mathcal{F}))) \subset q_S^{-1}(Z). \tag{5.9.1}$$

Moreover, the complex

$$j^*(d_Sf)^*\mathscr{F}_{\psi,T(X/S)}\left(i_{h,0}^*\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k}\otimes^L i_{h,0}^!\Lambda\right) \tag{5.9.2}$$

is supported on $(\pi^\times)^{-1}(Z)$.

Proof. Let $p_{Y/S}: T^*Y \to T^*(Y/S)$ be the natural surjection of bundles. Using the cartesian square

$$\begin{array}{ccc} T^*Y\times_Y X & \xrightarrow{df} & T^*X \\ {\scriptstyle p_{Y/S}\times_Y \mathrm{id}_X}\downarrow & & \downarrow{\scriptstyle p_{X/S}} \\ f^*T^*(Y/S)=T^*(Y/S)\times_Y X & \xrightarrow[d_Sf]{} & T^*(X/S), \end{array} \tag{5.9.3}$$

we have

$$(d_Sf)^{-1}p_{X/S}SS(\mathcal{F}) = (p_{Y/S}\times_Y X)(df)^{-1}SS(\mathcal{F}). \tag{5.9.4}$$

The $SS(\mathcal{F})$-transversality of f over $X\setminus Z$ implies $(df)^{-1}SS(\mathcal{F})\subset T^*_YY\times_Y X$ over $X\setminus Z$, hence taking projectivization we get $\mathbb{P}((d_Sf)^{-1}(p_{X/S}SS(\mathcal{F})))$ is included in $q_S^{-1}(Z)$.

The second part follows from the isomorphism (see Corollary 3.5)

$$\mathscr{F}_{\psi,T(X/S)}\left(i_{h,0}^*\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k}\otimes^L i_{h,0}^!\Lambda\right)\simeq p_{X/S,!}\mu\mathrm{Hom}_X(\mathcal{F},\mathcal{F}) \tag{5.9.5}$$

and Theorem 4.21. □

5.10. Retain the previous setting. Recall that we have

$$cc^Z_{X/Y/S}(\mathcal{F}) = (-1)^s q_*\left[c_{r-1}\left(\frac{q^*f^*T^*(Y/S)}{\mathcal{O}(-1)}\right)\cap \overline{df}^!CC(\mathcal{F})_{\mathcal{O}(1)}\right] \tag{5.10.1}$$

$$= (q_S)_*\left[c_{r-1}\left(\frac{q_S^*f^*T^*(Y/S)}{\mathcal{O}(-1)}\right)\cap \overline{d_Sf}^!((-1)^s p_{X/S*}CC(\mathcal{F}))_{\mathcal{O}(1)}\right] \tag{5.10.2}$$

by (2.14.8) and (2.16.1). Applying (2.17.5) and (2.17.9), we have

$$(\pi^\times)^*cc^Z_{X/Y/S}(\mathcal{F}) = (-1)^s p^*\overline{d_Sf}^!(p_{X/S*}CC(\mathcal{F}))_{\mathcal{O}(1)}. \tag{5.10.3}$$

Using the compatibility of cl with pullback and pushforward, we get

$$\mathrm{cl}\left((\pi^\times)^*cc^Z_{X/Y/S}(\mathcal{F})\right) = (-1)^s\,\mathrm{cl}\left(p^*\overline{d_Sf}^!(p_{X/S*}CC(\mathcal{F}))_{\mathcal{O}(1)}\right) \tag{5.10.4}$$

$$= j^*d_Sf^*\left((-1)^s p_{X/S*}CC_\mu(\mathcal{F})\right). \tag{5.10.5}$$

By diagram (5.7.2), $j^*d_Sf^*\left((-1)^s p_{X/S*}CC_\mu(\mathcal{F})\right)$ is represented by the composition

$$\Lambda\xrightarrow{\mathrm{coev}} j^*(d_Sf)^*\mathscr{F}_{\psi,T(X/S)}\left(i_{h,0}^*\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k}\otimes^L i_{h,0}^!\Lambda\right)\xrightarrow{\mathrm{ev}}(\pi^\times)^*\mathcal{K}_{X/S} \tag{5.10.6}$$

which is equal to

$$\begin{aligned}\Lambda &\xrightarrow{\mathrm{coev}^\vee} R\mathcal{H}om\left(j^*(d_Sf)^*\mathscr{F}_{\psi,T(X/S)}\left(i_{h,0}^*\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k}\otimes^L i_{h,0}^!\Lambda\right),(\pi^\times)^*\mathcal{K}_{X/S}\right)\\ &\xrightarrow{\mathrm{ev}^\vee}(\pi^\times)^*\mathcal{K}_{X/S}\end{aligned} \tag{5.10.7}$$

by Lemma 2.11. Combining (5.10.4), (5.10.6) and (5.10.7), we get the following result.

Proposition 5.11. *The composition*

$$\begin{aligned}\Lambda &\xrightarrow{\text{adj.}} \pi^{\times}_*\Lambda\\ &\longrightarrow \pi^{\times}_* R\mathcal{H}om\Big(j^*(d_Sf)^*\mathscr{F}_{\psi,T(X/S)}\left(i^*_{h,0}\nu_{\delta_{X/k}}\mathcal{H}^{\vee}_{\mathcal{F}/k}\otimes^L i^!_{h,0}\Lambda\right),(\pi^{\times})^*\mathcal{K}_{X/S}\Big)\\ &\longrightarrow \pi^{\times}_*(\pi^{\times})^*\mathcal{K}_{X/S}.\end{aligned}\tag{5.11.1}$$

is equal to $\pi^{\times}_*(\pi^{\times})^*\operatorname{cl}\left(cc^Z_{X/Y/S}(\mathcal{F})\right)$ *as elements in* $H^0_Z(X,\pi^{\times}_*\pi^{\times *}\mathcal{K}_{X/S})$.

5.12. Retain the previous setting. Let

$$\begin{gathered}j_Y\colon Y\times_S Y\setminus\delta_{Y/S}(Y)\hookrightarrow Y\times_S Y,\qquad j_{Y,0}\colon T(Y/S)\setminus 0_{Y/S}(Y)\hookrightarrow T(Y/S),\\ j_{f,0}\colon T(X/S)\setminus T(X/Y)\hookrightarrow T(X/S)\end{gathered}\tag{5.12.1}$$

be the natural open immersions, let $T_Sf: T(X/S)\to f^*T(Y/S)$ be the relative tangential map which is the dual of d_Sf, let $\operatorname{pr}_{T(Y/S)}: f^*T(Y/S)\to T(Y/S)$ be the natural projection. For $\mathcal{G}\in D_{\mathrm{ctf}}(X\times_S X,\Lambda)$ and $\mathcal{L}\in D_{\mathrm{ctf}}(Y\times_S Y,\Lambda)$, we have the following morphism

$$\begin{aligned}b_{\mathcal{G},\mathcal{L}}:\nu_{\delta_{X/S}}\mathcal{G}\otimes^L(T_Sf)^*\operatorname{pr}^*_{T(Y/S)}\nu_{\delta_{Y/S}}\mathcal{L}&\xrightarrow{\text{Proposition 4.10}}\nu_{\delta_{X/S}}\mathcal{G}\otimes^L\nu_{\delta_{X/S}}((f\times f)^*\mathcal{L})\\ &\xrightarrow{\text{Proposition 4.11}}\nu_{\delta_{X/S}}\left(\mathcal{G}\otimes^L(f\times f)^*\mathcal{L}\right)\end{aligned}\tag{5.12.2}$$

in $D_{\mathrm{ctf}}(T(X/S)\overline{\times}_s\eta_0,\Lambda)$, which is natural with respect to both $\mathcal{G}$ and $\mathcal{L}$.

Lemma 5.13. *Retain the above constructions.*

1. *The morphism* $j^*_{f,0}b_{\mathcal{G},j_{Y*}\Lambda}$ *is an isomorphism in* $D_{\mathrm{ctf}}((T(X/S)\setminus T(X/Y))\overline{\times}_s\eta_0,\Lambda)$.
2. *The morphism* $0^*_{X/S}i_{f,0*}i^!_{f,0}b_{\mathcal{G},j_{Y*}\Lambda}$ *is an isomorphism in* $D_{\mathrm{ctf}}(X\overline{\times}_s\eta_0,\Lambda)$.

Proof. (1). Notice that $b_{\mathcal{G},\Lambda}$ is an isomorphism. Consider the localization triangle

$$\delta_{Y/S*}\delta^!_{Y/S}\Lambda\longrightarrow\Lambda\longrightarrow j_{Y*}\Lambda\xrightarrow{+1}.\tag{5.13.1}$$

Applying $j^*_{f,0}b_{\mathcal{G},-}$, it suffices to show the cofibre of

$$\begin{aligned}j^*_{f,0}b_{\mathcal{G},\delta_{Y/S*}\delta^!_{Y/S}\Lambda}: j^*_{f,0}&\left(\nu_{\delta_{X/S}}\mathcal{G}\otimes^L(T_Sf)^*\operatorname{pr}^*_{T(Y/S)}\nu_{\delta_{Y/S}}(\delta_{Y/S*}\delta^!_{Y/S}\Lambda)\right)\\ &\longrightarrow j^*_{f,0}\nu_{\delta_{X/S}}\left(\mathcal{G}\otimes^L(f\times f)^*\delta_{Y/S*}\delta^!_{Y/S}\Lambda\right)\end{aligned}\tag{5.13.2}$$

is 0. By Proposition 4.10 we have

$$\nu_{\delta_{Y/S}}(\delta_{Y/S*}\delta^!_{Y/S}\Lambda)\simeq 0_{Y/S*}\delta^!_{Y/S}\Lambda.\tag{5.13.3}$$

Then use proper base change in the following diagram with cartesian squares:

$$\begin{array}{ccccc}T(X/Y) & \xrightarrow{\pi_{X/Y}} & X & \xrightarrow{f} & Y\\ {\scriptstyle i_{f,0}}\downarrow & & {\scriptstyle 0_{Y/S}\times\mathrm{id}_X}\downarrow & & \downarrow{\scriptstyle 0_{Y/S}}\\ T(X/S) & \xrightarrow[T_Sf]{} & f^*T(Y/S) & \xrightarrow[\operatorname{pr}_{T(Y/S)}]{} & T(Y/S),\end{array}\tag{5.13.4}$$

we have

$$\begin{aligned}\nu_{\delta_{X/S}}\mathcal{G}\otimes^L(T_Sf)^*\operatorname{pr}^*_{T(Y/S)}\nu_{\delta_{Y/S}}(\delta_{Y/S*}\delta^!_{Y/S}\Lambda)&\simeq\nu_{\delta_{X/S}}\mathcal{G}\otimes^L(T_Sf)^*\operatorname{pr}^*_{T(Y/S)}0_{Y/S*}\delta^!_{Y/S}\Lambda\\ &\simeq\nu_{\delta_{X/S}}\mathcal{G}\otimes^L(T_Sf)^*(0_{Y/S}\times\mathrm{id}_X)_*f^*\delta^!_{Y/S}\Lambda\\ &\simeq\nu_{\delta_{X/S}}\mathcal{G}\otimes^L i_{f,0*}\pi^*_{X/Y}f^*\delta^!_{Y/S}\Lambda\\ &\simeq i_{f,0*}\left(i^*_{f,0}\nu_{\delta_{X/S}}\mathcal{G}\otimes^L\pi^*_{X/Y}f^*\delta^!_{Y/S}\Lambda\right),\end{aligned}\tag{5.13.5}$$

where the last isomorphism is induced by projection formula. Hence the source of $j_{f,0}^* b_{\mathcal{G},\delta_{Y/S*}\delta_{Y/S}^!\Lambda}$ is 0 by $j_{f,0}^* i_{f,0*} = 0$. Similarly, for the target of $j_{f,0}^* b_{\mathcal{G},\delta_{Y/S*}\delta_{Y/S}^!\Lambda}$, using the following diagram with cartesian square (see (2.13.2)):

$$\begin{array}{ccc} X \times_Y X & \xrightarrow{i_f} & X \times_S X \\ {\scriptstyle p_f}\downarrow & & \downarrow{\scriptstyle f\times f} \\ Y & \xrightarrow[\delta_{Y/S}]{} & Y \times_S Y, \end{array} \tag{5.13.6}$$

we have

$$\begin{aligned} \nu_{\delta_{X/S}}\left(\mathcal{G} \otimes^L (f\times f)^* \delta_{Y/S*}\delta_{Y/S}^!\Lambda\right) &\simeq \nu_{\delta_{X/S}}\left(\mathcal{G} \otimes^L i_{f*}p_f^*\delta_{Y/S}^!\Lambda\right) \\ &\simeq \nu_{\delta_{X/S}}\left(i_{f*}(i_f^*\mathcal{G} \otimes^L p_f^*\delta_{Y/S}^!\Lambda)\right), \end{aligned} \tag{5.13.7}$$

where the last isomorphism is induced by projection formula. By Proposition 4.10(3), $\nu_{\delta_{X/S}}\left(i_{f*}\left(i_f^*\mathcal{G} \otimes^L p_f^*\delta_{Y/S}^!\Lambda\right)\right)$ is supported on $C_{\delta_{X/S}(X)}(X \times_Y X) \subset T(X/Y)$, hence the target of $j_{f,0}^* b_{\mathcal{G},\delta_{Y/S*}\delta_{Y/S}^!\Lambda}$ is 0. Hence we get the desired consequence.

(2). Apply the localization triangle

$$0_{X/S}^* i_{f,0*} i_{f,0}^!(-) \longrightarrow 0_{X/S}^*(-) \longrightarrow 0_{X/S}^* j_{f,0*} j_{f,0}^*(-) \xrightarrow{+1}. \tag{5.13.8}$$

to $b_{\mathcal{G},j_{Y*}\Lambda}$. By Lemma 4.12, the middle morphism $0_{X/S}^* b_{\mathcal{G},j_{Y*}\Lambda}$ is equal to

$$\delta_{X/S}^*\mathcal{G} \otimes^L f^*\delta_{Y/S}^* j_{Y*}\Lambda \xrightarrow{\simeq} \delta_{X/S}^*\left(\mathcal{G} \otimes^L (f\times f)^* j_{Y*}\Lambda\right). \tag{5.13.9}$$

By (1), the right morphism $0_{X/S}^* j_{f,0*} j_{f,0}^* b_{\mathcal{G},j_{Y*}\Lambda}$ is an isomorphism, hence the left one $0_{X/S}^* i_{f,0*} i_{f,0}^! b_{\mathcal{G},j_{Y*}\Lambda}$ is an isomorphism. □

5.14. Apply $\nu_{\delta_{X/S}}$ to the counit morphism we get

$$\nu_{\delta_{X/S}}\left(i_* i^!(\mathcal{G} \otimes^L (f\times f)^* j_{Y*}\Lambda)\right) \longrightarrow \nu_{\delta_{X/S}}\left(\mathcal{G} \otimes^L (f\times f)^* j_{Y*}\Lambda\right). \tag{5.14.1}$$

By Proposition 4.10(3), the source is supported on $C_{\delta_{X/S}(X)}(X \times_Y X) \subset T(X/Y)$, thus (5.14.1) lifts to $i_{f,0*} i_{f,0}^! \nu_{\delta_{X/S}}\left(\mathcal{G} \otimes^L (f\times f)^* j_{Y*}\Lambda\right)$, giving a natural morphism

$$\nu_{\delta_{X/S}}\left(i_* i^!(\mathcal{G} \otimes^L (f\times f)^* j_{Y*}\Lambda)\right) \longrightarrow i_{f,0*} i_{f,0}^! \nu_{\delta_{X/S}}\left(\mathcal{G} \otimes^L (f\times f)^* j_{Y*}\Lambda\right). \tag{5.14.2}$$

Combining the above discussion, we define the following natural morphism for $\mathcal{G} \in D_{\mathrm{ctf}}(X \times_S X, \Lambda)$:

$$\begin{aligned} \Theta_{\mathcal{G}}\colon \delta_{X/Y}^*\delta_{Y/S}^{\Delta}(\mathcal{G}) &\xrightarrow{\simeq} 0_{X/S}^*\nu_{\delta_{X/S}}(i_* i^!(\mathcal{G} \otimes^L (f\times f)^* j_{Y*}\Lambda)) \\ &\longrightarrow 0_{X/S}^* i_{f,0*} i_{f,0}^! \nu_{\delta_{X/S}}(\mathcal{G} \otimes^L (f\times f)^* j_{Y*}\Lambda) \\ &\xrightarrow{\simeq} 0_{X/S}^* i_{f,0*} i_{f,0}^!(\nu_{\delta_{X/S}}\mathcal{G} \otimes^L (T_S f)^* \operatorname{pr}_{T(Y/S)}^* \nu_{\delta_{Y/S}}(j_{Y*}\Lambda)) \\ &\xrightarrow{\simeq} 0_{X/S}^* i_{f,0*} i_{f,0}^!(\nu_{\delta_{X/S}}\mathcal{G} \otimes^L (T_S f)^* \operatorname{pr}_{T(Y/S)}^* j_{Y,0*}\Lambda) \\ &\xrightarrow{\simeq} 0_{X/Y}^* 0_{Y/S}^{\Delta}(\nu_{\delta_{X/S}}\mathcal{G}). \end{aligned} \tag{5.14.3}$$

Here, the first isomorphism is (2.5.2) and (4.10.7), the second morphism is (5.14.2), the third isomorphism is $\left(0_{X/S}^* i_{f,0*} i_{f,0}^! b_{\mathcal{G},j_{Y*}\Lambda}\right)^{-1}$ by Lemma 5.13(2), the fourth isomorphism is Lemma 4.13, the fifth isomorphism is (2.5.2). In particular, if $\mathcal{G} = \delta_{X/S*}\mathcal{G}_0$ for $\mathcal{G}_0 \in D_{\mathrm{ctf}}(X, \Lambda)$, then $\mathcal{G} \otimes^L$

$(f\times f)^*j_{Y*}\Lambda$ is supported on X and hence (5.14.2) is an isomorphism. In this case, $\Theta_{\delta_{X/S*}\mathcal{G}_0}$ is an isomorphism.

5.15. Retain the previous setting. Recall that (5.5.1) defines a natural morphism

$$a_{\mathcal{F}}: i_{h,0}^*\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k}\otimes^L i_{h,0}^!\Lambda\longrightarrow \nu_{\delta_{X/S}}\mathcal{H}^\vee_{\mathcal{F}/S} \tag{5.15.1}$$

for $\mathcal{F}\in D_{\mathrm{ctf}}(X,\Lambda)$ which satisfies the transversality condition in Section 5.1. Applying the duality functor $D_{T(X/S)/S}(-)$, we get a natural morphism

(5.15.2)
$$\nu_{\delta_{X/S}}\mathcal{H}_{\mathcal{F}/S}\simeq D_{T(X/S)/S}\left(\nu_{\delta_{X/S}}\mathcal{H}^\vee_{\mathcal{F}/S}\right)\xrightarrow{D_{T(X/S)/S}(a_{\mathcal{F}})} D_{T(X/S)/S}\left(i_{h,0}^*\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k}\otimes^L i_{h,0}^!\Lambda\right).$$

Let $\mathcal{N}\in D_{\mathrm{ctf}}(f^*T(Y/S)\overline{\times}_s\eta_0,\Lambda)$ be monodromic, we have

$$\begin{aligned}(0_{Y/S}\times\mathrm{id}_X)^\Delta D_{f^*T(Y/S)/S}\mathcal{N}&\simeq D_{X/S}((0_{Y/S}\times\mathrm{id}_X)^\Delta\mathcal{N})(-r)[1-2r]\\&\simeq D_{X/S}(\pi_!^\times j^*\mathscr{F}_{\psi,f^*T(Y/S)}\mathcal{N}[1])(-r)[1-2r]\\&\simeq \pi_*^\times D_{f^*T^*(Y/S)^\times/S}(j^*\mathscr{F}_{\psi,f^*T(Y/S)}\mathcal{N})(-r)[-2r]\\&\simeq \pi_*^\times R\mathcal{H}om(j^*\mathscr{F}_{\psi,f^*T(Y/S)}\mathcal{N},(\pi^\times)^*\mathcal{K}_{X/S}),\end{aligned}\tag{5.15.3}$$

where the first isomorphism is by (2.6.11), the second is Lemma 3.8, the third and the fourth are the properties of the duality functor. Here, we use the previous notation such that $j:(f^*T^*(Y/S))^\times\to f^*T^*(Y/S)$ is the open immersion and $\pi^\times:(f^*T^*(Y/S))^\times\to X$ is the projection.

Let $\mathcal{M}\in D_{\mathrm{ctf}}(T(X/S)\overline{\times}_s\eta_0,\Lambda)$ be a constructible complex. Using Lemma 2.6, we have

$$\begin{aligned}(0_{Y/S}\times\mathrm{id}_X)^\Delta(T_Sf)_*\mathcal{M}&\simeq \pi_{X/Y*}(0_{Y/S}\times\mathrm{id})^\Delta_{T_Sf}(\mathcal{M})\\&\simeq \pi_{X/Y*}0^\Delta_{Y/S}(\mathcal{M})\\&\simeq \pi_{X/S*}i_{f,0*}0^\Delta_{Y/S}(\mathcal{M})\\&\simeq 0^*_{X/S}i_{f,0*}0^\Delta_{Y/S}(\mathcal{M})\\&\simeq 0^*_{X/Y}i^*_{f,0}i_{f,0*}0^\Delta_{Y/S}(\mathcal{M})\\&\simeq 0^*_{X/Y}0^\Delta_{Y/S}(\mathcal{M}),\end{aligned}\tag{5.15.4}$$

here the fourth isomorphism is Lemma 3.7.

By Proposition 4.10, the complex $i_{h,0}^*\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k}\otimes^L i_{h,0}^!\Lambda$ is monodromic. Combining the calculation above, we have the following morphism for every $\mathcal{G}\in D_{\mathrm{ctf}}(X\times_S X,\Lambda)$:

$$\begin{aligned}\widetilde{\Theta_{\mathcal{G}}}:\delta^*_{X/Y}\delta^\Delta_{Y/S}\mathcal{G}&\xrightarrow{\Theta_{\mathcal{G}}}0^*_{X/Y}0^\Delta_{Y/S}\nu_{\delta_{X/S}}\mathcal{G}\\&\xrightarrow{\simeq}0^*_{X/Y}0^\Delta_{Y/S}D_{T(X/S)/S}(\nu_{\delta_{X/S}}D_{(X\times_SX)/S}\mathcal{G})\\&\xrightarrow{\simeq}(0_{Y/S}\times\mathrm{id}_X)^\Delta(T_Sf)_*D_{T(X/S)/S}(\nu_{\delta_{X/S}}D_{(X\times_SX)/S}\mathcal{G})\\&\xrightarrow{\simeq}(0_{Y/S}\times\mathrm{id}_X)^\Delta D_{f^*T(Y/S)/S}((T_Sf)_!\nu_{\delta_{X/S}}D_{(X\times_SX)/S}\mathcal{G})\\&\xrightarrow{\simeq}\pi_*^\times R\mathcal{H}om\Big(j^*\mathscr{F}_{\psi,f^*T(Y/S)}(T_Sf)_!\nu_{\delta_{X/S}}D_{(X\times_SX)/S}\mathcal{G},(\pi^\times)^*\mathcal{K}_{X/S}\Big)\\&\xrightarrow{\simeq}\pi_*^\times R\mathcal{H}om\Big(j^*(d_Sf)^*\mathscr{F}_{\psi,T(X/S)}\nu_{\delta_{X/S}}D_{(X\times_SX)/S}\mathcal{G},(\pi^\times)^*\mathcal{K}_{X/S}\Big).\end{aligned}\tag{5.15.5}$$

The first morphism is (5.14.3), the second isomorphism is (4.10.10) and biduality, the third is the inverse of (5.15.4), and the fourth is duality for direct image. The last two isomorphisms are

(5.15.3) and (3.2.6), respectively. In particular, for $\mathcal{G} = \delta_{X/S*}\mathcal{G}_0$ with $\mathcal{G}_0 \in D_{\mathrm{ctf}}(X, \Lambda)$, base change for duality and (4.10.8) give

$$\nu_{\delta_{X/S}} D_{(X\times_S X)/S}(\delta_{X/S*}\mathcal{G}_0) \simeq \nu_{\delta_{X/S}}(\delta_{X/S*} D_{X/S}\mathcal{G}_0) \simeq 0_{X/S*} D_{X/S}\mathcal{G}_0, \tag{5.15.6}$$

hence $j^*(d_S f)^* \mathscr{F}_{\psi, T(X/S)} \nu_{\delta_{X/S}} D_{(X\times_S X)/S} \delta_{X/S*}\mathcal{G}_0 \simeq (\pi^\times)^* D_{X/S}\mathcal{G}_0$ by (3.2.1), and hence the target of $\widetilde{\Theta}_{\delta_{X/S*}\mathcal{G}_0}$ can be calculated as follows:

$$\begin{aligned} &\pi^\times_* R\mathcal{H}om\Big((\pi^\times)^* D_{X/S}\mathcal{G}_0, (\pi^\times)^* \mathcal{K}_{X/S}\Big) \\ &\simeq \pi^\times_* D_{(f^*T^*(Y/S))^\times/S}((\pi^\times)^* D_{X/S}\mathcal{G}_0)(-r)[-2r] \\ &\simeq \pi^\times_* (\pi^\times)^! D_{X/S} D_{X/S}\mathcal{G}_0(-r)[-2r] \\ &\simeq \pi^\times_* (\pi^\times)^* \mathcal{G}_0, \end{aligned} \tag{5.15.7}$$

thus we have

$$\widetilde{\Theta}_{\delta_{X/S*}\mathcal{G}_0} : \quad \delta^*_{X/Y}\delta^\Delta_{Y/S}\delta_{X/S*}\mathcal{G}_0 \xrightarrow{\simeq} \pi^\times_*(\pi^\times)^*\mathcal{G}_0. \tag{5.15.8}$$

It is an isomorphism because $\Theta_{\delta_{X/S*}\mathcal{G}_0}$ is an isomorphism by the discussion following (5.14.3).

5.16. *Proof of Theorem 5.2.* Put $\mathscr{P} = i^*_{h,0}\nu_{\delta_{X/k}}\mathcal{H}^\vee_{\mathcal{F}/k} \otimes^L i^!_{h,0}\Lambda$, we verify the following diagram is commutative:

$$\begin{array}{ccccc} \Lambda & \xrightarrow{(2.13.5)} & \delta^*_{X/Y}\delta^\Delta_{Y/S}\mathcal{H}_{\mathcal{F}/S} & \xrightarrow{\mathrm{ev}} & \mathcal{K}_{X/Y/S} \\ {\scriptstyle (-1)^r}\downarrow & & \downarrow{\scriptstyle \widetilde{\Theta}_{\mathcal{H}_{\mathcal{F}/S}}} & & \downarrow{\scriptstyle \simeq} \\ \Lambda & \xrightarrow[\mathrm{coev}^\vee]{} & \pi^\times_* R\mathcal{H}om\Big(j^*(d_S f)^*\mathscr{F}_{\psi,T(X/S)} & \xrightarrow{\mathrm{ev}^\vee} & \pi^\times_*(\pi^\times)^*\mathcal{K}_{X/S} \\ \| & & \nu_{\delta_{X/S}}\mathcal{H}^\vee_{\mathcal{F}/S}, (\pi^\times)^*\mathcal{K}_{X/S}\Big) & & \| \\ & & \downarrow{\scriptstyle a_{\mathcal{F}}} & & \\ \Lambda & \xrightarrow[\mathrm{coev}^\vee]{} & \pi^\times_* R\mathcal{H}om\Big(j^*(d_S f)^*\mathscr{F}_{\psi,T(X/S)}\mathscr{P}, & \xrightarrow[\mathrm{ev}^\vee]{} & \pi^\times_*(\pi^\times)^*\mathcal{K}_{X/S}. \\ & & (\pi^\times)^*\mathcal{K}_{X/S}\Big) & & \end{array} \tag{5.16.1}$$

The two lower squares are commutative by commutativity of (5.7.1). The upper right square is commutative by the naturality of (5.15.5) , using (5.15.6)–(5.15.8) and the compatibility of evaluation and coevaluation with duality. The upper left square is commutative by the naturality of (5.15.5) together with the commutativity of the diagram obtained from (3.8.9) by taking $\mathcal{N} = (0_{Y/S} \times \mathrm{id}_X)_*\mathcal{K}_{X/S}$, applying $D_{X/S}(-)(-r)[-2r]$. Here we use (5.15.6)–(5.15.8) and the compatibility with the boundary given by (2.5.2) and (4.7.2); the vertical morphism on Λ is the dual of the first vertical morphism of (3.8.9), hence is $(-1)^r$.

For the final comparison, apply the same calculation to $(0_{Y/S} \times \mathrm{id}_X)_*\Lambda$. Since $D_{X/S}\mathcal{K}_{X/S} = \Lambda$, and the boundary for $\delta_{X/S*}\mathcal{K}_{X/S}$ is α by (2.13.3) and (2.13.4), we obtain a commutative diagram

$$\begin{array}{ccc} \mathcal{K}_{X/S} & \xrightarrow{\alpha} & \mathcal{K}_{X/Y/S} \\ {\scriptstyle (-1)^r}\downarrow & & \downarrow{\scriptstyle \simeq}\widetilde{\Theta}_{\delta_{X/S*}\mathcal{K}_{X/S}} \\ \mathcal{K}_{X/S} & \xrightarrow[\eta]{} & \pi^\times_*(\pi^\times)^*\mathcal{K}_{X/S}. \end{array} \tag{5.16.2}$$

The lower morphism is the adjunction unit because $D_{X/S}$ sends the counit $\pi_!^\times(\pi^\times)^!\Lambda \to \Lambda$ to

$$\mathcal{K}_{X/S} \xrightarrow{\eta} \pi_*^\times(\pi^\times)^*\mathcal{K}_{X/S} \simeq \mathcal{K}_{X/S} \otimes^L \pi_*^\times\Lambda. \tag{5.16.3}$$

The upper middle object of (5.16.1) is supported on Z by Section 2.13; the bottom middle object is also supported on Z, by Lemma 5.9. Thus both outer composites of (5.16.1) are classes supported on Z. By Proposition 5.11, its bottom composite represents the image of $\mathrm{cl}(cc^Z_{X/Y/S}(\mathcal{F}))$ under the adjunction unit. Combining this with (5.16.2), we obtain

$$\begin{aligned}\widetilde{\Theta}_{\delta_{X/S*}\mathcal{K}_{X/S}}\widetilde{C}^Z_{X/Y/S}(\mathcal{F}) &= (-1)^r\eta(\mathrm{cl}(cc^Z_{X/Y/S}(\mathcal{F}))) \\ &= \widetilde{\Theta}_{\delta_{X/S*}\mathcal{K}_{X/S}}(\alpha_Z\,\mathrm{cl}(cc^Z_{X/Y/S}(\mathcal{F}))).\end{aligned} \tag{5.16.4}$$

Since $\widetilde{\Theta}_{\delta_{X/S*}\mathcal{K}_{X/S}}$ is an isomorphism by (5.15.8), we have

$$p_X^*(\widetilde{C}^Z_{X/Y/S}(\mathcal{F})) = p_X^*(\alpha_Z\,\mathrm{cl}(cc^Z_{X/Y/S}(\mathcal{F}))) \qquad \text{in } H^0_Z(X\overline{\times}_s\eta_0, p_X^*\mathcal{K}_{X/Y/S}). \tag{5.16.5}$$

The inflation map (4.5.2) is injective by Proposition 4.4. Therefore, (5.1.2) yields

$$\widetilde{C}^Z_{X/Y/S}(\mathcal{F}) = \alpha_Z\,\mathrm{cl}(cc^Z_{X/Y/S}(\mathcal{F})) = \widetilde{\mathrm{cl}}(cc^Z_{X/Y/S}(\mathcal{F})) \quad \text{in } H^0_Z(X, \mathcal{K}_{X/Y/S}) \tag{5.16.6}$$

which completes the proof. □

6. Applications

Throughout this section, k is a perfect field of characteristic $p > 0$ and Λ is a finite local ring whose residue field has characteristic $\ell \neq p$. We give some applications of Theorem 5.2.

6.1. Let $h : X \to S$ be a separated morphism of smooth schemes of pure dimension over k, with S connected of dimension s. Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$ such that h is $SS(\mathcal{F})$-transversal. By Proposition 2.9(2), h is universally locally acyclic relatively to $\mathcal{F}$, so we may define the cohomological characteristic class $C_{X/S}(\mathcal{F}) \in H^0(X, \mathcal{K}_{X/S})$ as in (2.12.6). Recall the geometric relative characteristic class is defined as

$$cc_{X/S}(\mathcal{F}) := (-1)^s(dh)^!CC(\mathcal{F}) \in \mathrm{CH}_s(X), \qquad dh : X \times_S T^*S \longrightarrow T^*X, \tag{6.1.1}$$

cf. [3, Definition 3.5]. Here, by the transversality assumption, $(-1)^s(dh)^!CC(\mathcal{F})$ is supported on the zero section of $X \times_S T^*S$, hence it can be viewed as a class on X.

Theorem 6.2. *Under the above assumptions, we have*

$$C_{X/S}(\mathcal{F}) = \mathrm{cl}(cc_{X/S}(\mathcal{F})) \quad \textit{in } H^0(X, \mathcal{K}_{X/S}). \tag{6.2.1}$$

Proof. Put $Y = \mathbb{A}^1_S = \mathbb{A}^1_k \times_k S$, let $g : Y \to S$ be the projection, let $f = (0, h) : X \to Y$. We apply Theorem 5.2 in this case with $Z = X$.

We first calculate the geometric class. Denote the projection on $Y = \mathbb{A}^1_S = \mathbb{A}^1_k \times_k S$ to $\mathbb{A}^1_k$ and S by p_1 and p_2, respectively. By the construction of Y and identification $T^*\mathbb{A}^1_k \simeq \mathcal{O}_{\mathbb{A}^1_k}$, we have $T^*Y \simeq p_1^*\mathcal{O}_{\mathbb{A}^1_k} \oplus p_2^*T^*S$, and hence $f^*T^*Y = h^*T^*S \oplus \mathcal{O}_X$. The $SS(\mathcal{F})$-transversality of h implies that $\mathbb{P}((df)^{-1}SS(\mathcal{F}))$ is contained in the open subscheme $\mathbb{P}(h^*T^*S \oplus \mathcal{O}_X) \setminus \mathbb{P}(h^*T^*S) \simeq X \times_S T^*S$ of $\mathbb{P}(f^*T^*Y = h^*T^*S \oplus \mathcal{O}_X)$. On this open subscheme, the canonical line bundle $\mathcal{O}(-1)$ is trivial and the section $\overline{df} = \mathrm{id} \times dh$. Since $r = 1$, the Chern class in (2.14.8) is $c_0 = 1$, we have

$$cc^X_{X/Y/S}(\mathcal{F}) = (-1)^s(dh)^!CC(\mathcal{F}) = cc_{X/S}(\mathcal{F}). \tag{6.2.2}$$

We now calculate the cohomology side. Since the zero section $S \hookrightarrow Y = \mathbb{A}^1_S$ is a monomorphism, we have $X \times_Y X = X \times_S X$. Thus the immersion $i : X \times_Y X \to X \times_S X$ in (2.13.2) is the identity. Comparing the definitions (2.13.5) with (2.12.6), we obtain

$$\widetilde{C}^X_{X/Y/S}(\mathcal{F}) = \alpha(C_{X/S}(\mathcal{F})), \tag{6.2.3}$$

where $\alpha : \mathcal{K}_{X/S} \to \mathcal{K}_{X/Y/S}$ is the morphism in (2.13.4). By Lemma 2.6(4), in our situation the triangle (2.13.4) takes the form

$$\mathcal{K}_{X/S}(-1)[-2] \xrightarrow{\cap c_1(\mathcal{O}_X)} \mathcal{K}_{X/S} \xrightarrow{\alpha} \mathcal{K}_{X/Y/S} \xrightarrow{+1}, \tag{6.2.4}$$

hence $H^0(\alpha)$ is injective since $c_1(\mathcal{O}_X) = 0$. Theorem 5.2, together with (5.1.2), (6.2.2) and (6.2.3), now gives $\alpha(C_{X/S}(\mathcal{F})) = \widetilde{\mathrm{cl}}(cc^X_{X/Y/S}(\mathcal{F})) = \alpha(\mathrm{cl}(cc_{X/S}(\mathcal{F})))$. The injectivity of α implies (6.2.1). □

When h is smooth of relative dimension d, let $p_{X/S} : T^*X \to T^*(X/S)$ be the projection and $0_{X/S} : X \to T^*(X/S)$ the zero section. The restriction of $p_{X/S}$ to $SS(\mathcal{F})$ is finite, and the cartesian square

$$\begin{array}{ccc} X \times_S T^*S & \xrightarrow{dh} & T^*X \\ \downarrow & & \downarrow{\scriptstyle p_{X/S}} \\ X & \xrightarrow[0_{X/S}]{} & T^*(X/S) \end{array} \tag{6.2.5}$$

identifies (6.1.1) with

$$cc_{X/S}(\mathcal{F}) = (-1)^s 0^!_{X/S}(p_{X/S})_* CC(\mathcal{F}) \in \mathrm{CH}^d(X). \tag{6.2.6}$$

Thus (6.2.1) is an equality in $H^{2d}(X, \Lambda(d))$. It proves [22, Conjecture 3.8], and its formulation in [3, Conjecture 3.6].

6.3. Consider the setting of Section 5.1, we study when the equality of Theorem 5.2 lifts to an equality in $H^0_Z(X, \mathcal{K}_{X/S})$. Theorem 5.2 implies that, if the map

$$\alpha^0 : H^0_Z(X, \mathcal{K}_{X/S}) \longrightarrow H^0_Z(X, \mathcal{K}_{X/Y/S}) \tag{6.3.1}$$

is injective, then $\widetilde{C}^Z_{X/Y/S}(\mathcal{F})$ lifts canonically to $C^Z_{X/Y/S}(\mathcal{F}) \in H^0_Z(X, \mathcal{K}_{X/S})$ such that

$$C^Z_{X/Y/S}(\mathcal{F}) = \mathrm{cl}(cc^Z_{X/Y/S}(\mathcal{F})), \qquad \alpha^0(C^Z_{X/Y/S}(\mathcal{F})) = \widetilde{C}^Z_{X/Y/S}(\mathcal{F}). \tag{6.3.2}$$

By the distinguished triangle (2.13.4), injectivity of α^0 is equivalent to the image of

$$H^0_Z(X, \mathcal{K}_{X/Y}) \longrightarrow H^0_Z(X, \mathcal{K}_{X/S}) \tag{6.3.3}$$

being zero. We record some sufficient conditions for later use.

Lemma 6.4. *Retain the setting above, then each condition below implies α^0 is injective.*

(1) $H^0_Z(X, \mathcal{K}_{X/Y}) = 0$;
(2) $\dim Z < \dim Y$;
(3) $c_r(f^*T^*(Y/S))|_Z = 0$ *in* $H^{2r}(Z, \Lambda(r))$;
(4) $T^*(Y/S)$ *is a trivial vector bundle.*

Proof. (1). By the long exact sequence induced from (2.13.4) the consequence follows.
(2). We have $H^0_Z(X, \mathcal{K}_{X/Y}) = 0$ in this case, so (1) applies.
(3). Using Lemma 2.6(4), cf. [23, Lemma 2.6(2)].
(4). This follows directly from (3). □

6.5. Let Y be a smooth connected curve over k, let $y \in Y$ be a closed point, and let $f : X \to Y$ be a separated morphism with X smooth of pure dimension n over k. Put $X_y = f^{-1}(y)$, we have the following cohomological version of Bloch's conductor formula, cf. [23, Remark 6.6].

Theorem 6.6. *Assume that f is flat and that $f|_{X\setminus X_y} : X \setminus X_y \to Y \setminus \{y\}$ is smooth. Then $\widetilde{C}^{X_y}_{X/Y/k}(\Lambda)$ lifts uniquely to $C^{X_y}_{X/Y/k}(\Lambda) \in H^0_{X_y}(X, \mathcal{K}_{X/k})$, and we have*

$$C^{X_y}_{X/Y/k}(\Lambda) = (-1)^n \operatorname{cl}\left(c^X_{n,X_y}(\Omega^1_{X/Y}) \cap [X]\right) \quad \text{in } H^0(X_y, \mathcal{K}_{X_y/k}) \simeq H^0_{X_y}(X, \mathcal{K}_{X/k}). \tag{6.6.1}$$

Here $c^X_{n,X_y}(\Omega^1_{X/Y}) \in \mathrm{CH}^n(X_y \to X)$ is the localized Chern class, cf. [9, Chapter 18], [5].

Proof. Notice that we have $CC(\Lambda) = (-1)^n[T^*_X X]$ by [16, Lemma 5.11], hence smoothness of f outside X_y gives the transversality condition in Theorem 5.2. By the étale locality of non-acyclicity classes [23, Proposition 4.6], we may assume $Y = \mathbb{A}^1_k$ with coordinate t, $y = \{0\}$. In this case we have a global section of $\Omega^1_{Y/k}$ defined by dt which gives a section

$$f^*dt : X \longrightarrow X \times_Y T^*Y \longrightarrow T^*X \tag{6.6.2}$$

which we still denote by df. The zero scheme of df is supported on the nonsmooth locus of f, hence contained in X_y. Lemma 6.4(4) implies $\widetilde{C}^{X_y}_{X/Y/k}(\Lambda)$ uniquely lifts to $C^{X_y}_{X/Y/k}(\Lambda) \in H^0_{X_y}(X, \mathcal{K}_{X/k})$, and we have $C^{X_y}_{X/Y/k}(\Lambda) = \operatorname{cl}(cc^{X_y}_{X/Y/k}(\Lambda))$. It suffices to verify $\operatorname{cl}(cc^{X_y}_{X/Y/k}(\Lambda)) = (-1)^n \operatorname{cl}\left(c^X_{n,X_y}(\Omega^1_{X/Y}) \cap [X]\right)$.

By Proposition 2.15 we have $cc^{X_y}_{X/Y/k}(\Lambda) = (df)^! CC(\Lambda) = (-1)^n (df)^![T^*_X X]$. The smoothness of f outside X_y and flatness of f imply $df : f^*\Omega^1_{Y/k} \to \Omega^1_{X/k}$ is injective on every generic point of X, hence it is injective on all of X. Therefore, we have the following exact sequence

$$0 \longrightarrow f^*\Omega^1_{Y/k} \xrightarrow{df} \Omega^1_{X/k} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0. \tag{6.6.3}$$

It gives a locally free resolution of $\Omega^1_{X/Y}$. By [17, Lemma 2.1.4] we have

$$(df)^![T^*_X X] = c^X_{n,X_y}(\Omega^1_{X/Y}) \cap [X] \quad \text{in } \mathrm{CH}_0(X_y), \tag{6.6.4}$$

hence the consequence follows. □

We record the relation of Theorem 6.6 with the conductor formula, cf. [23, Section 6].

Corollary 6.7. *In the setting of Theorem 6.6, assume in addition that f is proper. Let $\bar{\eta}$ be a geometric generic point of the strict localization of Y at a geometric point $\bar{y}$ above y. Consider the Artin conductor*

$$a_y(Rf_*\Lambda) := \operatorname{rank}(Rf_*\Lambda)_{\bar{\eta}} - \operatorname{rank}(Rf_*\Lambda)_{\bar{y}} + \operatorname{Sw}_y(Rf_*\Lambda), \tag{6.7.1}$$

then we have

$$f_* C^{X_y}_{X/Y/k}(\Lambda) = (-1)^n \deg\left(c^X_{n,X_y}(\Omega^1_{X/Y}) \cap [X]\right) = -a_y(Rf_*\Lambda) \tag{6.7.2}$$

in $H^0_y(Y, \mathcal{K}_{Y/k}) = \Lambda$.

Proof. Compatibility of the cycle class map with proper pushforward and Theorem 6.6 give the first equality. For the second, we denote map $H^0_y(Y, \mathcal{K}_{Y/k}) \xrightarrow{\alpha} H^0_y(Y, \mathcal{K}_{Y/Y/k})$ by α_y. Proper pushforward of non-acyclicity classes gives $\alpha_y(f_* C^{X_y}_{X/Y/k}(\Lambda)) = \widetilde{C}^{\{y\}}_{Y/Y/k}(Rf_*\Lambda)$. Absolute purity gives $H^0_y(Y, \Lambda) = H^1_y(Y, \Lambda) = 0$, so α_y is an isomorphism and $f_* C^{X_y}_{X/Y/k}(\Lambda) = C^{\{y\}}_{Y/Y/k}(Rf_*\Lambda)$. The cohomological Milnor formula [23, Section 6] identifies this term with $-\operatorname{dimtot} R\Phi_{\bar{y}}(Rf_*\Lambda, \mathrm{id}_Y) = -a_y(Rf_*\Lambda)$. □

6.8. Let X be a smooth connected scheme of dimension $n > 0$ over k, let $Z \subsetneq X$ be a closed subscheme, and let $j : U = X \setminus Z \hookrightarrow X$ be the open immersion. We recall the localized characteristic classes of Abbes–Saito [1, Section 5.2] in terms of the construction in Section 2.13. Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$ be a constructible complex such that $\mathcal{F}|_U$ is locally constant. For $S = \operatorname{Spec} k$, $Y = X$ and $f = \mathrm{id}_X$, the triangle (2.13.4) becomes

$$\Lambda \longrightarrow \mathcal{K}_{X/k} \xrightarrow{\alpha} \mathcal{K}_{X/X/k} \xrightarrow{+1} . \tag{6.8.1}$$

Since U is dense, purity gives $H^0_Z(X, \Lambda) = H^1_Z(X, \Lambda) = 0$. Thus the morphism

$$\alpha^0 : H^0_Z(X, \mathcal{K}_{X/k}) \xrightarrow{\simeq} H^0_Z(X, \mathcal{K}_{X/X/k}) \tag{6.8.2}$$

is an isomorphism. The *localized characteristic class* is defined as

$$C^0_Z(\mathcal{F}) := (\alpha^0)^{-1}(\widetilde{C}^Z_{X/X/k}(\mathcal{F})) = C^Z_{X/X/k}(\mathcal{F}) \in H^0_Z(X, \mathcal{K}_{X/k}). \tag{6.8.3}$$

By [1, Proposition 5.2.3], the image of $C^0_Z(\mathcal{F})$ under the natural map $H^0_Z(X, \mathcal{K}_{X/k}) \to H^0(X, \mathcal{K}_{X/k})$ is

$$C_{X/k}(\mathcal{F}) - \operatorname{rank}(\mathcal{F}|_U) \cdot C_{X/k}(\Lambda) \quad \text{in } H^0(X, \mathcal{K}_{X/k}). \tag{6.8.4}$$

For a locally constant sheaf $\mathcal{L}$ of finite free Λ-modules of rank r on U, we put

$$C^{00}_Z(j_!\mathcal{L}) := C^0_Z(j_!\mathcal{L}) - r \cdot C^0_Z(j_!\Lambda) \in H^0_Z(X, \mathcal{K}_{X/k}), \tag{6.8.5}$$

cf. [19, Definition 3.4 and Remark 3.5]. Its image in $H^0(X, \mathcal{K}_{X/k})$ is $C_{X/k}(j_!\mathcal{L}) - r \cdot C_{X/k}(j_!\Lambda)$.

Let $0_X : X \hookrightarrow T^*X$ be the zero section, recall that we have $CC(\Lambda) = (-1)^n[T^*_X X]$. Since $CC(\mathcal{F})|_{T^*U} = r \cdot CC(\Lambda)$, the cycle $CC(\mathcal{F}) - \operatorname{rank}(\mathcal{F}|_U) \cdot CC(\Lambda)$ is supported on $T^*X \times_X Z$.

Theorem 6.9. *Under the above assumptions, we have*

$$C^0_Z(\mathcal{F}) = \mathrm{cl}\left(0^!_X(CC(\mathcal{F}) - \operatorname{rank}(\mathcal{F}|_U) \cdot CC(\Lambda))\right) \quad \textit{in } H^0_Z(X, \mathcal{K}_{X/k}). \tag{6.9.1}$$

Proof. We apply Theorem 5.2 with $S = \operatorname{Spec} k$, $Y = X$ and $f = \mathrm{id}_X$. The local constancy of $\mathcal{F}$ on U implies $SS(\mathcal{F}|_U) \subset T^*_U U$, hence $f|_U$ is $SS(\mathcal{F}|_U)$-transversal.

We calculate the geometric class in this case. Put $A = CC(\mathcal{F}) - \operatorname{rank}(\mathcal{F}|_U) \cdot CC(\Lambda) \in \mathrm{Z}_n(X)$, let $q : \mathbb{P}(T^*X) \to X$ be the projection. The section $\overline{df} = \overline{d(\mathrm{id}_X)}$ is the nowhere vanishing section of $q^*T^*X \otimes \mathcal{O}(1)$ induced by $\mathcal{O}(-1) \hookrightarrow q^*T^*X$. In this case we have $\overline{df}^!(A) = \mathbb{P}(A)$, therefore (2.14.8), with $r = n$ and $s = 0$, gives

$$cc^Z_{X/X/k}(\mathcal{F}) = q_{Z*}\left[c_{n-1}\left(\frac{q^*T^*X}{\mathcal{O}(-1)}\right) \cap \mathbb{P}(A)\right]. \tag{6.9.2}$$

Put $\xi = c_1(\mathcal{O}(1))$, we have

$$c_{n-1}\left(\frac{q^*T^*X}{\mathcal{O}(-1)}\right) = \sum_{i=0}^{n-1} q^* c_{n-1-i}(T^*X)\, \xi^i. \tag{6.9.3}$$

Let $\overline{A}$ be the componentwise closure of A in $\mathbb{P}(T^*X \oplus \mathcal{O}_X)$, with projection $\overline{q}_Z : \overline{A} \to Z$. Since $\dim Z < n$, we have $\overline{q}_{Z*}\overline{A} = 0$ and $\mathbb{P}(E) \cdot \overline{A} = \mathbb{P}(A)$. Writing $\{-\}_0$ for the zero-dimensional component of a cycle, the Segre class description of intersection with the zero section [9, Example 4.1.8]

gives

$$
\begin{aligned}
0_X^! A &= \left\{ c(T^*X|_Z) \cap \overline{q}_{Z*} \left(\sum_{j=0}^{n} \xi^j \cap \overline{A} \right) \right\}_0 \\
&= \sum_{i=0}^{n-1} c_{n-1-i}(T^*X|_Z) \cap q_{Z*}(\xi^i \cap \mathbb{P}(A)) \\
&= q_{Z*} \left[c_{n-1} \left(\frac{q^*T^*X}{\mathcal{O}(-1)} \right) \cap \mathbb{P}(A) \right],
\end{aligned} \tag{6.9.4}
$$

which is the right-hand side of (6.9.2). Hence we obtain

$$
cc^Z_{X/X/k}(\mathcal{F}) = 0_X^!(CC(\mathcal{F}) - \operatorname{rank}(\mathcal{F}) \cdot CC(\Lambda)) \quad \text{in } \mathrm{CH}_0(Z). \tag{6.9.5}
$$

By Theorem 5.2 and (5.1.2), we have

$$
\alpha^0(C^0_Z(\mathcal{F})) = \widetilde{C}^Z_{X/X/k}(\mathcal{F}) = \alpha^0 \left(\mathrm{cl}(cc^Z_{X/X/k}(\mathcal{F})) \right). \tag{6.9.6}
$$

Now (6.8.2) and (6.9.5) imply the assertion. □

For $\mathcal{L}$ as in (6.8.5), define

$$
\mathrm{Sw}^{cc}_X(\mathcal{L}) := 0_X^!(CC(j_!\mathcal{L}) - r \cdot CC(j_!\Lambda)) \in \mathrm{CH}_0(Z), \tag{6.9.7}
$$

see [16, Definition 6.7(3)], [20, (6.1.1)].

Corollary 6.10. *With the above notation, we have*

$$
C^{00}_Z(j_!\mathcal{L}) = \mathrm{cl}(\mathrm{Sw}^{cc}_X(\mathcal{L})) \quad \textit{in } H^0_Z(X, \mathcal{K}_{X/k}). \tag{6.10.1}
$$

Proof. Applying Theorem 6.9 to $j_!\mathcal{L}$ and $j_!\Lambda$, and using (6.8.5) and (6.9.7), the assertion follows. □

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